

JESS Thermodynamic Database v8.9

Neptunium

24-Jan-23

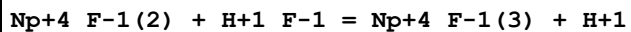
Reaction No. 2704							
F-1 + NpO ₂ +1 = NpO ₂ +1_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.60 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:1.2 (0.3SD)	5	CRV	25812
3	25	0	Inf. Dilution/m	lgK:0.92 (2SF)	1	CRV	24953
4	25	0.1	Unknown	lgK:1.6 (0.1SD)	0	CRV	2556
5	25	1	Unknown	lgK:1.3 (0.1SD)	3	CRV	2556
6	25	2	NaClO ₄	lgK:1. (0.07SD)	6	CMN	437
7	25	2	NaClO ₄	lgK:1. (0.04SD)	6	MSE	5597
8	25	2	Unknown/m	lgK:0.28 (2SF)	0	CRV	24953
9	50	0	Inf. Dilution/m	lgK:1.41 (3SF)	2	CRV	24953
10	75	0	Inf. Dilution/m	lgK:1.86 (3SF)	1	CRV	24953
11	100	0	Inf. Dilution/m	lgK:2.28 (3SF)	0	CRV	24953
12	150	0	Inf. Dilution/m	lgK:3.04 (3SF)	0	CRV	24953
13	200	0	Inf. Dilution/m	lgK:3.71 (3SF)	0	CRV	24953
14	25	0	Inf. Dilution	dH:40.77 (30.04SD) kJ	4	CRV	32222

Reaction No. 2765							
Np+4 + H+1_F-1 = Np+4_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO ₄	lgR:4.8 (0.05SD)	5	MIX	437
2	25	1	HClO ₄	lgR:4.70 (3SF)	4	MCE	437
3	25	1	HClO ₄	lgR:4.60 (3SF)	4	MDS	437
4	25	1	HClO ₄	lgR:4.56 (3SF)	5	MCE	5398
5	25	1	HNO ₃	lgR:4.23 (3SF)	3	MIX	437
6	25	2	HClO ₄	lgR:4.60 (3SF)	3	MDS	437
7	25	2	HClO ₄	lgR:4.70 (3SF)	5	MCE	5398
8	25	2	HNO ₃	lgR:4.11 (3SF)	3	MIX	437

Reaction No. 2766							
Np+4_F-1 + H+1_F-1 = Np+4_F-1(2) + H+1							

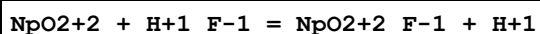
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgR:2.7 (0.05SD)	4	MAX	437
2	25	2	HClO4	lgR:2.89 (3SF)	3	MDS	437

Reaction No. 2767



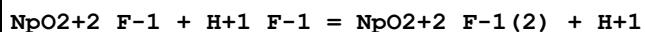
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgR:2.34 (3SF)	4	MAX	437
2	25	4	HClO4	lgR:2.34 (3SF)	4	MRX	931

Reaction No. 2768



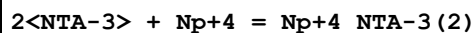
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.41 (3SF)	3	MRX	437
2	21	1	HClO4	lgR:0.9 (0.07SD)	5	MDS	437
3	25	0.5	HClO4	lgR:1.11 (3SF)	3	MRX	437
4	25	0.5	HClO4	lgR:1.1 (0.05SD)	5	MPT	5595
5	25	1	HClO4	lgR:0.9 (0.05SD)	5	MSE	5595
6	25	1.04	HClO4	lgR:2.20 (3SF)	3	MCE	931
7	25	2	HClO4	lgR:1.12 (3SF)	3	MDS	437
8	25	2	HClO4	lgR:1.1 (0.05SD)	3	MSE	5595
9	25	2.1	HClO4	lgR:2.1 (0.1SD)	5	MIX	5595
10	25	2.11	HClO4	lgR:2.11 (3SF)	5	MCE	931

Reaction No. 2769



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.04 (1SF)	3	MRX	437
2	21	1	HClO4	lgR:0.2 (0.07SD)	5	MDS	437
3	25	0.5	HClO4	lgR:0.03 (1SF)	3	MRX	437

Reaction No. 3070



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	ClO4-1 salt	lgK:32.06 (4SF)	5	MSP	1118
2	25	1	HClO4	lgK:32.06 (4SF)	5	MSP	1118
3	25	1	NaClO4	lgK:32.06 (4SF)	6	CMN	1118

Reaction No. 3076							
NTA-3 + Np+4 = Np+4_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	ClO4-1 salt	lgK:17.28 (4SF)	5	MSP	1118
2	25	1	HClO4	lgK:17.28 (4SF)	5	MSP	1118
3	25	1	NaClO4	lgK:17.28 (4SF)	6	CMN	1118

Reaction No. 3160							
HOEDTA-3 + Np+4 = Np+4_HOEDTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Na+1 salt	lgK:20.82 (4SF)	6	CRV	552
2	25	1	NaClO4	lgK:12.97 (4SF)	0	MSP	906
3	25	1	Unknown	lgK:20.82 (4SF)	3	ENS	773

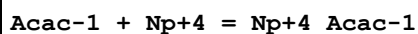
Reaction No. 3161							
2<HOEDTA-3> + Np+4 = Np+4_HOEDTA-3 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Na+1 salt	lgK:33.59 (4SF)	6	CRV	552
2	25	1	NaClO4	lgK:23.72 (4SF)	0	MSP	906

Reaction No. 4211							
EDTA-4 + Np+4 = Np+4_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.2:1	Unknown	lgK:22.9 (3SF)	0	MDS	903
2	22	0.5	HCl	lgK:24.4 (3SF)	0	MDS	903
3	25	0	Inf. Dilution	lgK:31.2 (4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:31.200 (0.3SD)	6	CRV	20830
5	25	1	HClO4	lgK:24.55 (4SF)	7	EIU	903
6	25	1	NaClO4	lgK:24.6 (0.03SD)	6	MSP	906

Reaction No. 4275							
DTPA-5 + Np+4 = Np+4_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	Unknown	lgK:29.29 (0.02SD) w	2	MPH	906
2	20	0.5	Unknown	lgK:29.29 (0.02SD) w	2	MPH	999
3	23	1	NH4Cl	lgK:29.80 (4SF)	5	MIX	999

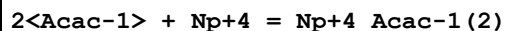
4	23	1	Unknown	lgK:29.8 (3SF)	5	MSC	906
5	25	1	NaClO4	lgK:30.33 (4SF)	0	MSP	999

Reaction No. 4555



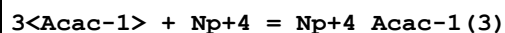
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:8.6 (0.07SD)	4	MDS	999

Reaction No. 4556



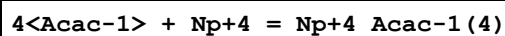
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:17.23 (4SF)	6	ENS	773

Reaction No. 4557



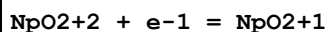
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:23.94 (4SF)	6	ENS	773

Reaction No. 4558



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.2 (3SF)	5	MGL	2360
2	25	1	Unknown	lgK:30.22 (4SF)	6	ENS	773

Reaction No. 4808



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.236 (4SF)	5	CRV	7761
2	25	0	Inf. Dilution	E:1.24 (3SF)	4	CRV	22551
3	25	1	HClO4	E:1.14 (0.01SD)	6	MEF	775
4	25	0	Inf. Dilution	dH:-117.6 (4SF) kJ	5	CRV	7761

Reaction No. 4809



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.91 (4SF)	0	CRV	32224
2	25	0	CHEMVAL2	lgK:10.99 (0.01SD)	0	ENS	5156

3	25	0	CHEMVAL3	lgK:10.99(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:11.00(0.01SD)	3	ENS	5156
5	25	0	Inf. Dilution	E:0.567(3SF)	0	CRV	7761
6	25	0	Inf. Dilution	E:0.66(2SF)	4	CRV	22551
7	25	1	HClO4	E:0.74(0.01SD)	3	MEF	775
8	25	0	Inf. Dilution	dH:-149.6(4SF)kJ	3	CRV	7761

Reaction No. 4810							
Np+4 + e-1 = Np+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.57(3SF)	4	CRV	32224
2	25	0	CHEMVAL2	lgK:2.54(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:2.54(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:2.54(0.01SD)	3	ENS	5156
5	25	0	Inf. Dilution	E:0.147(3SF)	0	CRV	6330
Note (5): Low wgt for internal consistency							
6	25	0	Inf. Dilution	E:0.157(3SF)	5	CRV	7761
7	25	0	Inf. Dilution	E:0.18(2SF)	2	CRV	22551
8	25	1	HCl	E:0.14(0.01SD)	4	MPL	775
9	25	1	HClO4	E:0.15(0.01SD)	4	MEF	775
10	25	1	HNO3	E:0.11(0.01SD)	0	MEF	775
11	25	0	Inf. Dilution	dH:28.9(3SF)kJ	5	CRV	7761

Reaction No. 4909							
Np+3 + 3<e-1> = Np(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-88.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:517.(3SF)kJ	0	CRV	7421
3	25	0	Inf. Dilution	E:-1.85(0.01SD)	0	MDG	775
4	25	0	Inf. Dilution	E:-1.856(4SF)	0	CRV	6330
5	25	0	Inf. Dilution	E:-1.79(3SF)	0	ENS	7276
6	25	0	Inf. Dilution	E:-1.768(4SF)	0	CRV	7761
7	25	0	Inf. Dilution	E:-1.79(3SF)	0	CRV	22551
8	25	0	Inf. Dilution	dH:527.3(4SF)kJ	2	CRV	7761

Reaction No. 6607							
Oxalic-2 + NpO2+1 = NpO2+1_Oxalic-2							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.5	NH4ClO4	lgK:4.04(3SF)	0	MIX	931
2	20	1	NaClO4	lgK:3.74(0.05SD)	3	MGL	3675
3	23	1	NaClO4	lgK:3.5(0.06SD)	0	MPS	3899
4	25	0	Inf. Dilution	lgK:3.757(4SF)	1	EOR	
5	25	0	Inf. Dilution	lgK:3.900(0.05SD)	5	CRV	20830
6	25	0.5	HClO4	lgK:3.29(3SF)	3	MPS	5737
7	25	1	NaClO4	lgK:3.44(3SF)	3	MDS	1586
8	25	1	NaClO4	lgK:3.6(2SF)	5	MSE	1586

Reaction No. 6608							
2<Oxalic-2> + NpO2+1 = NpO2+1_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.5	NH4ClO4	lgK:7.36(3SF)	0	MIX	931
2	20	0.05	NH4ClO4	lgK:11.40(4SF)	0	MIX	11516
Note (2): Zolotov Y et al., Zhur. Neorg. Khim., 1961, 6, 539							
3	20	1	NaClO4	lgK:6.31(0.1SD)	5	MGL	3675
4	23	1	NaClO4	lgK:6.1(0.06SD)	5	MPS	3899
5	25	0	Inf. Dilution	lgK:5.8(4SF)	1	EOR	
6	25	0	Inf. Dilution	lgK:5.800(0.1SD)	5	CRV	20830
7	25	0.5	Unknown	lgK:7.01(3SF)	0	MSP	140
8	25	0.5	Unknown	lgK:7.07(3SF)	0	MSP	5737
9	25	1	NaClO4	lgK:5.83(3SF)	0	MDS	1586
10	25	1	NaClO4	lgK:6.1(2SF)	5	MSE	1586

Reaction No. 6609							
NpO2+1 + H+1_Oxalic-2 = H+1_NpO2+1_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.5	NH4ClO4	lgK:2.70(3SF)	3	MIX	931
2	25	0	Inf. Dilution	lgK:2.7(2SF)	1	ESC	

Reaction No. 6842							
ClAcet-1 + NpO2+2 = NpO2+2_ClAcet-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:1.33(3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:1.3(2SF)	1	ESC	

Reaction No. 6843							
2<ClAcet-1> + NpO2+2 = NpO2+2_ClAcet-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:2.10 (3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:2.1 (2SF)	1	ESC	

Reaction No. 6844							
3<ClAcet-1> + NpO2+2 = NpO2+2_ClAcet-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:2.78 (3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:2.8 (2SF)	1	ESC	

Reaction No. 7088							
Malonic-2 + NpO2+1 = NpO2+1_Malonic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:2.75 (0.02SD)	5	MGL	3675
2	23	1	NaClO4	lgK:2.5 (0.04SD)	5	MPS	3899
3	25	1	NaClO4	lgK:2.3 (2SF)	5	MSE	1586

Reaction No. 7326							
Lactic-1 + NpO2+1 = NpO2+1_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.75 (0.02SD)	4	MSP	999
2	25	1	NaClO4	lgK:1.1 (2SF)	5	MSE	1586
3	25	1	NaClO4	lgK:1.40 (3SF)	5	MDS	7703

Reaction No. 9205							
NTA-3 + NpO2+1 = NpO2+1_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:6.80 (3SF)	6	CMN	1118
2	25	0.1	NH4ClO4	lgK:6.80 (3SF)	6	MSP	1118
3	25	0.5	NaClO4	lgK:7.51 (3SF)	0	MGL	11516
Note (3): Rizkalla E et al., Radiochim. Acta, 1990, 51, 151							
4	25	1	ClO4-1 salt	lgK:5.85 (3SF)	5	MSP	1118
5	25	1	NaClO4	lgK:6.2 (2SF)	5	MSE	1586
6	25	1	NaClO4	lgK:6.08 (3SF)	5	MDS	11516
Note (6): Inoue Y et al., Polyhedron, 1983, 2, 627							

Reaction No. 9217							
$\text{NpO}_2+1_ \text{OH}-1(2)_ \text{NTA}-3 + \text{H}+1 = \text{NpO}_2+1_ \text{OH}-1_ \text{NTA}-3 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	$\lg K: 11.5(0.1\text{SD})$	3	MSP	1118

Reaction No. 10249							
$3\text{ClPropan}-1 + \text{NpO}_2+2 = \text{NpO}_2+2_3\text{ClPropan}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	$\lg K: 1.88(3\text{SF})$	6	MGL	2500
2	25	0	Inf. Dilution	$\lg K: 1.9(2\text{SF})$	1	ESC	

Reaction No. 10250							
$2<3\text{ClPropan}-1> + \text{NpO}_2+2 = \text{NpO}_2+2_3\text{ClPropan}-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	$\lg K: 3.30(3\text{SF})$	6	MGL	2500
2	25	0	Inf. Dilution	$\lg K: 3.3(2\text{SF})$	1	ESC	

Reaction No. 10251							
$3<3\text{ClPropan}-1> + \text{NpO}_2+2 = \text{NpO}_2+2_3\text{ClPropan}-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	$\lg K: 3.60(3\text{SF})$	6	MGL	2500
2	25	0	Inf. Dilution	$\lg K: 3.6(2\text{SF})$	1	ESC	

Reaction No. 10418							
$\text{Maleic}-2 + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{Maleic}-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	$\lg K: 2.20(0.02\text{SD})$	5	MGL	3675
2	23	1	NaClO ₄	$\lg K: 1.9(0.03\text{SD})$	5	MPS	3899

Reaction No. 10498							
$\text{Succinic}-2 + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{Succinic}-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	$\lg K: 1.7(0.03\text{SD})$	2	MGL	3675
2	23	1	NaClO ₄	$\lg K: 1.5(0.05\text{SD})$	0	MPS	3899
3	25	0	Inf. Dilution	$\lg K: 2.0(2\text{SF})$	1	EDH	
4	25	1	NaClO ₄	$\lg K: 1.1(2\text{SF})$	2	MSE	1586

Reaction No. 10538							
Diglycol-2 + NpO ₂ +2 = NpO ₂ +2_Diglycol-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:5.16(0.01SD)	6	MGL	999
2	25	0	Inf. Dilution	lgK:5.2(2SF)	1	ESC	

Reaction No. 10765							
2OHButan-1 + NpO ₂ +1 = NpO ₂ +1_2OHButan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO ₄	lgK:1.62(0.02SD)	6	MSP	481
2	25	1	NaClO ₄	lgK:1.1(2SF)	3	MSE	1586
3	25	1	NaClO ₄	lgK:1.64(3SF)	5	MDS	7703

Reaction No. 11181							
IDA-2 + NpO ₂ +1 = NpO ₂ +1_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:8.72(0.02SD)	0	MGL	906
2	25	0.1	NH ₄ ClO ₄	lgK:6.3(0.1SD)	6	MSP	906
3	25	1	NaClO ₄	lgK:5.7(2SF)	5	MSE	1586

Reaction No. 11182							
NpO ₂ +1 + H+1_IDA-2 = H+1_NpO ₂ +1_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:1.(0.5SD)	4	MSP	999

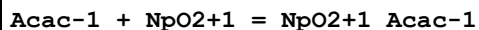
Reaction No. 11369							
Np+4_Acac-1 + Acac-1 = Np+4_Acac-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:8.7(0.05SD)	4	MDS	999

Reaction No. 11370							
Np+4_Acac-1(2) + Acac-1 = Np+4_Acac-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:6.71(0.02SD)	4	MDS	999

Reaction No. 11371							
Np+4_Acac-1(3) + Acac-1 = Np+4_Acac-1(4)							

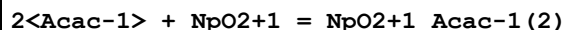
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(2SF)	1	ELC	
2	25	1	NaClO4	lgK:6.28(0.02SD)	2	MDS	999

Reaction No. 11372



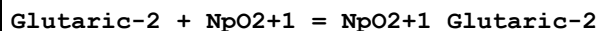
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.1	NaClO4	lgK:4.33(0.01SD)	6	MPS	5767
2	25	0.1	NaClO4	lgK:4.08(0.01SD)	4	MPS	5767
3	25	1	NaClO4	lgK:4.08(0.02SD)	4	MPS	5767
4	32	0.1	NaClO4	lgK:4.01(0.01SD)	4	MPS	5767
5	25	0.1	NaClO4	dH:-6.3(1.0SD)	5	MTD	5767

Reaction No. 11373



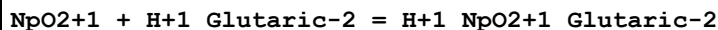
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.1	NaClO4	lgK:7.56(0.01SD)	2	MPS	5767
2	25	0	Inf. Dilution	lgK:7.3(2SF)	1	EDH	
3	25	0.1	NaClO4	lgK:7.00(0.02SD)	4	MPS	5767
4	25	1	NaClO4	lgK:7.07(0.02SD)	4	MPS	5767
5	32	0.1	NaClO4	lgK:6.96(0.05SD)	2	MSP	5767

Reaction No. 11483



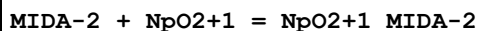
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.4(0.03SD)	3	ENS	999
2	25	1	NaClO4	lgK:1.3(2SF)	5	MSE	1586

Reaction No. 11484



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.87(2SF)	6	ENS	999

Reaction No. 11662



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:7.4(0.05SD)	4	MSP	999

Reaction No. 11663							
$\text{NpO2+1} + \text{H+1_MIDA-2} = \text{H+1_NpO2+1_MIDA-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:1.3(0.05SD)	3	MSP	999

Reaction No. 12559							
$\text{NpO2+1} + \text{H+1_NTA-3} = \text{H+1_NpO2+1_NTA-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:1.77(3SF)	6	MIX	999

Reaction No. 12560							
$\text{NpO2+1_NTA-3} + \text{H2O} = \text{NpO2+1_OH-1_NTA-3} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:-11.46(4SF)	6	MIX	999

Reaction No. 12951							
$\text{Tropolone-1} + \text{NpO2+1} = \text{NpO2+1_Tropolone-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:5.45(0.01SD)	6	MGL	3109

Reaction No. 12952							
$2<\text{Tropolone-1}> + \text{NpO2+1} = \text{NpO2+1_Tropolone-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:9.8(0.03SD)	6	MGL	3109

Reaction No. 13075							
$\text{HIMDA-2} + \text{NpO2+1} = \text{NpO2+1_HIMDA-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:20.82(4SF)	0	MSP	906
2	25	0.1	Unknown	lgK:6.1(0.05SD)	4	MSP	999

Reaction No. 13076							
$\text{NpO2+1} + \text{H+1_HIMDA-2} = \text{H+1_NpO2+1_HIMDA-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.45(3SF)	6	MSP	999

Reaction No. 13077							
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NpO2+1_HIMDA-2 + H2O = NpO2+1_OH-1_HIMDA-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-11.4(0.03SD)	4	MSP	999

Reaction No. 13466							
Phthalic-2 + NpO2+1 = NpO2+1_Phthalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.1	HClO4	lgK:3.32(0.22SD)	4	MIX	2395
2	20	1	NaClO4	lgK:2.22(0.02SD)	4	MGL	3675
3	25	0.1	HClO4	lgK:3.43(0.33SD)	4	MIX	2395
4	25	2	Unknown	lgK:1.68(3SF)	3	MSP	11516
Note (4): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113							
5	35	0.1	HClO4	lgK:3.83(0.19SD)	4	MIX	2395
6	10	0.1	HClO4	dG:-18.0(3SF) kJ	4	EFE	2395
7	25	0.1	HClO4	dG:-19.7(3SF) kJ	4	EFE	2395
8	35	0.1	HClO4	dG:-22.6(3SF) kJ	4	EFE	2395
9	25	0.1	HClO4	dH:33.5(3SF) kJ	4	MTD	2395

Reaction No. 14011							
EDTA-4 + NpO2+1 = NpO2+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.05	NH4ClO4	lgK:9.7(0.1SD)	0	MIX	931
Note (1): Low wgt for internal consistency							
2	22	1	Unknown	lgK:9.05(3SF)	0	MDS	903
3	25	0	Inf. Dilution	lgK:8.225(4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:8.54(3SF)	0	EOD	8120
5	25	0	Inf. Dilution	lgK:9.230(0.07SD)	3	CRV	20830
6	25	0.1	ClO4-1 salt	lgK:7.3(0.06SD)	4	EIU	903
7	25	0.1	NH4ClO4	lgK:7.3(0.06SD)	3	MIX	906

Reaction No. 14012							
NpO2+1 + H+1_EDTA-4 = H+1_NpO2+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.98(3SF)	0	EOD	8120
2	25	0	Inf. Dilution	lgK:5.820(0.06SD)	6	CRV	20830
3	25	0.1	ClO4-1 salt	lgK:5.3(0.08SD)	7	EIU	903
4	25	0.1	NH4ClO4	lgK:5.3(0.08SD)	4	MIX	906

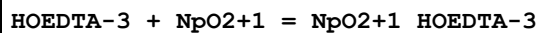
5	25	1	NaClO ₄	lgK:4.5 (2SF)	5	MSE	1586
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Reaction No. 14013



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	ClO ₄ -1 salt	lgK:11.5 (0.08SD)	7	EIU	903
2	25	0.1	NH ₄ ClO ₄	lgK:11.5 (0.08SD)	4	MIX	999

Reaction No. 14128



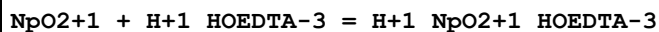
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:6.87 (3SF)	2	MIX	999
2	25	0.1	NaClO ₄	lgK:6.1 (0.05SD)	2	MSP	999

Reaction No. 14129



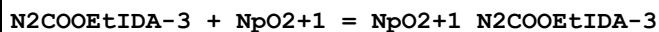
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:-11.37 (4SF)	6	MIX	999
2	25	0.1	NaClO ₄	lgK:-11.4 (0.03SD)	4	MSP	999

Reaction No. 14130



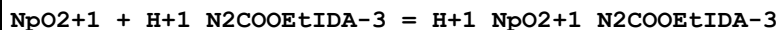
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:4.06 (0.01SD)	4	MIX	999
2	25	0.1	NaClO ₄	lgK:1. (0.3SD)	0	MSP	906
3	25	0.1	Unknown	lgK:4.06 (3SF)	6	CRV	552

Reaction No. 19084



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:7.0 (0.09SD)	4	MIX	999

Reaction No. 19085



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH ₄ ClO ₄	lgK:2.4 (0.1SD)	3	MIX	999

Reaction No. 19086

NpO2+1_OH-1_N2COOEtIDA-3 + H+1 = NpO2+1_N2COOEtIDA-3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:11.6(0.09SD)	4	MIX	999

Reaction No. 19487							
Acetic-1 + NpO2+2 = NpO2+2_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:2.31(3SF)	6	MGL	2107
2	25	0	Inf. Dilution	lgK:2.885(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:2.98(3SF)	4	MDG	999
4	25	0	Inf. Dilution	lgK:2.79(3SF)	4	EBU	6372
5	25	1	NaClO4	lgK:2.31(3SF)	6	MGL	4387

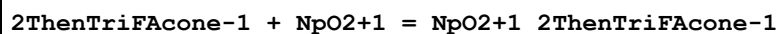
Reaction No. 19488							
2<Acetic-1> + NpO2+2 = NpO2+2_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:4.23(3SF)	6	MGL	2107
2	25	0	Inf. Dilution	lgK:5.06(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:5.51(3SF)	4	MDG	999
4	25	0	Inf. Dilution	lgK:4.61(3SF)	4	EBU	6372

Reaction No. 19489							
3<Acetic-1> + NpO2+2 = NpO2+2_Acetic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:6.0(2SF)	4	MGL	2107
2	25	0	Inf. Dilution	lgK:6.831(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:7.41(3SF)	4	MDG	999
4	25	0	Inf. Dilution	lgK:5.72(3SF)	0	EBU	6372

Reaction No. 19826							
NpO2+1_Oxalic-2 + Oxalic-2 = NpO2+1_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.05	NH4ClO4	lgK:7.36(3SF)	0	MIX	11516
Note (1): Zolotov Y et al., Zhur. Neorg. Khim., 1961, 6, 539							
2	20	1	NaClO4	lgK:2.57(3SF)	5	MEF	11516
Note (2): Magon L et al., Radiochim. Acta, 1972, 17, 164							
3	23	1	NaClO4	lgK:2.57(3SF)	5	MSP	3899

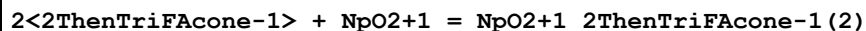
4	25	0.5	HClO4	lgK:3.77 (3SF)	0	MPS	5737
5	25	1	NaClO4	lgK:2.39 (3SF)	5	MDS	1586

Reaction No. 21553



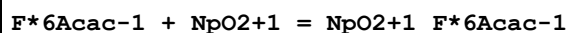
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.89 (0.02SD)	4	MPS	5767

Reaction No. 21554



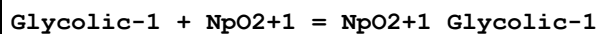
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:5.48 (0.09SD)	4	MSP	5767

Reaction No. 22789



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.94 (0.15SD)	6	MPS	5767

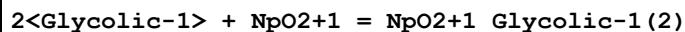
Reaction No. 26480



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.71 (2SF)	0	EBU	6372
2	25	0.1	NaClO4	lgK:1.5 (0.03SD)	5	MSP	906
3	25	1	NaClO4	lgK:1.21 (3SF)	5	MSE	1586
4	25	2	NaClO4	lgK:1.43 (3SF)	5	MSP	11516

Note (4): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113

Reaction No. 26481



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.05 (1SF)	0	EBU	6372
2	25	1	NaClO4	lgK:1.70 (3SF)	5	MDS	1586
3	25	1	NaClO4	lgK:1.8 (2SF)	5	MSE	1586
4	25	2	NaClO4	lgK:1.90 (3SF)	5	MSP	11516

Note (4): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113

Reaction No. 26482

2<Lactic-1> + NpO2+1 = NpO2+1_Lactic-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.7 (2SF)	5	MSE	1586
2	25	1	NaClO4	lgK:2.01 (3SF)	5	MDS	7703

Reaction No. 26483							
2<2OHButan-1> + NpO2+1 = NpO2+1_2OHButan-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9 (2SF)	1	EDH	
2	25	1	NaClO4	lgK:1.5 (2SF)	2	MSE	1586
3	25	1	NaClO4	lgK:2.13 (3SF)	2	MDS	7703

Reaction No. 26484							
3OHButan-1 + NpO2+1 = NpO2+1_3OHButan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.6 (1SF)	5	MSE	1586

Reaction No. 26485							
2<3OHButan-1> + NpO2+1 = NpO2+1_3OHButan-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.0 (2SF)	5	MSE	1586

Reaction No. 26486							
2OH2MePropan-1 + NpO2+1 = NpO2+1_2OH2MePropan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.05	NaClO4	lgK:1.99 (3SF)	5	MIX	906
2	25	1	NaClO4	lgK:1.5 (2SF)	5	MSE	1586
3	25	1	NaClO4	lgK:1.73 (3SF)	5	MDS	7703
4	25	2	NaClO4	lgK:1.80 (3SF)	5	MSP	7703

Reaction No. 26487							
2<2OH2MePropan-1> + NpO2+1 = NpO2+1_2OH2MePropan-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.05	NaClO4	lgK:2.90 (3SF)	5	MIX	906
2	25	1	NaClO4	lgK:2.3 (2SF)	5	MSE	1586
3	25	1	NaClO4	lgK:2.46 (3SF)	5	MDS	7703

Reaction No. 26488							
2<Malonic-2> + NpO2+1 = NpO2+1_Malonic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaClO4	lgK:4.20 (0.02SD)	0	MPS	3899
2	25	1	NaClO4	lgK:3.6 (2SF)	3	MSE	1586

Reaction No. 26489							
2<Succinic-2> + NpO2+1 = NpO2+1_Succinic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaClO4	lgK:2.4 (0.2SD)	0	MPS	3899
Note (1): Low wgt for internal consistency							
2	25	1	NaClO4	lgK:1.5 (2SF)	5	MSE	1586

Reaction No. 26490							
3<Succinic-2> + NpO2+1 = NpO2+1_Succinic-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.4 (2SF)	5	MSE	1586

Reaction No. 26491							
2<Glutaric-2> + NpO2+1 = NpO2+1_Glutaric-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.4 (2SF)	5	MSE	1586

Reaction No. 26492							
3<Glutaric-2> + NpO2+1 = NpO2+1_Glutaric-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.4 (2SF)	5	MSE	1586

Reaction No. 26493							
Gly-1 + NpO2+1 = NpO2+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.515 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.45 (3SF)	0	EBU	6372
3	25	1	NaClO4	lgK:3.2 (2SF)	5	MSE	1586
4	25	1	Unknown	lgK:3.31 (3SF)	6	CRV	552

Reaction No. 26494							
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2<Gly-1> + NpO2+1 = NpO2+1_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.791(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:1.28(3SF)	0	EBU	6372
3	25	1	NaClO4	lgK:5.5(2SF)	5	MSE	1586
4	25	1	Unknown	lgK:5.44(3SF)	6	CRV	552

Reaction No. 26495							
Ala-1 + NpO2+1 = NpO2+1_Ala-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.599(4SF)	1	EOR	
2	25	1	NaClO4	lgK:3.3(2SF)	5	MSE	1586

Reaction No. 26496							
2<Ala-1> + NpO2+1 = NpO2+1_Ala-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.999(4SF)	1	EOR	
2	25	1	NaClO4	lgK:5.7(2SF)	5	MSE	1586

Reaction No. 26497							
Asp-2 + NpO2+1 = NpO2+1_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.6(2SF)	5	MSE	1586

Reaction No. 26498							
NpO2+1 + H+1_Asp-2 = H+1_NpO2+1_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.7(1SF)	5	MSE	1586

Reaction No. 26499							
2<Asp-2> + NpO2+1 = NpO2+1_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.3(2SF)	5	MSE	1586

Reaction No. 26500							
NpO2+1 + 2<H+1_Asp-2> = H+1(2)_NpO2+1_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	NaClO4	lgK:1.3 (2SF)	5	MSE	1586
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Reaction No. 26501							
Glu-2 + NpO2+1 = NpO2+1_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.7 (2SF)	5	MSE	1586

Reaction No. 26502							
NpO2+1 + H+1_Glu-2 = H+1_NpO2+1_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.7 (1SF)	5	MSE	1586

Reaction No. 26503							
2<Glu-2> + NpO2+1 = NpO2+1_Glu-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.1 (2SF)	5	MSE	1586

Reaction No. 26504							
NpO2+1 + 2<H+1_Glu-2> = H+1 (2)_NpO2+1_Glu-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.4 (2SF)	5	MSE	1586

Reaction No. 26505							
Picolinic-1 + NpO2+1 = NpO2+1_Picolinic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:3.5 (2SF)	5	MSE	1586

Reaction No. 26506							
2<Picolinic-1> + NpO2+1 = NpO2+1_Picolinic-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:6.0 (2SF)	5	MSE	1586

Reaction No. 28075							
Propanoic-1 + NpO2+2 = NpO2+2_Propanoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:2.44 (3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:2.5 (2SF)	1	ESC	

Reaction No. 28076							
2<Propanoic-1> + NpO2+2 = NpO2+2_Propanoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:4.45(3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:4.5(2SF)	1	ESC	

Reaction No. 28077							
3<Propanoic-1> + NpO2+2 = NpO2+2_Propanoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:6.49(3SF)	6	MGL	2500
2	25	0	Inf. Dilution	lgK:6.5(2SF)	1	ESC	

Reaction No. 28945							
Oxalic-2 + NpO2+2 = NpO2+2_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:6.0(0.1SD)	4	MSP	906
2	20	1	Unknown	lgK:3.38(3SF)	0	MSL	2261
3	25	0	Inf. Dilution	lgK:6.0(2SF)	1	ESC	

Reaction No. 28946							
2<Oxalic-2> + NpO2+2 = NpO2+2_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:10.10(4SF)	3	MSP	11516
Note (1): Mefodeva M et al., Radiokhim., 1969, 11, 193							
2	20	1	Unknown	lgK:5.65(3SF)	0	MSL	2261
3	25	0	Inf. Dilution	lgK:10.1(3SF)	1	ESC	

Reaction No. 29592							
5AMP-2 + NpO2+1 = NpO2+1_5AMP-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.51(0.02SD)	5	MGL	2439

Reaction No. 29593							
5ADP-3 + NpO2+1 = NpO2+1_5ADP-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.0(0.04SD)	5	MGL	2439

Reaction No. 29594							
$H+1 + 5ADP-3 + NpO2+1 = H+1_NpO2+1_5ADP-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:7.5(0.1SD)	5	MGL	2439

Reaction No. 29595							
$NpO2+1 + H+1_5ADP-3 = H+1_NpO2+1_5ADP-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.1(0.1SD)	5	MGL	2439

Reaction No. 29596							
$5ATP-4 + NpO2+1 = NpO2+1_5ATP-4$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.5(0.04SD)	5	MGL	2439

Reaction No. 29597							
$H+1 + 5ATP-4 + NpO2+1 = H+1_NpO2+1_5ATP-4$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.9(0.06SD)	5	MGL	2439

Reaction No. 29598							
$NpO2+1 + H+1_5ATP-4 = H+1_NpO2+1_5ATP-4$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.4(0.06SD)	5	MGL	2439

Reaction No. 31824							
$2OHPentan-1 + NpO2+1 = NpO2+1_2OHPentan-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.59(0.03SD)	5	MSP	999

Reaction No. 31954							
$2<ThioSemicarbDiEthan-2> + Np+4 = Np+4_ThioSemicarbDiEthan-2(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.05	NaClO4	lgK:7.11(3SF)	5	MSP	2261

Reaction No. 31955							
$NpO2+1 + H+1(-1)_ThioSemicarbDiEthan-2 = H+1(-1)_NpO2+1_ThioSemicarbDiEthan-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.05	NaClO ₄	lgK:3.36 (3SF)	0	MSP	2261
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Reaction No. 32210

Citric-3 + NpO₂+1 = NpO₂+1_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.1	HClO ₄	lgK:4.70 (0.53SD)	0	MIX	2395
2	20	0.05	Unknown	lgK:3.67 (3SF)	3	MIX	11516
Note (2): Moskvin A et al., Zhur. Neorg. Khim., 1961, 6, 926							
3	25	0	Inf. Dilution	lgK:3.777 (4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:3.68 (3SF)	6	CRV	20830
5	25	0.05	Unknown	lgK:2.87 (3SF)	0	MSP	2261
6	25	0.1	HClO ₄	lgK:4.84 (0.72SD)	0	MIX	2395
7	25	2	NaClO ₄	lgK:2.49 (3SF)	5	MSP	11516
Note (7): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113							
8	35	0.1	HClO ₄	lgK:4.93 (0.11SD)	0	MIX	2395
9	10	0.1	HClO ₄	dG:-25.6 (3SF) kJ	0	EFE	2395
10	25	0.1	HClO ₄	dG:-27.6 (3SF) kJ	0	EFE	2395
11	35	0.1	HClO ₄	dG:-29.0 (3SF) kJ	0	EFE	2395
12	25	0.1	HClO ₄	dH:14.0 (3SF) kJ	0	MTD	2395

Reaction No. 34837

Glycolic-1 + NpO₂+2 = NpO₂+2_Glycolic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:2.37 (0.03SD)	5	MQH	3659
2	25	0	Inf. Dilution	lgK:2.38 (3SF)	4	EBU	6372

Reaction No. 34838

2<Glycolic-1> + NpO₂+2 = NpO₂+2_Glycolic-1 (2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:3.95 (0.04SD)	5	MQH	3659
2	25	0	Inf. Dilution	lgK:3.72 (3SF)	4	EBU	6372

Reaction No. 34839

3<Glycolic-1> + NpO₂+2 = NpO₂+2_Glycolic-1 (3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:5.0 (0.08SD)	5	MQH	3659
2	25	0	Inf. Dilution	lgK:4.29 (3SF)	4	EBU	6372

Reaction No. 34842							
2OH2MePropan-1 + NpO2+2 = NpO2+2_2OH2MePropan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:3.15(0.02SD)	5	MPT	3659
2	25	0	Inf. Dilution	lgK:3.2(2SF)	1	ESC	

Reaction No. 34843							
2<2OH2MePropan-1> + NpO2+2 = NpO2+2_2OH2MePropan-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:5.25(0.1SD)	5	MPT	3659
2	25	0	Inf. Dilution	lgK:5.3(2SF)	1	ESC	

Reaction No. 35378							
Np(s) + O2(g) - 2<e-1> = NpO2+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:137.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-795.8(4SF) kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-819.66(5SF) kJ	0	CRV	29375

Reaction No. 36319							
Oxalic-2 + Np+4 = Np+4_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	24:28	1	HClO4	lgK:9.63(0.01SD)	0	MSL	3261
2	25	0	Inf. Dilution	lgK:11.62(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:10.29(4SF)	4	CRV	31301
4	25	1	NaClO4	lgK:9.220(4SF)	5	MDS	4706
5	26	1	(H,Na)ClO4	lgK:7.40(0.025SD)	0	EOD	20830
Note (5): p. 626							
6	26	1	HClO4	lgK:7.47(0.005SD)	0	EOD	20830
Note (6): p. 626							
7	26	1	HClO4	lgK:8.19(0.03SD)	0	EOD	20830
Note (7): p. 626							

Reaction No. 36320							
2<Oxalic-2> + Np+4 = Np+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	24:28	1	HClO ₄	lgK:16.88 (0.01SD)	0	MSL	3261
2	25	0	Inf. Dilution	lgK:20.15 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:17.71 (4SF)	4	CRV	31301
4	25	1	NaClO ₄	lgK:16.627 (5SF)	5	MDS	4706
5	26	1	(H,Na)ClO ₄	lgK:13.62 (0.02SD)	0	EOD	20830
Note (5): p. 626							
6	26	1	HClO ₄	lgK:13.69 (0.01SD)	0	EOD	20830
Note (6): p. 626							
7	26	1	HClO ₄	lgK:16.21 (0.065SD)	0	EOD	20830
Note (7): p. 626							

Reaction No. 36321							
3<Oxalic-2> + Np+4 = Np+4_Oxalic-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	24:28	1	HClO ₄	lgK:23.69 (0.02SD)	3	MSL	3261
2	25	0	Inf. Dilution	lgK:27.29 (4SF)	1	EOR	
3	26	1	(H,Na)ClO ₄	lgK:19.45 (0.045SD)	0	EOD	20830
Note (3): p. 626							
4	26	1	HClO ₄	lgK:19.37 (0.035SD)	0	EOD	20830
Note (4): p. 626							

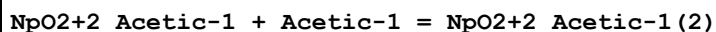
Reaction No. 36632							
2<Maleic-2> + NpO ₂ +1 = NpO ₂ +1_Maleic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaClO ₄	lgK:3.1 (0.08SD)	5	MPS	3899
2	25	0	Inf. Dilution	lgK:3.3 (2SF)	1	ESC	

Reaction No. 37508							
Np+4_Oxalic-2(2)_(s) = Np+4 + 2<Oxalic-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5:1	Unknown	lgK:-22.07 (4SF)	0	MSL	931
2	25	0	Inf. Dilution	lgK:-22.0 (3SF)	1	ESC	

Reaction No. 41398							
DTPA-5 + Np+3 = Np+3_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:25.2 (3SF)	4	MSC	906

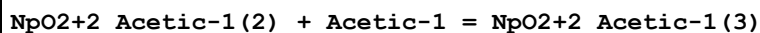
2	25	1	Unknown	lgK:30.3 (3SF)	4	CRV	2556
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Reaction No. 42103



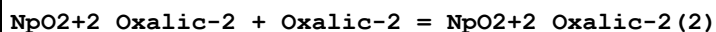
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:1.919 (4SF)	5	MGL	4387

Reaction No. 42104



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:1.771 (4SF)	5	MGL	4387

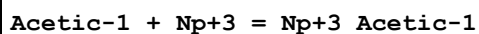
Reaction No. 42858



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO ₄	lgK:4.1 (0.1SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:4.1 (2SF)	1	ESC	
3	25	0.5	Unknown	lgK:2.27 (3SF)	0	MSL	11516

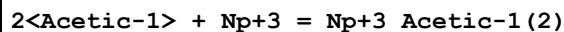
Note (3): Moskvina A et al., Zhur. Neorg. Khim., 1979, 24, 3076

Reaction No. 42902



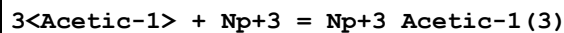
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:2.77 (3SF)	3	MSC	906
2	25	0	Inf. Dilution	lgK:2.968 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:3.08 (3SF)	4	EBU	6372

Reaction No. 42903



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:5.04 (3SF)	3	MSC	906
2	25	0	Inf. Dilution	lgK:4.803 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:4.67 (3SF)	4	EBU	6372

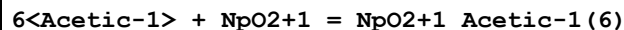
Reaction No. 42904



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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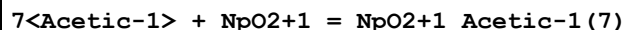
1	23	0	Inf. Dilution	lgK:6.58 (3SF)	3	MSC	906
2	25	0	Inf. Dilution	lgK:5.569 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:5.00 (3SF)	4	EBU	6372

Reaction No. 42934



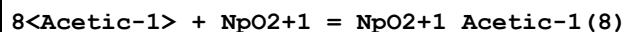
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1	23	0.5	Unknown	lgK:14.7 (3SF)	4	MSC	906

Reaction No. 42935



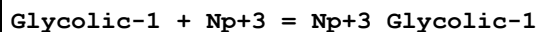
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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Reaction No. 42936



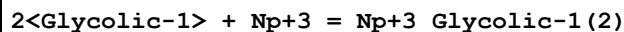
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	Unknown	lgK:20.2 (3SF)	4	MSC	906

Reaction No. 42946



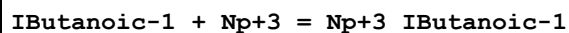
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:3.60 (3SF)	3	MSC	906
2	25	0	Inf. Dilution	lgK:2.77 (3SF)	4	EBU	6372

Reaction No. 42947



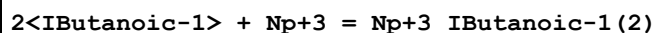
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:6.15 (3SF)	0	MSC	906
2	25	0	Inf. Dilution	lgK:3.99 (3SF)	4	EBU	6372

Reaction No. 43278



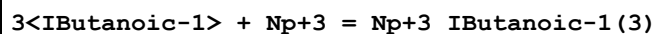
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:3.60 (3SF)	4	MSC	906

Reaction No. 43279



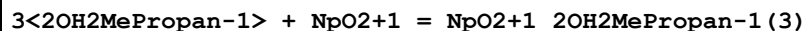
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1	23	0	Inf. Dilution	lgK:6.10 (3SF)	4	MSC	906

Reaction No. 43280



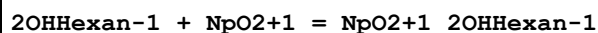
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:7.30 (3SF)	4	MSC	906

Reaction No. 43289



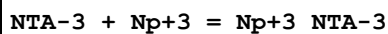
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.05	NaClO4	lgK:3.53 (3SF)	5	MIX	906

Reaction No. 43771



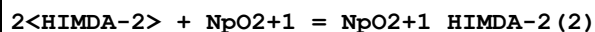
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.63 (0.02SD)	5	MSP	906

Reaction No. 44038



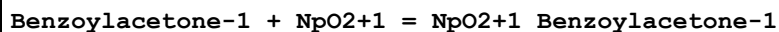
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:12.7 (3SF)	4	EOD	906
2	25	0	Inf. Dilution	lgK:12.7 (3SF)	1	ESC	

Reaction No. 44085



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:33.59 (4SF)	0	MSP	906

Reaction No. 44749

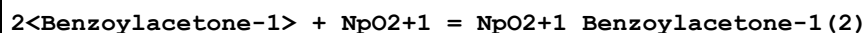


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:4.44 (3SF)	5	ENS	2556

Note (1): NIST: 'Value adjusted to listed protonation constant'

2	25	0.1	NaClO4	lgK:4.99 (0.01SD)	3	MPS	5767
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Reaction No. 44750



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.86(0.03SD)	1	MPS	906
2	25	0.1	NaClO4	lgK:7.76(3SF)	1	ENS	2556
Note (2): NIST: 'Value adjusted to listed protonation constant'							

Reaction No. 44780							
F*3Benzoylacetone-1 + NpO2+1 = NpO2+1_F*3Benzoylacetone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.84(3SF)	5	MPS	2556
Note (1): NIST: 'Value adjusted to listed protonation constant'							
2	25	0.1	NaClO4	lgK:4.11(0.03SD)	4	MPS	5767

Reaction No. 44781							
2<F*3Benzoylacetone-1> + NpO2+1 = NpO2+1_F*3Benzoylacetone-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.8(2SF)	1	EDH	
2	25	0.1	NaClO4	lgK:7.38(3SF)	2	MPS	2556
Note (2): NIST: 'Value adjusted to listed protonation constant'							
3	25	0.1	NaClO4	lgK:7.86(0.07SD)	3	MPS	5767

Reaction No. 44866							
EDTA-4 + Np+3 = Np+3_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:20.5(3SF)	2	MSC	906
2	25	0	Inf. Dilution	lgK:20.5(3SF)	1	ESC	

Reaction No. 45250							
CDTA-4 + Np+3 = Np+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:21.2(3SF)	0	MSC	906
2	25	0	Inf. Dilution	lgK:21.2(3SF)	1	ESC	

Reaction No. 45273							
DTPA-5 + NpO2+1 = NpO2+1_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.05	NH4Cl	lgK:10.8(0.03SD)	4	MIX	906
2	25	0	Inf. Dilution	lgK:10.9(3SF)	1	ESC	

Reaction No. 45963							
HOEDTA-3 + Np+3 = Np+3_HOEDTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:12.97(4SF)	3	MSP	906
2	25	1	Na+1 salt	lgK:20.82(4SF)	0	CRV	2556

Reaction No. 45964							
2<HOEDTA-3> + Np+3 = Np+3_HOEDTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:23.72(4SF)	3	MSP	906
2	25	1	Na+1 salt	lgK:33.59(4SF)	0	CRV	2556

Reaction No. 52440							
2ThenTriFAcone-1 + Np+4 = Np+4_2ThenTriFAcone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO ₄	lgK:1.681(4SF)	5	MDS	4680

Reaction No. 52751							
IO3-1 + NpO2+2 = NpO2+2_IO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.23(3SF)	1	EBU	6372
2	25	0.3	HClO ₄	lgK:0.61(2SF)	5	MSP	5398

Reaction No. 53008							
Cl-1 + Np+4 = Np+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	HClO ₄	lgK:0.15(2SF)	5	MDS	931
2	20	1	HClO ₄	lgK:-0.04(1SF)	5	MDS	931
3	20	2	HClO ₄	lgK:0.04(1SF)	5	MDS	931
4	25	0	Inf. Dilution	lgK:1.29(3SF)	4	EBU	6372
5	25	0	Inf. Dilution	lgK:-0.04(1SF)	2	CRV	32224
6	25	0	CHEMVAL2	lgK:-0.04(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:1.02(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:1.02(0.01SD)	3	ENS	5156
9	25	0	Inf. Dilution/m	lgK:0.33(2SF)	2	CRV	24953
10	25	1	H(ClO ₄)	lgK:-0.3(1SF)	2	MRX	140

11	25	1	Unknown/m	lgK:-0.04 (1SF)	2	CRV	24953
12	25	2	NaClO ₄	lgK:-0.28 (2SF)	5	MSP	931
13	25	2	Unknown	lgK:-0.05 (1SF)	6	CRV	552
14	25	4	NaClO ₄	lgK:-0.11 (0.06SD)	4	MDS	5657
15	25	9	HClO ₄	lgK:2.12 (3SF)	4	MSP	5398
16	50	0	Inf. Dilution/m	lgK:0.67 (2SF)	1	CRV	24953
17	75	0	Inf. Dilution/m	lgK:1.09 (3SF)	1	CRV	24953
18	100	0	Inf. Dilution/m	lgK:1.57 (3SF)	1	CRV	24953
19	150	0	Inf. Dilution/m	lgK:2.64 (3SF)	0	CRV	24953
20	200	0	Inf. Dilution/m	lgK:3.78 (3SF)	0	CRV	24953
21	25	0	Inf. Dilution	dH:24.17 (44.91SD) kJ	4	CRV	32222

Reaction No. 53009							
2<Cl-1> + Np+4 = Np+4_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO ₄	lgK:-0.24 (2SF)	5	MDS	931
2	20	2	HClO ₄	lgK:-0.15 (2SF)	5	MDS	931
3	25	0	Inf. Dilution	lgK:2.27 (3SF)	3	EBU	6372
4	25	0	Inf. Dilution	lgK:-0.24 (2SF)	0	CRV	32224
5	25	0	CHEMVAL2	lgK:-0.24 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:1.72 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:1.73 (0.01SD)	4	ENS	5156
8	25	4	NaClO ₄	lgK:-0.10 (0.06SD)	4	MDS	5657
9	25	9	HClO ₄	lgK:3.04 (3SF)	4	MSP	5398

Reaction No. 53010							
NpO ₂ +1 + Cl-1 = NpO ₂ +1_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.15 (2SF)	0	EBU	6372
2	25	0	Inf. Dilution/m	lgK:-0.40 (2SF)	2	CRV	24953
3	25	2	HClO ₄	lgK:-0.29 (2SF)	5	MCE	931
4	25	2	HClO ₄	lgK:-0.3 (0.05SD)	5	MIX	5597
5	25	2	NaClO ₄	lgK:-0.4 (0.05SD)	4	MSE	5597
6	25	2	Unknown/m	lgK:0.38 (2SF)	2	CRV	24953
7	25	4	NaClO ₄	lgK:-0.046 (2SF)	4	MGL	2348
8	25	4	NaClO ₄	lgK:-2.5 (0.003SD)	0	MRX	5657

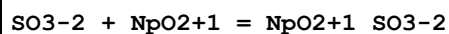
9	100	0	Inf. Dilution/m	lgK:0.00 (2SF)	1	CRV	24953
10	150	0	Inf. Dilution/m	lgK:0.00 (2SF)	0	CRV	24953
11	25	0.5	NaClO ₄	D:0.2 (0.05SD)	5	MSE	5597
12	25	1	NaClO ₄	D:0.2 (0.05SD)	5	MSE	5597

Reaction No. 53011							
2<Cl-1> + NpO ₂ +1 = NpO ₂ +1_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.03 (1SF)	4	EBU	6372
2	25	4	NaClO ₄	lgK:-1.55 (0.003SD)	5	MRX	5657

Reaction No. 53012							
Cl-1 + NpO ₂ +2 = NpO ₂ +2_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	3	H(ClO ₄)	lgK:0.21 (2SF)	3	MKN	140
2	4.78	3	H(ClO ₄)	lgK:0.06 (1SF)	3	MKN	140
3	9.84	3	H(ClO ₄)	lgK:-0.06 (1SF)	3	MKN	140
4	10	3	Unknown	lgK:-0.09 (1SF)	6	CRV	552
5	25	0.3:0.5	HClO ₄	lgK:-0.34 (2SF)	5	MRX	5398
6	25	0	Inf. Dilution	lgK:0.44 (2SF)	4	EBU	6372
7	25	0	Inf. Dilution/m	lgK:-0.30 (2SF)	2	CRV	24953
8	25	0.01	Unknown/m	lgK:-0.48 (2SF)	2	CRV	24953
9	25	0.1	Unknown/m	lgK:-0.77 (2SF)	2	CRV	24953
10	25	0.2	Unknown/m	lgK:-0.88 (2SF)	2	CRV	24953
11	25	0.5	Unknown/m	lgK:-1.02 (3SF)	2	CRV	24953
12	25	1	Unknown/m	lgK:-1.07 (3SF)	2	CRV	24953
13	25	2	HClO ₄	lgK:-0.21 (2SF)	3	MSP	931
14	25	2	Unknown/m	lgK:-0.98 (2SF)	2	CRV	24953
15	25	3	Unknown/m	lgK:-0.82 (2SF)	0	CRV	24953
16	25	4	NaClO ₄	lgK:-0.05 (1SF)	4	MDS	5398
17	25	4	NaClO ₄	lgK:-0.16 (0.05SD)	4	MDS	5657
18	50	0	Inf. Dilution/m	lgK:-0.04 (1SF)	2	CRV	24953
19	75	0	Inf. Dilution/m	lgK:0.34 (2SF)	1	CRV	24953
20	100	0	Inf. Dilution/m	lgK:0.81 (2SF)	0	CRV	24953
21	150	0	Inf. Dilution/m	lgK:1.91 (3SF)	0	CRV	24953
22	200	0	Inf. Dilution/m	lgK:3.14 (3SF)	0	CRV	24953

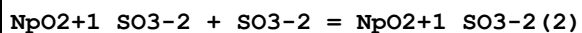
23	0	3	H(ClO ₄)	dH:-8.7 (2SF)	2	MTD	140
24	25	0	Inf. Dilution	dH:8.39 (44.84SD) kJ	2	CRV	32222

Reaction No. 53276



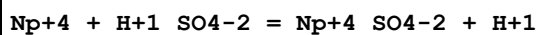
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:1.50 (0.02SD)	5	MSL	2261
2	23	0	Inf. Dilution	lgK:2.15 (3SF)	3	MCE	931
3	25	0	Inf. Dilution	lgK:2.15 (4SF)	1	EOR	
4	25	1	NaNO ₃	lgK:2.6 (2SF)	5	MSP	5398

Reaction No. 53277



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaNO ₃	lgK:1.0 (2SF)	4	MSP	5398

Reaction No. 53337



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10.2	2	NaClO ₄	lgK:2.47 (3SF)	3	MDS	140
2	10.2	2	NaClO ₄	lgK:2.39 (3SF)	0	MDS	5398
3	25	0	Inf. Dilution	lgK:3.6 (2SF)	1	ELC	
4	25	2	Unknown	lgK:2.4 (0.1SD)	2	MSE	5273
5	25	3	NaClO ₄	lgK:2.49 (3SF)	0	MPL	931
6	25.2	2	NaClO ₄	lgK:2.43 (3SF)	2	MDS	140
7	25.2	2	NaClO ₄	lgK:2.47 (3SF)	2	MDS	5398
8	35.3	2	NaClO ₄	lgK:2.40 (3SF)	2	MDS	140
9	35.3	2	NaClO ₄	lgK:2.49 (3SF)	0	MDS	5398
10	10	2	HClO ₄	lgR:2.45 (3SF)	5	MDS	5398
11	20	4	HClO ₄	lgR:2.70 (3SF)	5	MCE	931
12	23	2	HClO ₄	lgR:2.52 (3SF)	5	MDS	5398
13	25	2	HClO ₄	lgR:2.5 (0.05SD)	5	MIX	5273
14	25	2	HClO ₄	lgR:2.5 (2SF)	5	MDS	5398
15	40	2	HClO ₄	lgR:2.54 (3SF)	5	MDS	5398
16	25	2	HClO ₄	dH:1.25 (3SF)	5	MCL	5398

Reaction No. 53338

Np+4 + 2<H+1_SO4-2> = Np+4_SO4-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.9(2SF)	1	ELC	
2	25	2	Unknown	lgK:4.0(0.05SD)	0	MIX	5273
3	25	2	Unknown	lgK:4.0(0.05SD)	0	MSE	5273
4	25	3	NaClO4	lgK:3.58(3SF)	2	MPL	931
5	23	2	HClO4	lgR:4.01(3SF)	2	MDS	5398

Reaction No. 53339							
Np+4_SO4-2 + H+1_SO4-2 = Np+4_SO4-2(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10.2	2	NaClO4	lgK:0.91(2SF)	3	MDS	140
2	10.2	2	NaClO4	lgK:1.44(3SF)	2	MDS	5398
3	25	0	Inf. Dilution	lgK:3.3(2SF)	1	ELC	
4	25.2	2	NaClO4	lgK:1.04(3SF)	3	MDS	140
5	25.2	2	NaClO4	lgK:1.36(3SF)	2	MDS	5398
6	35.3	2	NaClO4	lgK:1.14(3SF)	2	MDS	140
7	35.3	2	NaClO4	lgK:1.32(3SF)	2	MDS	5398
8	10	2	HClO4	lgR:1.5(2SF)	2	MDS	5398
9	20	4	HClO4	lgR:1.56(3SF)	2	MCE	931
10	25	2	HClO4	lgR:1.55(3SF)	3	MDG	5398
11	40	2	HClO4	lgR:1.55(3SF)	2	MDG	5398

Reaction No. 53340							
NpO2+2 + H+1_SO4-2 = NpO2+2_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	21	1	NaClO4	lgK:0.79(2SF)	5	MDS	931
2	25	0	Inf. Dilution	lgK:1.28(3SF)	4	MEF	5398
3	25	0.5	HClO4	lgR:0.8(0.05SD)	5	MPT	5595
4	25	1	HClO4	lgR:1.1(0.05SD)	5	MPT	5595
5	25	1	HClO4	lgR:0.8(0.05SD)	5	MSE	5595
6	25	2	HClO4	lgR:1.1(0.05SD)	5	MSE	5595

Reaction No. 53380							
PO4-3 + Np+3 = Np+3_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:2.40(3SF)	4	MSC	5398

Reaction No. 53381							
2<PO4-3> + Np+3 = Np+3_PO4-3 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:3.73 (3SF)	4	MSC	5398

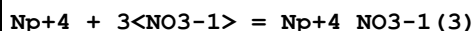
Reaction No. 53382							
3<PO4-3> + Np+3 = Np+3_PO4-3 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:5.64 (3SF)	4	MSC	5398

Reaction No. 53671							
NO3-1 + Np+4 = Np+4_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:1.68 (3SF)	4	MDS	931
2	20	0.5	HClO4	lgK:0.45 (2SF)	5	MDS	931
3	20	1	HClO4	lgK:0.34 (2SF)	5	MDS	931
4	20	2	HClO4	lgK:0.30 (2SF)	5	MDS	931
5	20	8	HClO4	lgK:-1.52 (3SF)	5	MDS	5398
6	23	2	Unknown	lgK:0.83 (2SF)	2	MSC	5398
7	23	9	HClO4	lgK:0.90 (2SF)	5	MSP	5398
8	25	0	Inf. Dilution	lgK:1.6 (2SF)	1	EDH	
9	25	0	Inf. Dilution	lgK:1.86 (3SF)	4	EBU	6372
10	25	0	Inf. Dilution	lgK:0.38 (2SF)	0	CRV	32224
11	25	0	CHEMVAL4	lgK:1.90 (0.01SD)	4	ENS	5156
12	25	1	HClO4	lgK:0.38 (2SF)	4	MRX	140
13	25	2	HClO4	lgK:0.34 (2SF)	5	MSP	931
14	25	2	LiClO4	lgK:0.83 (2SF)	0	MDS	5398
15	25	2	NaClO4	lgK:0.11 (2SF)	3	MSP	931
16	25	4	LiClO4	lgK:0.72 (2SF)	5	MDS	5398
17	25	4	NaClO4	lgK:-0.15 (0.2SD)	0	MDS	5657

Reaction No. 53672							
2<NO3-1> + Np+4 = Np+4_NO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:0.08 (1SF)	5	MDS	931
2	20	2	HClO4	lgK:0.34 (2SF)	5	MDS	931

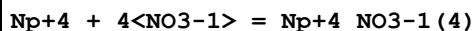
3	20	8	HClO4	lgK:-0.17 (2SF)	5	MDS	5398
4	23	2	Unknown	lgK:1.30 (3SF)	0	MSC	5398
5	23	9	HClO4	lgK:2.06 (3SF)	4	MSP	5398
6	25	0	Inf. Dilution	lgK:2.96 (3SF)	4	EBU	6372
7	25	0	Inf. Dilution	lgK:0.08 (1SF)	0	CRV	32224
8	25	0	CHEMVAL4	lgK:3.2 (2SF)	3	ENS	5156
9	25	1	Unknown	lgK:0.1 (1SF)	6	CRV	552
10	25	2	HClO4	lgK:0.18 (2SF)	5	MSP	931
11	25	2	LiClO4	lgK:1.30 (3SF)	0	MDS	5398
12	25	2	Unknown	lgK:0.2 (1SF)	6	CRV	552
13	25	4	LiClO4	lgK:1.08 (3SF)	3	MDS	5398
14	25	4	NaClO4	lgK:-0.74 (0.05SD)	0	MDS	5657

Reaction No. 53673



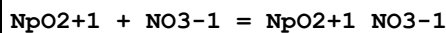
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:-0.26 (2SF)	5	MDS	931
2	20	8	HClO4	lgK:-0.82 (2SF)	0	MDS	5398
3	23	2	Unknown	lgK:1.55 (3SF)	0	MSC	5398
4	25	0	Inf. Dilution	lgK:3.57 (3SF)	2	EBU	6372
5	25	0	CHEMVAL4	lgK:3.96 (3SF)	0	ENS	5156
6	25	2	LiClO4	lgK:1.55 (3SF)	0	MDS	5398
7	25	4	LiClO4	lgK:1.23 (3SF)	5	MDS	5398

Reaction No. 53674



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	8	HClO4	lgK:-0.89 (2SF)	5	MDS	5398
2	23	2	Unknown	lgK:1.55 (3SF)	0	MSC	5398
3	25	0	Inf. Dilution	lgK:3.77 (3SF)	4	EBU	6372
4	25	0	CHEMVAL4	lgK:4.23 (0.01SD)	3	ENS	5156
5	25	2	LiClO4	lgK:1.55 (3SF)	0	MDS	5398
6	25	4	LiClO4	lgK:1.16 (3SF)	3	MDS	5398

Reaction No. 53675



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	20	8	HClO4	lgK:-0.28 (2SF)	5	MDS	5398
2	25	0	Inf. Dilution	lgK:0.19 (2SF)	4	EBU	6372
3	25	2	HClO4	lgK:-0.25 (2SF)	4	MCE	931
4	25	2	Unknown	lgK:-0.6 (1SF)	6	CRV	552
5	25	4	NaClO4	lgK:-1.60 (0.001SD)	5	MDS	5657
6	25	8	HClO4	lgK:-0.3 (0.05SD)	5	ENS	5597
7	25	0.5	NaClO4	D:0.2 (0.05SD)	5	MSE	5597
8	25	1	NaClO4	D:0.2 (0.05SD)	5	MSE	5597

Reaction No. 53676							
2<NO3-1> + NpO2+1 = NpO2+1_NO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.51 (2SF)	4	EBU	6372
2	25	4	NaClO4	lgK:-1.37 (0.001SD)	5	MDS	5657

Reaction No. 53677							
NO3-1 + NpO2+2 = NpO2+2_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	8	HClO4	lgK:-0.24 (2SF)	5	MDS	5398
2	25	0	Inf. Dilution	lgK:0.30 (0.15SD)	4	CRV	32222
3	25	0.4	HClO4	lgK:-0.98 (2SF)	4	MRX	5398
4	25	0.6	HClO4	lgK:-0.89 (2SF)	4	MRX	5398
5	25	2	HClO4	lgK:-0.4 (1SF)	0	MSP	931
6	25	4	NaClO4	lgK:-0.7 (0.03SD)	5	MDS	5657

Reaction No. 53678							
2<NO3-1> + NpO2+2 = NpO2+2_NO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	8	HClO4	lgK:0.20 (2SF)	5	MDS	5398
2	25	0	Inf. Dilution	lgK:0.2 (1SF)	1	ESC	

Reaction No. 53744							
Nicotinic-1 + NpO2+1 = NpO2+1_Nicotinic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.6 (0.03SD)	5	MSE	1586

Reaction No. 53745							
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26PyridineDiCOO-2 + NpO2+1 = NpO2+1_26PyridineDiCOO-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:7.1(2SF)	5	MSE	1586

Reaction No. 53746							
Sulfox-2 + NpO2+1 = NpO2+1_Sulfox-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.42(0.01SD)	5	MSE	1586

Reaction No. 53747							
2<Sulfox-2> + NpO2+1 = NpO2+1_Sulfox-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:10.21(0.01SD)	5	MSE	1586

Reaction No. 53904							
H+1_PO4-3 + NpO2+1 = H+1_NpO2+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:3.38(3SF)	2	MCE	931
2	20	0.2	NH4ClO4	lgK:2.85(3SF)	5	MCE	931
3	25	0	Inf. Dilution	lgK:3.28(3SF)	4	EBU	6372
4	25	0	Inf. Dilution	lgK:2.95(0.10SD)	5	CRV	25812

Reaction No. 53905							
NpO2+1 + H+1(2)_PO4-3 = H+1(2)_NpO2+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NH4ClO4	lgK:0.81(2SF)	5	MCE	931
2	25	0	Inf. Dilution	lgK:0.61(2SF)	4	EBU	6372

Reaction No. 54017							
2<SO3-2> + NpO2+1 = NpO2+1_SO3-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:3.01(0.02SD)	5	MSL	2261
2	23	0	Inf. Dilution	lgK:3.00(3SF)	3	MCE	931
3	25	0	Inf. Dilution	lgK:2.551(4SF)	1	ELC	

Reaction No. 54069							
SO4-2 + Np+4 = Np+4_SO4-2							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.462 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.02 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:5.60 (3SF)	3	CRV	9402
4	25	0	Inf. Dilution	lgK:5.5 (2SF)	5	MEF	11516
Note (4): Lemaire R, A. E. C. L., 1984, 7817							
5	25	0	Inf. Dilution	lgK:6.85 (0.15SD)	3	EOD	31756
6	25	0	Inf. Dilution	lgK:2.43 (3SF)	0	CRV	32224
7	25	0	CHEMVAL2	lgK:3.41 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL3	lgK:3.41 (0.01SD)	0	ENS	5156
9	25	0	CHEMVAL4	lgK:6.03 (0.01SD)	3	ENS	5156
10	25	0	Inf. Dilution/m	lgK:5.49 (3SF)	3	CRV	24953
11	25	0.01	Unknown/m	lgK:4.75 (3SF)	2	CRV	24953
12	25	0.1	Unknown/m	lgK:3.56 (3SF)	2	CRV	24953
13	25	0.2	Unknown/m	lgK:3.04 (3SF)	2	CRV	24953
14	25	0.5	Unknown/m	lgK:2.28 (3SF)	2	CRV	24953
15	25	1	NaClO4	lgK:2.56 (3SF)	5	MKN	11516
Note (15): Newton T et al., Inorg. Chem., 1976, 15, 2856							
16	25	1	Unknown/m	lgK:1.75 (3SF)	2	CRV	24953
17	25	2	NaClO4	lgK:3.51 (3SF)	0	MSP	931
18	25	2	NaClO4	lgK:2.43 (3SF)	5	MDS	11516
Note (18): Sullivan J et al., J. Am. Chem. Soc., 1954, 76, 5931							
19	25	2	Unknown/m	lgK:1.39 (3SF)	0	CRV	24953
20	25	3	Unknown	lgK:3.41 (3SF)	0	CRV	2556
21	25	3	Unknown/m	lgK:1.34 (3SF)	0	CRV	24953
22	25	4	Unknown	lgK:3.53 (3SF)	3	CRV	2556
23	50	0	Inf. Dilution/m	lgK:5.84 (3SF)	2	CRV	24953
24	75	0	Inf. Dilution/m	lgK:6.26 (3SF)	1	CRV	24953
25	100	0	Inf. Dilution/m	lgK:6.72 (3SF)	1	CRV	24953
26	150	0	Inf. Dilution/m	lgK:7.73 (3SF)	0	CRV	24953
27	200	0	Inf. Dilution/m	lgK:8.79 (3SF)	0	CRV	24953
28	25	0	Inf. Dilution	dH:2.00 (3SF)	3	CRV	9402
29	25	0	NaClO4	dH:-4.4 (0.05SD)	0	MCL	7427
30	25	2	NaClO4	dH:-4.0 (2SF)	4	MCL	140

Reaction No. 54070

SO4-2 + NpO2+2 = NpO2+2_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.205 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:3.27 (3SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:2.60 (3SF)	3	EBU	6372
4	25	0	Inf. Dilution/m	lgK:3.28 (3SF)	3	CRV	24953
5	25	0.01	Unknown/m	lgK:2.90 (3SF)	2	CRV	24953
6	25	0.1	Unknown	lgK:2.20 (3SF)	5	CRV	2556
7	25	0.1	Unknown/m	lgK:2.26 (3SF)	2	CRV	24953
8	25	0.2	Unknown/m	lgK:1.94 (3SF)	2	CRV	24953
9	25	0.5	Unknown	lgK:2.07 (3SF)	5	CRV	2556
10	25	0.5	Unknown/m	lgK:1.40 (3SF)	2	CRV	24953
11	25	1	HClO4	lgK:1.11 (3SF)	0	MRX	140
12	25	1	Unknown	lgK:1.82 (3SF)	5	CRV	2556
13	25	1	Unknown/m	lgK:0.86 (2SF)	2	CRV	24953
14	25	2	HClO4	lgK:2.08 (3SF)	5	CRV	2556
15	25	2	NaClO4	lgK:1.64 (3SF)	3	MSP	931
16	25	2	NaClO4	lgK:1.07 (3SF)	3	MDS	5595
17	25	2	Unknown/m	lgK:0.12 (2SF)	1	CRV	24953
18	25	3	Unknown/m	lgK:-0.45 (2SF)	1	CRV	24953
19	50	0	Inf. Dilution/m	lgK:3.65 (3SF)	2	CRV	24953
20	75	0	Inf. Dilution/m	lgK:4.15 (3SF)	1	CRV	24953
21	100	0	Inf. Dilution/m	lgK:4.75 (3SF)	0	CRV	24953
22	150	0	Inf. Dilution/m	lgK:6.11 (3SF)	0	CRV	24953
23	200	0	Inf. Dilution/m	lgK:7.60 (3SF)	0	CRV	24953
24	25	0	Inf. Dilution	dH:4.0 (2SF)	4	CRV	2556

Reaction No. 54071							
NpO2+2 + 2<H+1_SO4-2> = NpO2+2_SO4-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	21	1	NaClO4	lgK:0.56 (2SF)	5	MDS	931
2	25	0	Inf. Dilution	lgK:0.6 (1SF)	1	ESC	

Reaction No. 54144							
Np+4_F-1(3) + H+1_F-1 = Np+4_F-1(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20	4	HClO4	lgR:1.30 (3SF)	4	MAX	437
2	25	4	HClO4	lgR:1.3 (2SF)	4	MRX	931

Reaction No. 54145							
NpO2+2 + 2<H+1_F-1> = NpO2+2_F-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.1 (2SF)	1	ESC	
2	21	1	HClO4	lgR:1.11 (3SF)	5	MDS	931

Reaction No. 54146							
F-1 + NpO2+2 = NpO2+2_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:3.9 (0.05SD)	3	CRV	2556
2	25	0	Inf. Dilution	lgK:4.6 (2SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:4.80 (3SF)	0	EBU	6372
4	25	0	Inf. Dilution/m	lgK:4.07 (3SF)	3	CRV	24953
5	25	0.01	Unknown/m	lgK:3.89 (3SF)	2	CRV	24953
6	25	0.1	Unknown	lgK:4.2 (0.03SD)	5	CRV	2556
7	25	0.1	Unknown/m	lgK:3.62 (3SF)	2	CRV	24953
8	25	0.2	Unknown/m	lgK:3.52 (3SF)	2	CRV	24953
9	25	0.5	Unknown	lgK:4.04 (3SF)	5	CRV	2556
10	25	0.5	Unknown/m	lgK:3.43 (3SF)	2	CRV	24953
11	25	1	Unknown/m	lgK:3.47 (3SF)	2	CRV	24953
12	25	1.04	HClO4	lgK:5.37 (3SF)	0	MCE	931
13	25	2	HCl	lgK:4.25 (3SF)	0	CRV	2556
14	25	2	Unknown/m	lgK:3.73 (3SF)	1	CRV	24953
15	25	2.11	HClO4	lgK:5.29 (3SF)	5	MCE	931
16	25	3	Unknown/m	lgK:4.06 (3SF)	1	CRV	24953
17	50	0	Inf. Dilution/m	lgK:4.16 (3SF)	2	CRV	24953
18	75	0	Inf. Dilution/m	lgK:4.29 (3SF)	2	CRV	24953
19	100	0	Inf. Dilution/m	lgK:4.46 (3SF)	1	CRV	24953
20	150	0	Inf. Dilution/m	lgK:4.85 (3SF)	1	CRV	24953
21	200	0	Inf. Dilution/m	lgK:5.29 (3SF)	1	CRV	24953
22	250	0	Inf. Dilution/m	lgK:5.75 (3SF)	0	CRV	24953
23	300	0	Inf. Dilution/m	lgK:6.22 (3SF)	0	CRV	24953
24	25	0	Inf. Dilution	dH:1.40 (24.0SD) kJ	4	CRV	32222

Reaction No. 54190							
Cl-1 + Np+3 = Np+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-2.42 (3SF)	0	MSP	931
2	25	0	Inf. Dilution	lgK:-2.0 (2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:0.72 (2SF)	0	EBU	6372

Reaction No. 54191							
2<Cl-1> + Np+3 = Np+3_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-4.96 (3SF)	0	MSP	931
2	25	0	Inf. Dilution	lgK:-4.3 (2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:1.04 (3SF)	0	EBU	6372

Reaction No. 54192							
3<Cl-1> + Np+4 = Np+4_Cl-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:-0.48 (2SF)	4	MDS	931
2	25	0	Inf. Dilution	lgK:3.07 (3SF)	0	EBU	6372
3	25	0	CHEMVAL2	lgK:-0.50 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:1.91 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:1.91 (0.01SD)	4	ENS	5156

Reaction No. 54287							
Br-1 + Np+3 = Np+3_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-3.39 (3SF)	0	MSP	931
2	25	0	Inf. Dilution	lgK:0.46 (2SF)	4	EBU	6372

Reaction No. 54288							
2<Br-1> + Np+3 = Np+3_Br-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-6.48 (3SF)	0	MSP	931
2	25	0	Inf. Dilution	lgK:0.59 (2SF)	4	EBU	6372

Reaction No. 54438							
NpO2+1 + H+1_CO3-2 = H+1_NpO2+1_CO3-2							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:2.43 (3SF)	3	ENS	140
2	23	0.15	Unknown	lgK:2.17 (3SF)	3	ENS	140
3	25	0	Inf. Dilution	lgK:1.14 (3SF)	0	EBU	6372

Reaction No. 55164							
F-1 + Np+4 = Np+4_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:8.40 (3SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:8.45 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:8.60 (3SF)	3	CRV	9402
4	25	0	Inf. Dilution	lgK:4.82 (3SF)	0	CRV	32224
5	25	0	CHEMVAL2	lgK:8.30 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:8.60 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:8.60 (0.01SD)	4	ENS	5156
8	25	0	Inf. Dilution/m	lgK:7.55 (3SF)	0	CRV	24953
9	25	1	HClO4	lgK:7.55 (3SF)	5	CRV	2556
10	25	1	Unknown/m	lgK:7.45 (3SF)	2	CRV	24953
11	25	2	HClO4	lgK:7.77 (3SF)	5	CRV	2556
12	50	0	Inf. Dilution/m	lgK:7.62 (3SF)	1	CRV	24953
13	75	0	Inf. Dilution/m	lgK:7.73 (3SF)	1	CRV	24953
14	100	0	Inf. Dilution/m	lgK:7.86 (3SF)	0	CRV	24953
15	150	0	Inf. Dilution/m	lgK:8.17 (3SF)	0	CRV	24953
16	200	0	Inf. Dilution/m	lgK:8.51 (3SF)	0	CRV	24953
17	10:40	1	HClO4	dH:1. (1SF)	6	CRV	552
18	25	0	Inf. Dilution	dH:5.50 (3SF)	2	CRV	9402
19	25	0	Inf. Dilution	dH:15.93 (30.10SD) kJ	4	CRV	32222
20	25	1	NaClO4	dH:0.76 (5.SD)	5	EVK	437

Reaction No. 55647							
Np+3 + e-1 = Np+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	E:-4. (0.5SD)	4	ENS	5256

Reaction No. 55949							
Np+4 + 2<H2O> = NpO2+2 + 4<H+1> + 2<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-32.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-30.33 (4SF)	0	CRV	32224
3	25	0	CHEMVAL2	lgK:-31.78 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-31.78 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-31.80 (0.01SD)	0	ENS	5156

Reaction No. 55950							
Np+4 + H2O = Np+4_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.96 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-1.49 (3SF)	0	CRV	6478
4	25	0	Inf. Dilution	lgK:-1.50 (3SF)	0	CRV	9402
5	25	0	Inf. Dilution	lgK:-0.09 (0.15SD)	0	CRV	31301
6	25	0	Inf. Dilution	lgK:0.55 (0.1SD)	0	EOD	31301
7	25	0	Inf. Dilution	lgK:0.55 (0.20SD)	0	CRV	32222
8	25	0	Inf. Dilution	lgK:-2.05 (3SF)	4	CRV	32224
9	25	0	CHEMVAL2	lgK:-1.50 (0.01SD)	0	ENS	5156
10	25	0	CHEMVAL3	lgK:-1.50 (0.01SD)	0	ENS	5156
11	25	0	CHEMVAL4	lgK:-1.50 (0.01SD)	0	ENS	5156
12	25	0	Inf. Dilution/m	lgK:-1.46 (3SF)	0	CRV	24953
13	25	0.01	Unknown/m	lgK:-1.74 (3SF)	0	CRV	24953
14	25	0.1	Unknown/m	lgK:-2.19 (3SF)	0	CRV	24953
15	25	0.2	Unknown/m	lgK:-2.38 (3SF)	0	CRV	24953
16	25	0.5	Unknown/m	lgK:-2.67 (3SF)	0	CRV	24953
17	25	1	Unknown/m	lgK:-2.88 (3SF)	2	CRV	24953
18	25	2	Unknown/m	lgK:-3.02 (3SF)	0	CRV	24953
19	25	3	Unknown/m	lgK:-3.05 (3SF)	0	CRV	24953
20	50	0	Inf. Dilution/m	lgK:-0.86 (2SF)	0	CRV	24953
21	75	0	Inf. Dilution/m	lgK:-0.36 (2SF)	0	CRV	24953
22	100	0	Inf. Dilution/m	lgK:0.07 (1SF)	0	CRV	24953
23	150	0	Inf. Dilution/m	lgK:0.75 (2SF)	0	CRV	24953
24	200	0	Inf. Dilution/m	lgK:1.26 (3SF)	0	CRV	24953
25	250	0	Inf. Dilution/m	lgK:1.65 (3SF)	0	CRV	24953
26	300	0	Inf. Dilution/m	lgK:1.96 (3SF)	0	CRV	24953
27	25	0	Inf. Dilution	dH:10.7 (3SF)	2	CRV	6478

28	25	0	Inf. Dilution	dH:11.60 (4SF)	0	CRV	9402
29	25	0	Inf. Dilution	dH:39.19 (9.74SD) kJ	4	CRV	32222

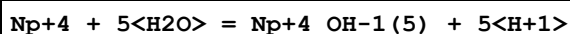
Reaction No. 55951							
Np+4 + 2<H2O> = Np+4_OH-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.50 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:-3.00 (3SF)	0	CRV	9402
3	25	0	Inf. Dilution	lgK:0.87 (0.075SD)	3	CRV	31301
4	25	0	Inf. Dilution	lgK:0.35 (0.15SD)	3	EOD	31301
5	25	0	Inf. Dilution	lgK:0.35 (0.30SD)	4	CRV	32222
6	25	0	CHEMVAL2	lgK:-3.00 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-3.00 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-2.80 (0.01SD)	0	ENS	5156
9	25	0	Inf. Dilution	dH:17.80 (4SF)	0	CRV	9402
10	25	0	Inf. Dilution	dH:54.53 (9.83SD) kJ	2	CRV	32222

Reaction No. 55952							
Np+4 + 3<H2O> = Np+4_OH-1(3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.48 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-6.00 (3SF)	0	CRV	9402
4	25	0	Inf. Dilution	lgK:4.30 (0.15SD)	0	CRV	31301
5	25	0	Inf. Dilution	lgK:-2.80 (1.00SD)	0	CRV	32222
6	25	0	CHEMVAL2	lgK:-6.00 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-6.00 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-5.59 (0.01SD)	0	ENS	5156
9	25	0	Inf. Dilution	dH:23.00 (4SF)	0	CRV	9402
10	25	0	Inf. Dilution	dH:74.93 (11.24SD) kJ	2	CRV	32222

Reaction No. 55953							
Np+4 + 4<H2O> = Np+4_OH-1(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.87 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:-10.00 (4SF)	0	CRV	9402
3	25	0	Inf. Dilution	lgK:-9.60 (0.55SD)	0	CRV	31301

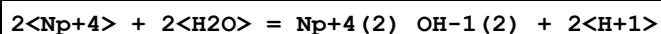
4	25	0	Inf. Dilution	lgK:-8.30 (0.15SD)	2	EOD	31301
5	25	0	Inf. Dilution	lgK:-8.30 (1.10SD)	2	CRV	32222
6	25	0	CHEMVAL2	lgK:-10.00 (0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-10.00 (0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-9.89 (0.01SD)	0	ENS	5156
9	25	0	Inf. Dilution	dH:25.50 (4SF)	3	CRV	9402
10	25	0	Inf. Dilution	dH:101.44 (11.54SD) kJ	4	CRV	32222

Reaction No. 55954



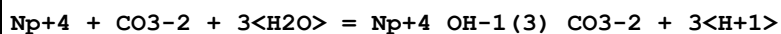
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.34 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-14.62 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-20.00 (4SF)	0	CRV	9402
4	25	0	CHEMVAL2	lgK:-20.00 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:-20.00 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:-17.00 (0.01SD)	0	ENS	5156
7	25	0	Inf. Dilution	dH:29.00 (4SF)	2	CRV	9402

Reaction No. 55955



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.00 (3SF)	3	CRV	9402
2	25	0	CHEMVAL2	lgK:-2.00 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-2.00 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-2.00 (0.01SD)	4	ENS	5156

Reaction No. 55956



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.48 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:0.0 (1SF)	0	CRV	9402
3	25	0	Inf. Dilution	lgK:3.82 (1.13SD)	4	CRV	32222
4	25	0	CHEMVAL2	lgK:-1.00 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:-1.00 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:0.01 (0.01SD)	0	ENS	5156

Reaction No. 55957							
PO4-3 + Np+4 + 2<H+1> = Np+4_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:24.10(4SF)	3	CRV	9402
3	25	0	CHEMVAL2	lgK:20.55(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:20.55(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:24.10(0.01SD)	4	ENS	5156
6	25	0	Inf. Dilution	dH:0.00(2SF)	3	CRV	9402

Reaction No. 55958							
2<PO4-3> + Np+4 + 4<H+1> = Np+4_H+1(4)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.0(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:48.00(4SF)	3	CRV	9402
3	25	0	CHEMVAL2	lgK:47.10(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:47.10(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:48.00(0.01SD)	4	ENS	5156
6	25	0	Inf. Dilution	dH:0.00(2SF)	3	CRV	9402

Reaction No. 55959							
2<SO4-2> + Np+4 = Np+4_SO4-2(2)							
Note: Not in full accord with other M+4 sulfates - see [19LFZ_31756] Table 1 Rxn is tricky due to large effect of ionic strength							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.805(4SF)	0	EOR	
2	25	0	Inf. Dilution	lgK:8.62(3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:10.30(4SF)	2	CRV	9402
4	25	0	Inf. Dilution	lgK:9.9(2SF)	0	MEF	11516
Note (4): Lemaire R, A. E. C. L., 1984, 7817							
5	25	0	Inf. Dilution	lgK:11.05(0.25SD)	3	EOD	31756
6	25	0	Inf. Dilution	lgK:3.47(3SF)	0	CRV	32224
7	25	0	CHEMVAL2	lgK:5.42(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL3	lgK:5.42(0.01SD)	0	ENS	5156
9	25	0	CHEMVAL4	lgK:9.59(0.01SD)	0	ENS	5156
10	25	1	NaClO4	lgK:3.75(3SF)	0	MKN	11516
Note (10): Newton T et al., Inorg. Chem., 1976, 15, 2856							

11	25	2	NaClO ₄	lgK:3.47 (3SF)	0	MDS	11516
Note (11): Sullivan J et al., J. Am. Chem. Soc., 1954, 76, 5931							
12	25	2	Unknown	lgK:6.06 (3SF)	0	CRV	2556
13	25	3	Unknown	lgK:5.42 (3SF)	4	CRV	2556
14	25	4	Unknown	lgK:5.92 (3SF)	3	CRV	2556
15	25	0	Inf. Dilution	dH:7.50 (3SF)	3	CRV	9402

Reaction No. 55960							
2<F-1> + Np+4 = Np+4_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO ₄	lgK:14.54 (4SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:15.66 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:14.50 (4SF)	3	CRV	9402
4	25	0	CHEMVAL2	lgK:14.50 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:15.17 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:15.20 (0.01SD)	4	ENS	5156
7	25	0	Inf. Dilution	dH:7.20 (3SF)	3	CRV	9402

Reaction No. 55961							
3<F-1> + Np+4 = Np+4_F-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO ₄	lgK:20.3 (3SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:21.77 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:19.10 (4SF)	3	CRV	9402
4	25	0	CHEMVAL2	lgK:20.30 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:20.30 (0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:19.90 (0.01SD)	4	ENS	5156
7	25	0	Inf. Dilution	dH:7.20 (3SF)	3	CRV	9402

Reaction No. 55962							
4<F-1> + Np+4 = Np+4_F-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO ₄	lgK:25.1 (3SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:26.81 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:23.60 (4SF)	3	CRV	9402
4	25	0	CHEMVAL2	lgK:25.10 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:25.20 (0.01SD)	0	ENS	5156

6	25	0	CHEMVAL4	lgK:25.50(0.01SD)	4	ENS	5156
7	25	0	Inf. Dilution	dH:4.50(3SF)	3	CRV	9402

Reaction No. 55963							
Np+4 + H2O + e-1 = Np+3_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.1(2SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-4.86(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-4.86(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-4.86(0.01SD)	0	ENS	5156

Reaction No. 55964							
Np+4 + 2<H2O> + e-1 = Np+3_OH-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-14.46(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL3	lgK:-14.46(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL4	lgK:-14.50(0.01SD)	3	ENS	5156

Reaction No. 55965							
Np+4 + 3<H2O> + e-1 = Np+3_OH-1(3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.2(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-24.46(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-24.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-24.40(0.01SD)	1	ENS	5156

Reaction No. 55966							
Np+4 + 4<H2O> + e-1 = Np+3_OH-1(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.2(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-35.46(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-35.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-35.40(0.01SD)	0	ENS	5156

Reaction No. 55967							
2<Np+4> + 2<H2O> + 2<e-1> = Np+3(2)_OH-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-8.5 (2SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-7.92 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-7.92 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-7.92 (0.01SD)	0	ENS	5156

Reaction No. 55968							
Np+4 + e-1 + CO3-2 = Np+3_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.4 (3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:9.04 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:9.04 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:9.04 (0.01SD)	0	ENS	5156

Reaction No. 55969							
Np+4 + e-1 + 2<CO3-2> = Np+3_CO3-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.4 (3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:13.54 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:13.54 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:13.50 (0.01SD)	0	ENS	5156

Reaction No. 55970							
Np+4 + e-1 + 3<CO3-2> = Np+3_CO3-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.5 (3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:17.05 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:17.05 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:17.00 (0.01SD)	0	ENS	5156

Reaction No. 55971							
Np+4 + e-1 + Cl-1 = Np+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:0.14 (0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:0.94 (0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:0.94 (0.01SD)	4	ENS	5156

Reaction No. 55972							
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Np+4 + e-1 + 2<Cl-1> = Np+3_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-2.46(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-1.08(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-1.08(0.01SD)	4	ENS	5156

Reaction No. 55973							
Np+4 + 3<H2O> = NpO2+1_OH-1 + 5<H+1> + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.91(4SF)	2	CRV	32224
2	25	0	CHEMVAL2	lgK:-19.89(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-19.89(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-19.80(0.01SD)	2	ENS	5156

Reaction No. 55974							
Np+4 + 4<H2O> = NpO2+1_OH-1(2) + 6<H+1> + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.1(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-34.35(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-34.35(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-29.00(0.01SD)	0	ENS	5156

Reaction No. 55975							
Np+4 + 2<H2O> + CO3-2 = H+1_NpO2+1_CO3-2 + 3<H+1> + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.85(3SF)	4	CRV	32224
2	25	0	CHEMVAL2	lgK:2.43(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:2.43(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:1.75(0.01SD)	4	ENS	5156

Reaction No. 55976							
Np+4 + 2<H2O> + CO3-2 = NpO2+1_CO3-2 + 4<H+1> + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-6.10(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-6.10(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-5.09(0.01SD)	4	ENS	5156

Reaction No. 55977							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + 2\langle\text{CO}_3-2\rangle = \text{NpO}_2+1_ \text{CO}_3-2(2) + 4\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.6(2\text{SF})$	1	ELC	
2	25	0	CHEMVAL2	$\lg K: 0.11(0.01\text{SD})$	0	ENS	5156
3	25	0	CHEMVAL3	$\lg K: 0.11(0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL4	$\lg K: -1.99(0.01\text{SD})$	0	ENS	5156

Reaction No. 55978							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + 3\langle\text{CO}_3-2\rangle = \text{NpO}_2+1_ \text{CO}_3-2(3) + 4\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.9(1\text{SF})$	1	ELC	
2	25	0	CHEMVAL2	$\lg K: -3.75(0.01\text{SD})$	0	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -3.75(0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL4	$\lg K: 5.31(0.01\text{SD})$	0	ENS	5156

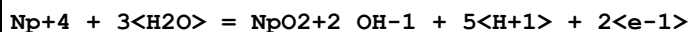
Reaction No. 55979							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{PO}_4-3 = \text{H}+1_ \text{NpO}_2+1_ \text{PO}_4-3 + 3\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 4.78(3\text{SF})$	4	CRV	32224
2	25	0	CHEMVAL2	$\lg K: 4.74(0.01\text{SD})$	0	ENS	5156
3	25	0	CHEMVAL3	$\lg K: 4.74(0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL4	$\lg K: 4.78(0.01\text{SD})$	4	ENS	5156

Reaction No. 55980							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{F}-1 = \text{NpO}_2+1_ \text{F}-1 + 4\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	$\lg K: -10.00(0.01\text{SD})$	4	ENS	5156
2	25	0	CHEMVAL3	$\lg K: -9.73(0.01\text{SD})$	4	ENS	5156
3	25	0	CHEMVAL4	$\lg K: -9.73(0.01\text{SD})$	4	ENS	5156

Reaction No. 55981							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{Cl}-1 = \text{NpO}_2+1_ \text{Cl}-1 + 4\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -11.21(4\text{SF})$	0	CRV	32224
2	25	0	CHEMVAL2	$\lg K: -11.29(0.01\text{SD})$	4	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -10.74(0.01\text{SD})$	4	ENS	5156

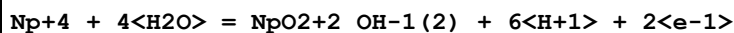
4	25	0	CHEMVAL4	lgK:-10.70 (0.01SD)	4	ENS	5156
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Reaction No. 55983



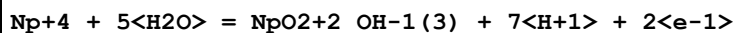
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-37.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-33.70 (4SF)	0	CRV	32224
3	25	0	CHEMVAL2	lgK:-36.88 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-36.88 (0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-36.70 (0.01SD)	0	ENS	5156

Reaction No. 55984



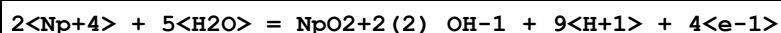
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-41.98 (0.01SD)	0	ENS	5156
2	25	0	CHEMVAL3	lgK:-41.98 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL4	lgK:-40.30 (0.01SD)	4	ENS	5156

Reaction No. 55985



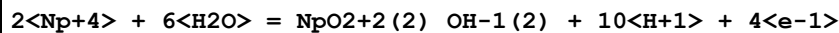
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-50.78 (0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-50.78 (0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-50.30 (0.01SD)	4	ENS	5156

Reaction No. 55986



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-67.56 (0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-67.56 (0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-67.60 (0.01SD)	4	ENS	5156

Reaction No. 55987



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-70.0 (3SF)	1	EOR	
2	25	0	CHEMVAL2	lgK:-69.96 (0.01SD)	0	ENS	5156

3	25	0	CHEMVAL3	lgK:-69.96(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-70.00(0.01SD)	3	ENS	5156

Reaction No. 55988							
3<Np+4> + 11<H2O> = NpO2+2(3)_OH-1(5) + 17<H+1> + 6<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-112.2(4SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-112.84(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-112.84(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-113.00(0.01SD)	0	ENS	5156

Reaction No. 55989							
Np+4 + 2<H2O> + CO3-2 = NpO2+2_CO3-2 + 4<H+1> + 2<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-21.78(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-21.78(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-21.70(0.01SD)	4	ENS	5156

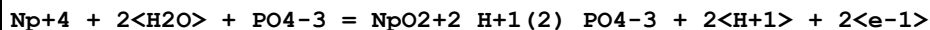
Reaction No. 55990							
Np+4 + 2<H2O> + 2<CO3-2> = NpO2+2_CO3-2(2) + 4<H+1> + 2<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.3(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-14.78(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-14.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-15.10(0.01SD)	0	ENS	5156

Reaction No. 55991							
Np+4 + 2<H2O> + 3<CO3-2> = NpO2+2_CO3-2(3) + 4<H+1> + 2<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.9(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-10.78(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-10.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-7.98(0.01SD)	0	ENS	5156

Reaction No. 55992							
Np+4 + 2<H2O> + PO4-3 = NpO2+2_H+1_PO4-3 + 3<H+1> + 2<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

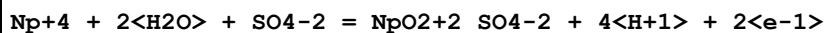
1	25	0	CHEMVAL2	lgK:-11.25(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-11.25(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-11.00(0.01SD)	4	ENS	5156

Reaction No. 55993



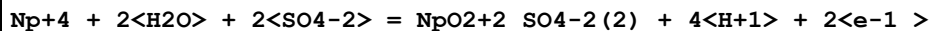
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-9.90(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-9.90(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-9.23(0.01SD)	4	ENS	5156

Reaction No. 55994



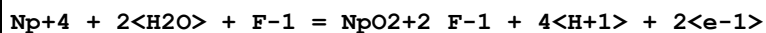
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.43(4SF)	4	CRV	32224
2	25	0	CHEMVAL2	lgK:-28.51(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-28.19(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-28.20(0.01SD)	4	ENS	5156

Reaction No. 55995



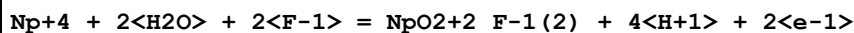
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-27.56(4SF)	0	CRV	32224
2	25	0	CHEMVAL2	lgK:-27.68(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-26.40(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-26.40(0.01SD)	4	ENS	5156

Reaction No. 55996



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.47(4SF)	2	CRV	32224
2	25	0	CHEMVAL2	lgK:-27.18(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-27.19(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-27.20(0.01SD)	2	ENS	5156

Reaction No. 55997



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.36(4SF)	2	CRV	32224
2	25	0	CHEMVAL2	lgK:-24.77(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-24.06(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-24.10(0.01SD)	2	ENS	5156

Reaction No. 55998							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{Cl}-1 = \text{NpO}_2+2_ \text{Cl}-1 + 4\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:-32.08(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:-31.58(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:-31.60(0.01SD)	4	ENS	5156

Reaction No. 55999							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + 2\langle\text{Cl}-1\rangle = \text{NpO}_2+2_ \text{Cl}-1(2) + 4\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.4(3SF)	1	ELC	
2	25	0	CHEMVAL2	lgK:-30.78(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-26.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-26.50(0.01SD)	0	ENS	5156

Reaction No. 56062							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{NO}_3-1 = \text{NpO}_2+1_ \text{NO}_3-1 + 4\langle\text{H}+1\rangle + \text{e}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-11.16(4SF)	0	CRV	32224
3	25	0	CHEMVAL4	lgK:-11.30(0.01SD)	0	ENS	5156

Reaction No. 56063							
$\text{Np}+4 + 2\langle\text{H}_2\text{O}\rangle + \text{NO}_3-1 = \text{NpO}_2+2_ \text{NO}_3-1 + 4\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.31(4SF)	2	CRV	32224
2	25	0	CHEMVAL4	lgK:-32.20(0.01SD)	2	ENS	5156

Reaction No. 56086							
$\text{Np}+4 + \text{e}-1 + 2\langle\text{H}+1\rangle + \text{PO}_4-3 = \text{Np}+3_ \text{H}+1(2) _ \text{PO}_4-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	CHEMVAL2	lgK:24.49(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:24.49(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:24.50(0.01SD)	4	ENS	5156

Reaction No. 56087							
$\text{Np}+4 + \text{e}-1 + 6\text{<H+1>} + 3\text{<PO4-3>} = \text{Np}+3_{\text{H}+1(6)}_{\text{PO4-3(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL2	lgK:27.73(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL3	lgK:27.73(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL4	lgK:27.70(0.01SD)	4	ENS	5156

Reaction No. 56110							
$\text{Np}+3_{\text{OH}-1(3)}_{\text{(s)}} + 3\text{<H+1>} = \text{Np}+3 + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.2(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:16.97(4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:18.00(2.00SD)	4	CRV	32222
4	25	0	CHEMVAL2	lgK:21.99(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:21.99(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:22.00(0.01SD)	0	ENS	5156

Reaction No. 56111							
$\text{NpO2(s)} + 4\text{<H+1>} = \text{Np}+4 + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.0(2.SD)	0	CRV	6478
3	25	0	Inf. Dilution	lgK:-9.75(1.07SD)	0	CRV	32222
4	25	0	Inf. Dilution	lgK:-2.68(3SF)	0	CRV	32224
5	25	0	CHEMVAL2	lgK:-2.68(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:-2.68(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:-4.00(0.01SD)	0	ENS	5156
8	25	0	Inf. Dilution/m	lgK:-7.80(3SF)	1	CRV	24953
9	50	0	Inf. Dilution/m	lgK:-8.52(3SF)	0	CRV	24953
10	75	0	Inf. Dilution/m	lgK:-9.13(3SF)	0	CRV	24953
11	100	0	Inf. Dilution/m	lgK:-9.65(3SF)	0	CRV	24953
12	150	0	Inf. Dilution/m	lgK:-10.49(4SF)	0	CRV	24953
13	200	0	Inf. Dilution/m	lgK:-11.14(4SF)	0	CRV	24953

14	250	0	Inf. Dilution/m	lgK:-11.64(4SF)	0	CRV	24953
15	300	0	Inf. Dilution/m	lgK:-12.05(4SF)	0	CRV	24953
16	25	0	Inf. Dilution	dH:-18.0(3.SD)	0	CRV	6478
17	25	0	Inf. Dilution	dH:-53.68(4.88SD)kJ	2	CRV	32222

Reaction No. 56112							
Np+4_OH-1(4)_(s) + 4<H+1> = Np+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.9(2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:0.75(2SF)	4	CRV	32224
3	25	0	CHEMVAL2	lgK:0.74(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:0.74(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:0.74(0.01SD)	3	ENS	5156
6	25	0	Inf. Dilution/m	lgK:0.70(2SF)	3	CRV	24953
7	50	0	Inf. Dilution/m	lgK:-0.34(2SF)	2	CRV	24953
8	75	0	Inf. Dilution/m	lgK:-1.20(3SF)	1	CRV	24953
9	100	0	Inf. Dilution/m	lgK:-1.91(3SF)	1	CRV	24953
10	150	0	Inf. Dilution/m	lgK:-3.01(3SF)	0	CRV	24953
11	200	0	Inf. Dilution/m	lgK:-3.81(3SF)	0	CRV	24953
12	250	0	Inf. Dilution/m	lgK:-4.39(3SF)	0	CRV	24953
13	300	0	Inf. Dilution/m	lgK:-4.82(3SF)	0	CRV	24953

Reaction No. 56113							
NpO2+1_OH-1_(s) = NpO2+1 + H2O - H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.95(3SF)	4	EBU	6372
2	25	0	CHEMVAL2	lgK:4.90(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:4.90(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:4.70(0.01SD)	4	ENS	5156

Reaction No. 56114							
NpO2+2_OH-1(2)_(s) + 2<H+1> = NpO2+2 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.29(3SF)	0	EBU	6372
2	25	0	CHEMVAL2	lgK:5.30(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:5.30(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:5.30(0.01SD)	0	ENS	5156

Note (4): Low wgt for inter-reaction consistency							
5	25	0	Inf. Dilution/m	lgK:5.97 (3SF)	2	CRV	24953
6	50	0	Inf. Dilution/m	lgK:5.23 (3SF)	1	CRV	24953
7	75	0	Inf. Dilution/m	lgK:4.61 (3SF)	1	CRV	24953
8	100	0	Inf. Dilution/m	lgK:4.08 (3SF)	0	CRV	24953
9	150	0	Inf. Dilution/m	lgK:3.25 (3SF)	0	CRV	24953
10	200	0	Inf. Dilution/m	lgK:2.62 (3SF)	0	CRV	24953
11	250	0	Inf. Dilution/m	lgK:2.14 (3SF)	0	CRV	24953
12	300	0	Inf. Dilution/m	lgK:1.76 (3SF)	0	CRV	24953
13	25	0	Inf. Dilution	dH:-52.24 (7.11SD) kJ	4	CRV	32222

Reaction No. 56269							
NpO2+2_H+1_PO4-3_(s) = NpO2+2 + H+1 + PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.23 (4SF)	1	EOR	
2	25	0	CHEMVAL2	lgK:-25.35 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-25.35 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-25.00 (0.01SD)	4	ENS	5156

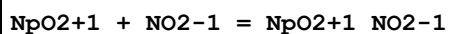
Reaction No. 56270							
NpO2+2 (3)_PO4-3 (2)_(s) = 3<NpO2+2> + 2<PO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.5 (4SF)	1	EOR	
2	25	0	CHEMVAL2	lgK:-50.00 (0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-50.00 (0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-23.50 (0.01SD)	3	ENS	5156

Reaction No. 57395							
Br-1 + Np+4 = Np+4_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.78 (2SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:1.55 (0.40SD)	4	CRV	32222
3	25	2	Unknown	lgK:-0.21 (0.02SD)	4	MSE	5259

Reaction No. 57396							
2<Br-1> + Np+4 = Np+4_Br-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

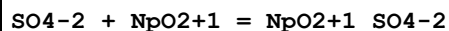
1	25	0	Inf. Dilution	lgK:1.26(3SF)	0	EBU	6372
2	25	2	Unknown	lgK:-0.6(0.07SD)	6	MSE	5259

Reaction No. 57929



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:-0.048(0.05SD)	5	MSE	5597

Reaction No. 57930



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.5	Unknown	lgK:1.04(3SF)	2	MSL	11516

Note (1): Moskvina A et al., Zhur. Neorg. Khim., 1979, 24, 2449

2	20	1.55	NH4Cl	lgK:1.04(3SF)	0	MSC	11516
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Note (2): Moskvina A et al., Koord. Khim., 1978, 4, 1065

3	25	0	Inf. Dilution	lgK:1.228(4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:1.43(3SF)	0	EBU	6372
5	25	0	Inf. Dilution	lgK:0.44(0.27SD)	5	CRV	25812
6	25	0	Inf. Dilution/m	lgK:0.10(2SF)	1	CRV	24953
7	25	0.01	Unknown/m	lgK:-0.08(1SF)	1	CRV	24953
8	25	0.1	Unknown/m	lgK:-0.35(2SF)	1	CRV	24953
9	25	0.2	Unknown/m	lgK:-0.44(2SF)	1	CRV	24953
10	25	0.5	Unknown/m	lgK:-0.54(2SF)	1	CRV	24953
11	25	1	Unknown/m	lgK:-0.51(2SF)	1	CRV	24953
12	25	2	NaClO4	lgK:0.53(2SF)	6	CRV	552
13	25	2	NaClO4	lgK:0.4(0.05SD)	2	MSE	5597
14	25	2	NaClO4	lgK:0.61(2SF)	5	MSP	11516

Note (14): Rizkalla E et al, Radiochim. Acta, 1990, 51, 113

15	25	2	Unknown/m	lgK:-0.27(2SF)	1	CRV	24953
16	25	3	Unknown/m	lgK:0.04(1SF)	0	CRV	24953
17	50	0	Inf. Dilution/m	lgK:0.37(2SF)	0	CRV	24953
18	75	0	Inf. Dilution/m	lgK:0.57(2SF)	0	CRV	24953
19	100	0	Inf. Dilution/m	lgK:0.72(2SF)	0	CRV	24953
20	150	0	Inf. Dilution/m	lgK:0.91(2SF)	0	CRV	24953
21	200	0	Inf. Dilution/m	lgK:1.00(3SF)	0	CRV	24953

Reaction No. 57931

NpO ₂ +1 + SCN-1 = NpO ₂ +1_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.24 (2SF)	4	EBU	6372
2	25	2	NaClO ₄	lgK:0.3 (0.05SD)	5	MSE	5597
3	25	5	Unknown	lgK:0.86 (2SF)	6	CRV	552
4	25	9	NaClO ₄	lgK:1.0 (0.05SD)	5	MEF	5597
5	25	9	NaClO ₄	lgK:1.2 (0.05SD)	5	MMR	5597
6	25	1	NaClO ₄	D:0.07 (0.05SD)	5	ENS	5597

Reaction No. 57932							
NpO ₂ +1 + IO ₃ -1 = NpO ₂ +1_IO ₃ -1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.35 (2SF)	4	EBU	6372
2	25	2	NaClO ₄	lgK:0.32 (0.02SD)	5	MSE	5597

Reaction No. 57934							
SCN-1 + Np+4 = Np+4_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	2	Unknown	lgK:1.50 (0.02SD)	5	MSE	5599
2	25	0	Inf. Dilution	lgK:2.42 (3SF)	4	EBU	6372
3	25	2	NaClO ₄	lgK:1.5 (2SF)	5	MDS	11516
Note (3): Rao P et al., , J. Inorg. Nucl. Chem., 1978, 40, 339							
4	25	2	Unknown	lgK:1.48 (0.01SD)	5	CMN	5599
5	40	2	Unknown	lgK:1.40 (0.02SD)	4	MSE	5599
6	25	2	Unknown	dH:-1.7 (0.1SD)	4	ENS	5599

Reaction No. 57935							
2<SCN-1> + Np+4 = Np+4_SCN-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	2	Unknown	lgK:2.3 (0.03SD)	5	MSE	5599
2	25	0	Inf. Dilution	lgK:4.39 (3SF)	0	EBU	6372
3	25	2	NaClO ₄	lgK:2.06 (3SF)	5	MDS	11516
Note (3): Rao P et al., , J. Inorg. Nucl. Chem., 1978, 40, 339							
4	25	2	Unknown	lgK:2.2 (0.08SD)	5	CMN	5599
5	40	2	Unknown	lgK:2.1 (0.03SD)	4	MSE	5599

Reaction No. 57936							
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3<SCN-1> + Np+4 = Np+4_SCN-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	2	Unknown	lgK:2.6(0.03SD)	5	MSE	5599
2	25	0	Inf. Dilution	lgK:6.06(3SF)	0	EBU	6372
3	25	2	Unknown	lgK:2.5(0.03SD)	5	CMN	5599
4	40	2	Unknown	lgK:2.3(0.04SD)	4	MSE	5599

Reaction No. 57937							
Np+4_SCN-1 + SCN-1 = Np+4_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:0.56(2SF)	5	MDS	11516
Note (1): Rao P et al., , J. Inorg. Nucl. Chem., 1978, 40, 339							
2	25	2	Unknown	dH:-0.5(0.04SD)	5	ENS	5599

Reaction No. 57938							
Np+4_SCN-1(2) + SCN-1 = Np+4_SCN-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	Unknown	dH:-0.9(0.1SD)	5	ENS	5599

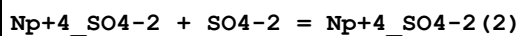
Reaction No. 58119							
3<PO4-3> + Np+4 + 3<H+1> = Np+4_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.5(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:70.50(4SF)	3	CRV	9402
3	25	0	Inf. Dilution	dH:-10.00(4SF)	3	CRV	9402

Reaction No. 58120							
2<PO4-3> + Np+4 + 2<H+1> = Np+4_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.5(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:48.50(4SF)	3	CRV	9402
3	25	0	Inf. Dilution	dH:0.50(2SF)	3	CRV	9402

Reaction No. 58121							
PO4-3 + Np+4 + H+1 = Np+4_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.4(4SF)	1	EOR	

2	25	0	Inf. Dilution	lgK:25.40 (4SF)	3	CRV	9402
3	25	0	Inf. Dilution	dH:7.00 (3SF)	3	CRV	9402

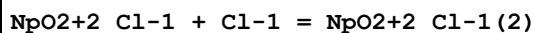
Reaction No. 58524



Note: Rxn is tricky due to large effect of ionic strength

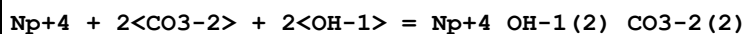
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.40 (3SF)	0	MEF	11516
Note (2): Lemaire R, A. E. C. L., 1984, 7817							
3	25	1	NaClO ₄	lgK:1.19 (3SF)	0	MKN	11516
Note (3): Newton T et al., Inorg. Chem., 1976, 15, 2856							
4	25	2	NaClO ₄	lgK:1.04 (3SF)	3	MDS	11516
Note (4): Sullivan J et al., J. Am. Chem. Soc., 1954, 76, 5931							
5	25	2	NaClO ₄	dH:6.22 (3SF)	4	MCL	140

Reaction No. 58590



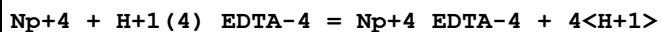
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	3	H(ClO ₄)	lgK:-0.80 (2SF)	3	MKN	140
2	4.78	3	H(ClO ₄)	lgK:-0.74 (2SF)	3	MKN	140
3	9.84	3	H(ClO ₄)	lgK:-0.70 (2SF)	3	MKN	140
4	25	0	Inf. Dilution	lgK:-0.5 (1SF)	1	ESC	
5	0	3	H(ClO ₄)	dH:3.5 (2SF)	3	MTD	140

Reaction No. 59318



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.17 (4SF)	2	EOD	29848
2	25	0	Inf. Dilution	lgK:44.39 (0.6SD)	3	CRV	31301

Reaction No. 60182



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.2:1	Unknown	lgK:1.8 (2SF)	0	MDS	903
2	25	0	Inf. Dilution	lgK:8.0 (2SF)	1	ELC	

Reaction No. 60419							
OH-1 + Np+3 = Np+3_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.147(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:6.5(2SF)	1	EOR	
3	25	0.5	Unknown	lgK:6.3(2SF)	4	CRV	2556

Reaction No. 60450							
Np+4 + OH-1 = Np+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:12.5(3SF)	2	CRV	2556
3	25	0	Inf. Dilution	lgK:14.55(4SF)	0	EOD	29848
4	25	1	Unknown	lgK:11.7(0.2SD)	4	CRV	2556
5	25	2	NaClO4	lgK:11.7(3SF)	4	CRV	2556

Reaction No. 60451							
Np+4_OH-1(4)_(s) = Np+4 + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-55.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-57.85(4SF)	0	CRV	2556

Reaction No. 60452							
NpO2(s) + 2<H2O> = Np+4 + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-60.0(3SF)	0	ENS	773
3	25	0	Inf. Dilution	lgK:-60.2(3SF)	0	CRV	2556
4	25	1	Unknown	lgK:-57.9(3SF)	3	ENS	773

Reaction No. 60455							
NpO2+1 + OH-1 = NpO2+1_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.14(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.7(0.7SD)	0	CRV	25812
3	25	0.1	Unknown	lgK:4.9(2SF)	0	CRV	2556
4	25	1	Unknown	lgK:4.7(2SF)	2	CRV	2556

Reaction No. 60456							
$\text{NpO}_2 + \text{OH}^- (\text{s}) = \text{NpO}_2 + \text{OH}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -9.372 (4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: -9.3 (2\text{SF})$	4	ENS	773
3	25	0.1	Unknown	$\lg K: -9.2 (2\text{SF})$	5	CRV	2556
4	25	1	Unknown	$\lg K: -9.1 (2\text{SF})$	4	ENS	773
5	25	1	Unknown	$\lg K: -8.81 (3\text{SF})$	5	CRV	2556

Reaction No. 60464							
$\text{OH}^- + \text{NpO}_2 + 2\text{H}^+ = \text{NpO}_2 + \text{OH}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 8.826 (4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: 8.9 (2\text{SF})$	4	CRV	2556
3	25	1	Unknown	$\lg K: 8.6 (2\text{SF})$	5	CRV	2556

Reaction No. 60465							
$2\text{OH}^- + 2\text{NpO}_2 + 2\text{H}^+ = \text{NpO}_2 + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 21.61 (4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: 21.6 (3\text{SF})$	4	CRV	2556
3	25	1	Unknown	$\lg K: 20.9 (3\text{SF})$	5	CRV	2556

Reaction No. 60466							
$5\text{OH}^- + 3\text{NpO}_2 + 2\text{H}^+ = \text{NpO}_2 + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 52.45 (4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: 52.5 (3\text{SF})$	4	CRV	2556
3	25	1	Unknown	$\lg K: 50.7 (3\text{SF})$	5	CRV	2556

Reaction No. 60626							
$2\text{SO}_4^{2-} + \text{NpO}_2 + 2\text{H}^+ = \text{NpO}_2 + 2\text{H}_2\text{SO}_4$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 4.91 (4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: 4.65 (3\text{SF})$	4	CRV	2556
3	25	0	Inf. Dilution	$\lg K: 3.89 (3\text{SF})$	3	EBU	6372

4	25	0.1	Unknown	lgK:3.8 (2SF)	5	CRV	2556
5	25	0.5	Unknown	lgK:3.4 (2SF)	5	CRV	2556
6	25	1	Unknown	lgK:2.61 (3SF)	5	CRV	2556
7	25	2	NaClO ₄	lgK:0.6 (1SF)	5	MDS	5595
8	25	0	Inf. Dilution	dH:6.2 (2SF)	4	CRV	2556

Reaction No. 60668							
2<F-1> + NpO ₂ +2 = NpO ₂ +2_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:6.9 (0.1SD)	3	CRV	2556
2	25	0	Inf. Dilution	lgK:8.32 (3SF)	2	EBU	6372
3	25	0	Inf. Dilution/m	lgK:7.70 (3SF)	3	CRV	24953
4	25	0.01	Unknown/m	lgK:7.43 (3SF)	2	CRV	24953
5	25	0.1	Unknown	lgK:7.0 (0.03SD)	5	CRV	2556
6	25	0.1	Unknown/m	lgK:7.06 (3SF)	2	CRV	24953
7	25	0.2	Unknown/m	lgK:6.94 (3SF)	2	CRV	24953
8	25	0.5	Unknown	lgK:7.00 (3SF)	5	CRV	2556
9	25	0.5	Unknown/m	lgK:6.89 (3SF)	2	CRV	24953
10	25	1	Unknown/m	lgK:7.10 (3SF)	2	CRV	24953
11	25	2	Unknown/m	lgK:7.76 (3SF)	2	CRV	24953
12	25	3	Unknown/m	lgK:8.54 (3SF)	1	CRV	24953
13	50	0	Inf. Dilution/m	lgK:7.78 (3SF)	2	CRV	24953
14	75	0	Inf. Dilution/m	lgK:7.92 (3SF)	2	CRV	24953
15	100	0	Inf. Dilution/m	lgK:8.10 (3SF)	1	CRV	24953
16	150	0	Inf. Dilution/m	lgK:8.55 (3SF)	1	CRV	24953
17	200	0	Inf. Dilution/m	lgK:9.05 (3SF)	0	CRV	24953
18	250	0	Inf. Dilution/m	lgK:9.58 (3SF)	0	CRV	24953
19	300	0	Inf. Dilution/m	lgK:10.11 (4SF)	0	CRV	24953
20	25	0	Inf. Dilution	dH:4.32 (24.06SD) kJ	4	CRV	32222

Reaction No. 60669							
3<F-1> + NpO ₂ +2 = NpO ₂ +2_F-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.7 (2SF)	5	CRV	2556
2	20	1	Unknown	lgK:8.5 (2SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:10.67 (4SF)	0	EBU	6372

Reaction No. 60826							
Salicylic-2 + NpO ₂ +1 = NpO ₂ +1_Salicylic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:0.84 (2SF)	6	MDS	5987
2	25	2	NaClO ₄	lgK:0.20 (2SF)	6	MSP	5987
3	25	2	NaClO ₄	lgK:0.28 (2SF)	5	MSP	11516
Note (3): Rizkalla E et al., Radiochim. Acta, 1990, 51, 1139							

Reaction No. 61090							
SulfSal-3 + NpO ₂ +1 = NpO ₂ +1_SulfSal-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO ₄	lgK:0.17 (2SF)	6	MSP	5987

Reaction No. 61490							
2<NTA-3> + NpO ₂ +1 = NpO ₂ +1_NTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	ClO ₄ -1 salt	lgK:6.77 (3SF)	5	MSP	1118

Reaction No. 62507							
NpO ₂ +1_IDA-2 + H+1 = H+1_NpO ₂ +1_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:4.5 (2SF)	6	CRV	552

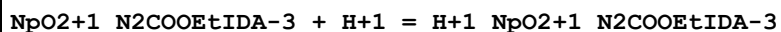
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IDA-2 + NpO ₂ +2 = NpO ₂ +2_IDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Na+1 salt	lgK:8.72 (3SF)	6	CRV	552
2	25	0	Inf. Dilution	lgK:8.7 (2SF)	1	ESC	

Reaction No. 62595							
NpO ₂ +1_MIDA-2 + H+1 = H+1_NpO ₂ +1_MIDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.54 (3SF)	6	CRV	552

Reaction No. 62701							
NpO ₂ +1_NTA-3 + H+1 = H+1_NpO ₂ +1_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

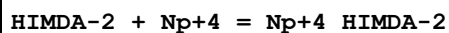
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Reaction No. 62716



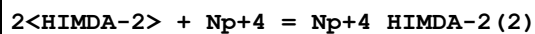
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Reaction No. 62962



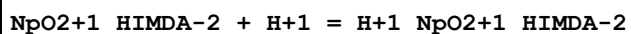
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Reaction No. 62963



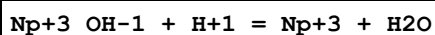
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1	25	1	Unknown	lgK:23.72 (4SF)	6	CRV	552

Reaction No. 62964



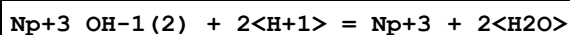
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1	25	0.1	Unknown	lgK:4.10 (3SF)	6	CRV	552

Reaction No. 63592



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.1 (2SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:7.77 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:7.00 (3SF)	4	CRV	9402
4	25	0	Inf. Dilution	lgK:6.80 (0.30SD)	4	CRV	32222
5	25	0	Inf. Dilution	dH:-77.09 (8.84SD) kJ	4	CRV	32222

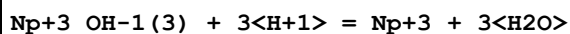
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No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.54 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:15.00 (4SF)	0	CRV	9402

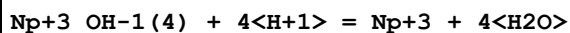
4	25	0	Inf. Dilution	lgK:17.00(1.00SD)	4	CRV	32222
5	25	0	Inf. Dilution	dH:-155.53(10.39SD)kJ	4	CRV	32222

Reaction No. 63594



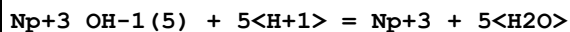
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.17(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:25.00(4SF)	0	CRV	9402
3	25	0	Inf. Dilution	lgK:27.00(2.00SD)	2	CRV	32222
4	25	0	Inf. Dilution	dH:-235.820(14.34SD)kJ	4	CRV	32222

Reaction No. 63595



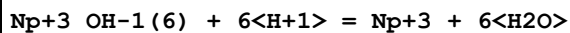
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.62(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:35.00(4SF)	4	CRV	9402

Reaction No. 63596



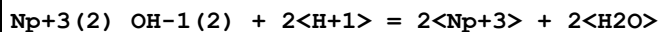
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.86(4SF)	1	EBU	6372

Reaction No. 63597



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:59.88(4SF)	1	EBU	6372

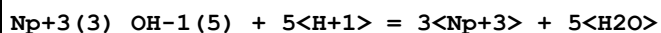
Reaction No. 63598



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.48(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:14.00(4SF)	3	CRV	9402

Note (2): inter-reaction consistency

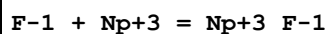
Reaction No. 63599



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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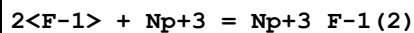
1	25	0	Inf. Dilution	lgK:29.52 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:32.00 (4SF)	3	CRV	9402

Reaction No. 63600



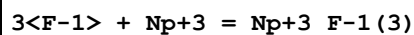
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.39 (3SF)	1	EBU	6372

Reaction No. 63601



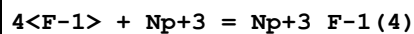
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.94 (3SF)	1	EBU	6372

Reaction No. 63602



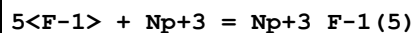
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.79 (4SF)	1	EBU	6372

Reaction No. 63603



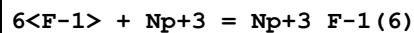
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.98 (4SF)	1	EBU	6372

Reaction No. 63604



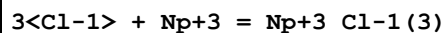
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.54 (4SF)	1	EBU	6372

Reaction No. 63605



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.49 (4SF)	1	EBU	6372

Reaction No. 63606



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.10 (3SF)	1	EBU	6372

Reaction No. 63607							
$4\text{Cl-1} + \text{Np+3} = \text{Np+3_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.94(2SF)	1	EBU	6372

Reaction No. 63608							
$5\text{Cl-1} + \text{Np+3} = \text{Np+3_Cl-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.60(2SF)	1	EBU	6372

Reaction No. 63609							
$6\text{Cl-1} + \text{Np+3} = \text{Np+3_Cl-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.09(1SF)	1	EBU	6372

Reaction No. 63610							
$3\text{Br-1} + \text{Np+3} = \text{Np+3_Br-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.53(2SF)	1	EBU	6372

Reaction No. 63611							
$4\text{Br-1} + \text{Np+3} = \text{Np+3_Br-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.32(2SF)	1	EBU	6372

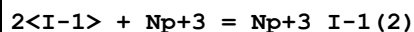
Reaction No. 63612							
$5\text{Br-1} + \text{Np+3} = \text{Np+3_Br-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.0(2SF)	1	EBU	6372

Reaction No. 63613							
$6\text{Br-1} + \text{Np+3} = \text{Np+3_Br-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.43(2SF)	1	EBU	6372

Reaction No. 63614							
$\text{I-1} + \text{Np+3} = \text{Np+3_I-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

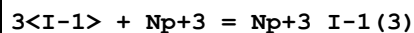
1	25	0	Inf. Dilution	lgK:0.30 (2SF)	1	EBU	6372
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Reaction No. 63615



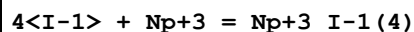
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.29 (2SF)	1	EBU	6372

Reaction No. 63616



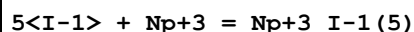
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.11 (2SF)	1	EBU	6372

Reaction No. 63617



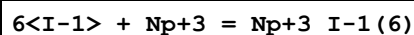
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.19 (2SF)	1	EBU	6372

Reaction No. 63618



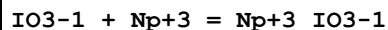
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.60 (2SF)	1	EBU	6372

Reaction No. 63619



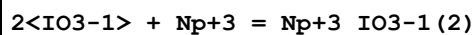
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.09 (3SF)	1	EBU	6372

Reaction No. 63620



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.72 (3SF)	1	EBU	6372

Reaction No. 63621



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.57 (3SF)	1	EBU	6372

Reaction No. 63622

3<IO3-1> + Np+3 = Np+3_IO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.79 (3SF)	1	EBU	6372

Reaction No. 63623							
4<IO3-1> + Np+3 = Np+3_IO3-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.47 (3SF)	1	EBU	6372

Reaction No. 63624							
5<IO3-1> + Np+3 = Np+3_IO3-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.65 (3SF)	1	EBU	6372

Reaction No. 63625							
6<IO3-1> + Np+3 = Np+3_IO3-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.47 (2SF)	1	EBU	6372

Reaction No. 63626							
SO4-2 + Np+3 = Np+3_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.09 (3SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:3.86 (0.66SD)	4	CRV	32222
3	25	0	Inf. Dilution	dH:17.89 (4.05SD) kJ	4	CRV	32222

Reaction No. 63627							
2<SO4-2> + Np+3 = Np+3_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.59 (3SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:5.56 (0.91SD)	4	CRV	32222

Reaction No. 63628							
3<SO4-2> + Np+3 = Np+3_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.75 (3SF)	1	EBU	6372

Reaction No. 63629							
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4<SO4-2> + Np+3 = Np+3_SO4-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.67 (3SF)	1	EBU	6372

Reaction No. 63630							
5<SO4-2> + Np+3 = Np+3_SO4-2 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.40 (3SF)	1	EBU	6372

Reaction No. 63631							
NH3 + Np+3 = Np+3_NH3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.34 (3SF)	1	EBU	6372

Reaction No. 63632							
2<NH3> + Np+3 = Np+3_NH3 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.82 (3SF)	1	EBU	6372

Reaction No. 63633							
3<NH3> + Np+3 = Np+3_NH3 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.58 (3SF)	1	EBU	6372

Reaction No. 63634							
4<NH3> + Np+3 = Np+3_NH3 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.66 (3SF)	1	EBU	6372

Reaction No. 63635							
5<NH3> + Np+3 = Np+3_NH3 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.10 (3SF)	1	EBU	6372

Reaction No. 63636							
6<NH3> + Np+3 = Np+3_NH3 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.91 (3SF)	1	EBU	6372

Reaction No. 63637							
$\text{NO3-1} + \text{Np+3} = \text{Np+3_NO3-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.01 (3SF)	1	EBU	6372

Reaction No. 63638							
$2\text{<NO3-1>} + \text{Np+3} = \text{Np+3_NO3-1 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.92 (2SF)	1	EBU	6372

Reaction No. 63639							
$3\text{<NO3-1>} + \text{Np+3} = \text{Np+3_NO3-1 (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.01 (1SF)	1	EBU	6372

Reaction No. 63640							
$4\text{<NO3-1>} + \text{Np+3} = \text{Np+3_NO3-1 (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.68 (3SF)	1	EBU	6372

Reaction No. 63641							
$5\text{<NO3-1>} + \text{Np+3} = \text{Np+3_NO3-1 (5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.03 (3SF)	1	EBU	6372

Reaction No. 63642							
$\text{Np+3} + \text{H+1_PO4-3} = \text{Np+3_H+1_PO4-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:7.40 (3SF)	0	EBU	6372

Reaction No. 63643							
$\text{Np+3} + 2\text{<H+1_PO4-3>} = \text{Np+3_H+1 (2)_PO4-3 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:13.50 (4SF)	0	EBU	6372

Reaction No. 63644							
$\text{Np}+3 + 3\text{<H+1_PO4-3>} = \text{Np}+3\text{_{H+1(3)_PO4-3(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.57 (4SF)	1	EBU	6372

Reaction No. 63645							
$\text{Np}+3 + 4\text{<H+1_PO4-3>} = \text{Np}+3\text{_{H+1(4)_PO4-3(4)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.70 (4SF)	1	EBU	6372

Reaction No. 63646							
$\text{Np}+3 + 5\text{<H+1_PO4-3>} = \text{Np}+3\text{_{H+1(5)_PO4-3(5)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.94 (4SF)	1	EBU	6372

Reaction No. 63647							
$\text{Np}+3 + 6\text{<H+1_PO4-3>} = \text{Np}+3\text{_{H+1(6)_PO4-3(6)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.34 (4SF)	1	EBU	6372

Reaction No. 63648							
$\text{Np}+3 + \text{H+1(2)_PO4-3} = \text{Np}+3\text{_{H+1(2)_PO4-3}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.56 (3SF)	1	EBU	6372

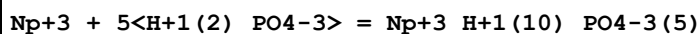
Reaction No. 63649							
$\text{Np}+3 + 2\text{<H+1(2)_PO4-3>} = \text{Np}+3\text{_{H+1(4)_PO4-3(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.75 (3SF)	1	EBU	6372

Reaction No. 63650							
$\text{Np}+3 + 3\text{<H+1(2)_PO4-3>} = \text{Np}+3\text{_{H+1(6)_PO4-3(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.82 (3SF)	0	EBU	6372

Reaction No. 63651							
$\text{Np}+3 + 4\text{<H+1(2)_PO4-3>} = \text{Np}+3\text{_{H+1(8)_PO4-3(4)}}$							

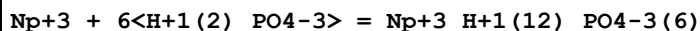
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.87(3SF)	1	EBU	6372

Reaction No. 63652



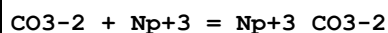
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.96(2SF)	1	EBU	6372

Reaction No. 63653



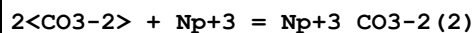
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.88(3SF)	1	EBU	6372

Reaction No. 63654



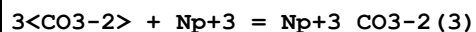
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.27(3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:9.60(3SF)	0	CRV	9402
3	25	0	Inf. Dilution	lgK:7.67(0.86SD)	4	CRV	32222
4	25	0	Inf. Dilution	dH:156.37(5.13SD)kJ	4	CRV	32222

Reaction No. 63655



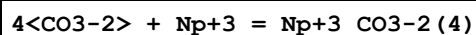
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.12(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:12.90(4SF)	2	CRV	9402
3	25	0	Inf. Dilution	lgK:12.60(0.85SD)	4	CRV	32222

Reaction No. 63656



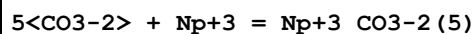
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.82(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:16.20(4SF)	2	CRV	9402
3	25	0	Inf. Dilution	lgK:15.66(2.51SD)	4	CRV	32222

Reaction No. 63657



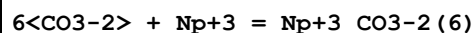
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.47 (4SF)	1	EBU	6372

Reaction No. 63658



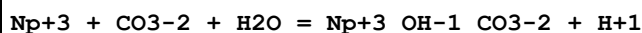
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.12 (4SF)	1	EBU	6372

Reaction No. 63659



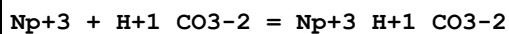
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.80 (4SF)	1	EBU	6372

Reaction No. 63660



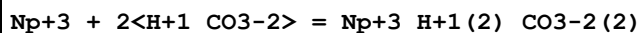
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.69 (3SF)	1	EBU	6372

Reaction No. 63661



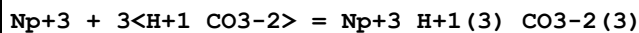
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.63 (3SF)	1	EBU	6372

Reaction No. 63662



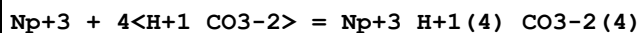
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.71 (3SF)	1	EBU	6372

Reaction No. 63663



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.49 (3SF)	1	EBU	6372

Reaction No. 63664



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.08 (3SF)	1	EBU	6372

Reaction No. 63665							
$\text{Np}+3 + 5\text{<H+1_CO3-2>} = \text{Np}+3\text{_{H+1(5)_CO3-2(5)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.52(3SF)	1	EBU	6372

Reaction No. 63666							
$\text{Np}+3 + 6\text{<H+1_CO3-2>} = \text{Np}+3\text{_{H+1(6)_CO3-2(6)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.86(3SF)	1	EBU	6372

Reaction No. 63667							
$\text{CN-1} + \text{Np}+3 = \text{Np}+3\text{_{CN-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.60(3SF)	1	EBU	6372

Reaction No. 63668							
$2\text{<CN-1>} + \text{Np}+3 = \text{Np}+3\text{_{CN-1(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.29(3SF)	1	EBU	6372

Reaction No. 63669							
$3\text{<CN-1>} + \text{Np}+3 = \text{Np}+3\text{_{CN-1(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.19(4SF)	1	EBU	6372

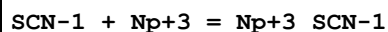
Reaction No. 63670							
$4\text{<CN-1>} + \text{Np}+3 = \text{Np}+3\text{_{CN-1(4)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.36(4SF)	1	EBU	6372

Reaction No. 63671							
$5\text{<CN-1>} + \text{Np}+3 = \text{Np}+3\text{_{CN-1(5)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.82(4SF)	1	EBU	6372

Reaction No. 63672							
$6\text{<CN-1>} + \text{Np}+3 = \text{Np}+3\text{_{CN-1(6)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

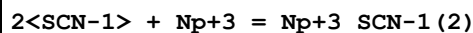
1	25	0	Inf. Dilution	lgK:15.60 (4SF)	1	EBU	6372
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Reaction No. 63673



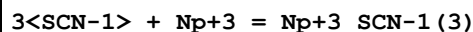
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.29 (3SF)	1	EBU	6372

Reaction No. 63674



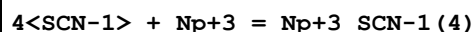
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.82 (3SF)	1	EBU	6372

Reaction No. 63675



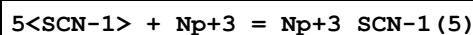
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.71 (3SF)	1	EBU	6372

Reaction No. 63676



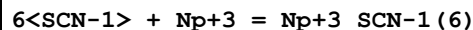
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.02 (3SF)	1	EBU	6372

Reaction No. 63677



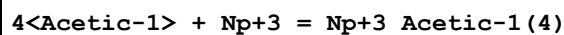
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.23 (2SF)	1	EBU	6372

Reaction No. 63678



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.03 (3SF)	1	EBU	6372

Reaction No. 63679



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.18 (3SF)	1	EBU	6372

Reaction No. 63680

5<Acetic-1> + Np+3 = Np+3_Acetic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.28(3SF)	1	EBU	6372

Reaction No. 63681							
6<Acetic-1> + Np+3 = Np+3_Acetic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.69(2SF)	1	EBU	6372

Reaction No. 63682							
Benzoic-1 + Np+3 = Np+3_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.43(3SF)	1	EBU	6372

Reaction No. 63683							
2<Benzoic-1> + Np+3 = Np+3_Benzoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.51(3SF)	1	EBU	6372

Reaction No. 63684							
3<Benzoic-1> + Np+3 = Np+3_Benzoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.47(3SF)	1	EBU	6372

Reaction No. 63685							
4<Benzoic-1> + Np+3 = Np+3_Benzoic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.42(3SF)	1	EBU	6372

Reaction No. 63686							
5<Benzoic-1> + Np+3 = Np+3_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.42(3SF)	1	EBU	6372

Reaction No. 63687							
6<Benzoic-1> + Np+3 = Np+3_Benzoic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.50(3SF)	1	EBU	6372

Reaction No. 63688							
Cat-2 + Np+3 = Np+3_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.75 (4SF)	1	EBU	6372

Reaction No. 63689							
2<Cat-2> + Np+3 = Np+3_Cat-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.66 (4SF)	1	EBU	6372

Reaction No. 63690							
3<Cat-2> + Np+3 = Np+3_Cat-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.43 (4SF)	1	EBU	6372

Reaction No. 63691							
EtDiAm + Np+3 = Np+3_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.70 (3SF)	1	EBU	6372

Reaction No. 63692							
2<EtDiAm> + Np+3 = Np+3_EtDiAm (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.85 (4SF)	1	EBU	6372

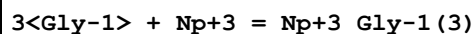
Reaction No. 63693							
3<EtDiAm> + Np+3 = Np+3_EtDiAm (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.13 (4SF)	1	EBU	6372

Reaction No. 63694							
Gly-1 + Np+3 = Np+3_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.24 (3SF)	1	EBU	6372

Reaction No. 63695							
2<Gly-1> + Np+3 = Np+3_Gly-1 (2)							

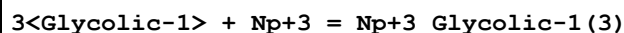
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.71(4SF)	1	EBU	6372

Reaction No. 63696



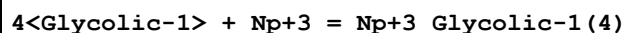
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.50(4SF)	1	EBU	6372

Reaction No. 63697



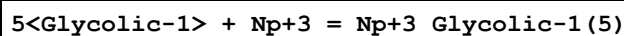
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.92(3SF)	1	EBU	6372

Reaction No. 63698



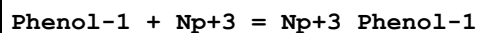
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.65(3SF)	1	EBU	6372

Reaction No. 63699



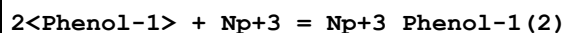
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.25(2SF)	1	EBU	6372

Reaction No. 63700



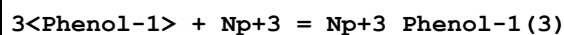
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.86(3SF)	1	EBU	6372

Reaction No. 63701



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.57(3SF)	1	EBU	6372

Reaction No. 63702



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.25(4SF)	1	EBU	6372

Reaction No. 63703							
4<Phenol-1> + Np+3 = Np+3_Phenol-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.97(4SF)	1	EBU	6372

Reaction No. 63704							
5<Phenol-1> + Np+3 = Np+3_Phenol-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.73(4SF)	1	EBU	6372

Reaction No. 63705							
6<Phenol-1> + Np+3 = Np+3_Phenol-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.57(4SF)	1	EBU	6372

Reaction No. 63706							
Np+4 + 6<H2O> = Np+4_OH-1(6) + 6<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.73(4SF)	1	EBU	6372

Reaction No. 63707							
Np+4 + 12<H2O> = Np+4_OH-1(12) + 12<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.46(4SF)	1	EBU	6372

Reaction No. 63708							
Np+4 + 15<H2O> = Np+4_OH-1(15) + 15<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.53(4SF)	1	EBU	6372

Reaction No. 63709							
5<F-1> + Np+4 = Np+4_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.83(4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:25.30(4SF)	3	CRV	9402
3	25	0	Inf. Dilution	dH:4.70(3SF)	3	CRV	9402

Reaction No. 63710							
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6<F-1> + Np+4 = Np+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.84 (4SF)	1	EBU	6372

Reaction No. 63711							
4<Cl-1> + Np+4 = Np+4_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.72 (3SF)	1	EBU	6372

Reaction No. 63712							
5<Cl-1> + Np+4 = Np+4_Cl-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.26 (3SF)	1	EBU	6372

Reaction No. 63713							
6<Cl-1> + Np+4 = Np+4_Cl-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.71 (3SF)	1	EBU	6372

Reaction No. 63714							
3<Br-1> + Np+4 = Np+4_Br-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.57 (3SF)	1	EBU	6372

Reaction No. 63715							
4<Br-1> + Np+4 = Np+4_Br-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.74 (3SF)	1	EBU	6372

Reaction No. 63716							
5<Br-1> + Np+4 = Np+4_Br-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.82 (3SF)	1	EBU	6372

Reaction No. 63717							
6<Br-1> + Np+4 = Np+4_Br-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.81 (3SF)	1	EBU	6372

Reaction No. 63718							
$I-1 + Np+4 = Np+4_{I-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.48 (2SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:1.50 (0.40SD)	4	CRV	32222

Reaction No. 63719							
$2<I-1> + Np+4 = Np+4_{I-1} (2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.65 (2SF)	1	EBU	6372

Reaction No. 63720							
$3<I-1> + Np+4 = Np+4_{I-1} (3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.66 (2SF)	1	EBU	6372

Reaction No. 63721							
$4<I-1> + Np+4 = Np+4_{I-1} (4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.53 (2SF)	1	EBU	6372

Reaction No. 63722							
$5<I-1> + Np+4 = Np+4_{I-1} (5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.31 (2SF)	1	EBU	6372

Reaction No. 63723							
$6<I-1> + Np+4 = Np+4_{I-1} (6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.01 (1SF)	1	EBU	6372

Reaction No. 63724							
$IO3-1 + Np+4 = Np+4_{IO3-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.26 (3SF)	1	EBU	6372

Reaction No. 63725							
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2<IO3-1> + Np+4 = Np+4_IO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.85 (3SF)	1	EBU	6372

Reaction No. 63726							
3<IO3-1> + Np+4 = Np+4_IO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.03 (3SF)	1	EBU	6372

Reaction No. 63727							
4<IO3-1> + Np+4 = Np+4_IO3-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.90 (3SF)	1	EBU	6372

Reaction No. 63728							
5<IO3-1> + Np+4 = Np+4_IO3-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.51 (4SF)	1	EBU	6372

Reaction No. 63729							
6<IO3-1> + Np+4 = Np+4_IO3-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.90 (4SF)	1	EBU	6372

Reaction No. 63730							
3<SO4-2> + Np+4 = Np+4_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.03 (4SF)	1	EBU	6372

Reaction No. 63731							
4<SO4-2> + Np+4 = Np+4_SO4-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.37 (4SF)	1	EBU	6372

Reaction No. 63732							
5<SO4-2> + Np+4 = Np+4_SO4-2 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.68 (4SF)	1	EBU	6372

Reaction No. 63733							
$6\text{SO}_4\text{-2} + \text{Np+4} = \text{Np+4_SO}_4\text{-2 (6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.01 (4SF)	1	EBU	6372

Reaction No. 63734							
$\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.60 (3SF)	1	EBU	6372

Reaction No. 63735							
$2\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3\text{ (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.72 (4SF)	1	EBU	6372

Reaction No. 63736							
$3\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3\text{ (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.48 (4SF)	1	EBU	6372

Reaction No. 63737							
$4\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3\text{ (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.93 (4SF)	1	EBU	6372

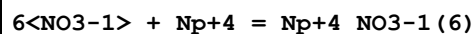
Reaction No. 63738							
$5\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3\text{ (5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.11 (4SF)	1	EBU	6372

Reaction No. 63739							
$6\text{NH}_3 + \text{Np+4} = \text{Np+4_NH}_3\text{ (6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.02 (4SF)	1	EBU	6372

Reaction No. 63740							
$5\text{NO}_3\text{-1} + \text{Np+4} = \text{Np+4_NO}_3\text{-1 (5)}$							

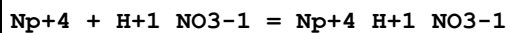
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.63(3SF)	1	EBU	6372

Reaction No. 63741



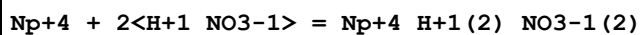
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.18(3SF)	1	EBU	6372

Reaction No. 63742



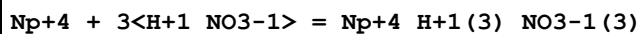
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.08(4SF)	1	EBU	6372

Reaction No. 63743



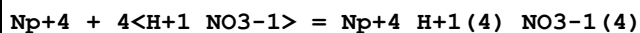
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.57(4SF)	1	EBU	6372

Reaction No. 63744



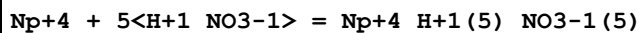
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.71(4SF)	1	EBU	6372

Reaction No. 63745



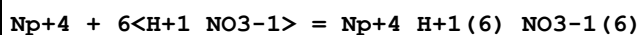
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.61(4SF)	1	EBU	6372

Reaction No. 63746



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.32(4SF)	1	EBU	6372

Reaction No. 63747



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.87(4SF)	1	EBU	6372

Reaction No. 63748							
$\text{Np}+4 + \text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(2)_{\text{NO}3-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.83(3SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63749							
$\text{Np}+4 + 2\text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(4)_{\text{NO}3-1(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.05(3SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63750							
$\text{Np}+4 + 3\text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(6)_{\text{NO}3-1(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.93(3SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63751							
$\text{Np}+4 + 4\text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(8)_{\text{NO}3-1(4)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.55(4SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63752							
$\text{Np}+4 + 5\text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(10)_{\text{NO}3-1(5)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.99(3SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63753							
$\text{Np}+4 + 6\text{H}+1(2)_{\text{NO}3-1} = \text{Np}+4_{\text{H}+1(12)_{\text{NO}3-1(6)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.26(3SF)	0	EBU	6372
Note (1): Spurious species H+1(2)_NO3-1							

Reaction No. 63754							
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CO3-2 + Np+4 = Np+4_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.89 (4SF)	1	EBU	6372

Reaction No. 63755							
2<CO3-2> + Np+4 = Np+4_CO3-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.81 (4SF)	1	EBU	6372

Reaction No. 63756							
3<CO3-2> + Np+4 = Np+4_CO3-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.01 (4SF)	1	EBU	6372

Reaction No. 63757							
4<CO3-2> + Np+4 = Np+4_CO3-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.58 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:38.91 (4SF)	2	EOD	29848

Reaction No. 63758							
5<CO3-2> + Np+4 = Np+4_CO3-2 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.59 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:36.50 (4SF)	2	CRV	9402
3	25	0	Inf. Dilution	lgK:37.84 (4SF)	2	EOD	29848
4	25	0	Inf. Dilution	dH:-1.61 (6.80SD) kJ	4	CRV	32222

Reaction No. 63759							
6<CO3-2> + Np+4 = Np+4_CO3-2 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.08 (4SF)	1	EBU	6372

Reaction No. 63760							
Np+4 + H+1_CO3-2 = Np+4_H+1_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.98 (3SF)	1	EBU	6372

Reaction No. 63761							
$\text{Np+4} + 2\langle\text{H+1_CO3-2}\rangle = \text{Np+4_H+1(2)_CO3-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.79(4SF)	1	EBU	6372

Reaction No. 63762							
$\text{Np+4} + 3\langle\text{H+1_CO3-2}\rangle = \text{Np+4_H+1(3)_CO3-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.70(4SF)	1	EBU	6372

Reaction No. 63763							
$\text{Np+4} + 4\langle\text{H+1_CO3-2}\rangle = \text{Np+4_H+1(4)_CO3-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.79(4SF)	1	EBU	6372

Reaction No. 63764							
$\text{Np+4} + 5\langle\text{H+1_CO3-2}\rangle = \text{Np+4_H+1(5)_CO3-2(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.14(4SF)	1	EBU	6372

Reaction No. 63765							
$\text{Np+4} + 6\langle\text{H+1_CO3-2}\rangle = \text{Np+4_H+1(6)_CO3-2(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.77(4SF)	1	EBU	6372

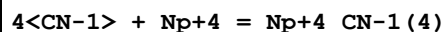
Reaction No. 63766							
$\text{CN-1} + \text{Np+4} = \text{Np+4_CN-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.86(3SF)	1	EBU	6372

Reaction No. 63767							
$2\langle\text{CN-1}\rangle + \text{Np+4} = \text{Np+4_CN-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.20(4SF)	1	EBU	6372

Reaction No. 63768							
$3\langle\text{CN-1}\rangle + \text{Np+4} = \text{Np+4_CN-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

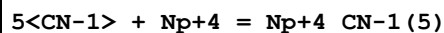
1	25	0	Inf. Dilution	lgK:25.15 (4SF)	1	EBU	6372
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Reaction No. 63769



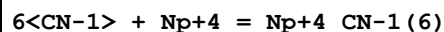
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.76 (4SF)	1	EBU	6372

Reaction No. 63770



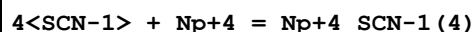
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Reaction No. 63771



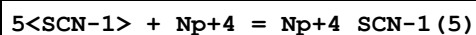
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.06 (4SF)	1	EBU	6372

Reaction No. 63772



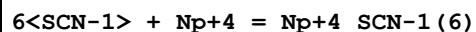
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1	25	0	Inf. Dilution	lgK:7.46 (3SF)	1	EBU	6372

Reaction No. 63773



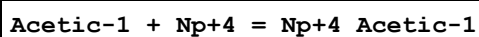
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.63 (3SF)	1	EBU	6372

Reaction No. 63774



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.59 (3SF)	1	EBU	6372

Reaction No. 63775



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.91 (3SF)	1	EBU	6372

Reaction No. 63776

2<Acetic-1> + Np+4 = Np+4_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.73 (4SF)	1	EBU	6372

Reaction No. 63777							
3<Acetic-1> + Np+4 = Np+4_Acetic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.73 (4SF)	1	EBU	6372

Reaction No. 63778							
4<Acetic-1> + Np+4 = Np+4_Acetic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.00 (4SF)	1	EBU	6372

Reaction No. 63779							
5<Acetic-1> + Np+4 = Np+4_Acetic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.60 (4SF)	1	EBU	6372

Reaction No. 63780							
6<Acetic-1> + Np+4 = Np+4_Acetic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.57 (4SF)	1	EBU	6372

Reaction No. 63781							
Benzoic-1 + Np+4 = Np+4_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.59 (3SF)	1	EBU	6372

Reaction No. 63782							
2<Benzoic-1> + Np+4 = Np+4_Benzoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.25 (4SF)	1	EBU	6372

Reaction No. 63783							
3<Benzoic-1> + Np+4 = Np+4_Benzoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.23 (4SF)	1	EBU	6372

Reaction No. 63784							
4<Benzoic-1> + Np+4 = Np+4_Benzoic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.64 (4SF)	1	EBU	6372

Reaction No. 63785							
5<Benzoic-1> + Np+4 = Np+4_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.52 (4SF)	1	EBU	6372

Reaction No. 63786							
6<Benzoic-1> + Np+4 = Np+4_Benzoic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.92 (4SF)	1	EBU	6372

Reaction No. 63787							
Cat-2 + Np+4 = Np+4_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.59 (4SF)	1	EBU	6372

Reaction No. 63788							
2<Cat-2> + Np+4 = Np+4_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.92 (4SF)	1	EBU	6372

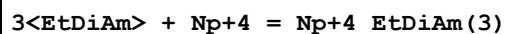
Reaction No. 63789							
3<Cat-2> + Np+4 = Np+4_Cat-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:73.65 (4SF)	1	EBU	6372

Reaction No. 63790							
EtDiAm + Np+4 = Np+4_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.70 (4SF)	1	EBU	6372

Reaction No. 63791							
2<EtDiAm> + Np+4 = Np+4_EtDiAm(2)							

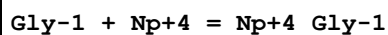
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Reaction No. 63792



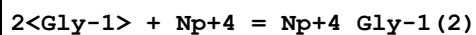
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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Reaction No. 63793



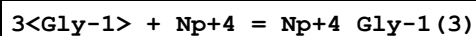
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.92 (4SF)	1	EBU	6372

Reaction No. 63794



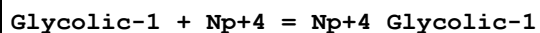
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.38 (4SF)	1	EBU	6372

Reaction No. 63795



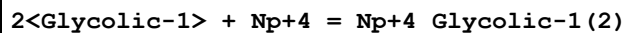
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.56 (4SF)	1	EBU	6372

Reaction No. 63796



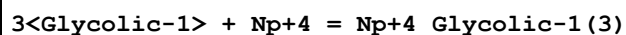
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.29 (3SF)	1	EBU	6372

Reaction No. 63797



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.43 (3SF)	1	EBU	6372

Reaction No. 63798



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.67 (4SF)	1	EBU	6372

Reaction No. 63799							
4<Glycolic-1> + Np+4 = Np+4_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.11(4SF)	1	EBU	6372

Reaction No. 63800							
5<Glycolic-1> + Np+4 = Np+4_Glycolic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.81(4SF)	1	EBU	6372

Reaction No. 63801							
6<Glycolic-1> + Np+4 = Np+4_Glycolic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.80(4SF)	1	EBU	6372

Reaction No. 63802							
Phenol-1 + Np+4 = Np+4_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.37(3SF)	1	EBU	6372

Reaction No. 63803							
2<Phenol-1> + Np+4 = Np+4_Phenol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.03(4SF)	1	EBU	6372

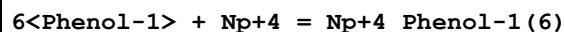
Reaction No. 63804							
3<Phenol-1> + Np+4 = Np+4_Phenol-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.09(4SF)	1	EBU	6372

Reaction No. 63805							
4<Phenol-1> + Np+4 = Np+4_Phenol-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.61(4SF)	1	EBU	6372

Reaction No. 63806							
5<Phenol-1> + Np+4 = Np+4_Phenol-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

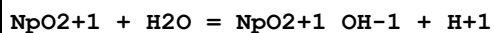
1	25	0	Inf. Dilution	lgK:40.62 (4SF)	1	EBU	6372
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Reaction No. 63807



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.13 (4SF)	1	EBU	6372

Reaction No. 63808



Note: See comments in Altmaier et al. [13AGF_25812], p. 910

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.9 (2SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-10.90 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-11.31 (4SF)	0	EOD	8120
4	25	0	Inf. Dilution/m	lgK:-8.87 (3SF)	3	CRV	24953
5	25	0.01	Unknown/m	lgK:-8.87 (3SF)	2	CRV	24953
6	25	0.1	Unknown/m	lgK:-8.89 (3SF)	2	CRV	24953
7	25	0.2	Unknown/m	lgK:-8.92 (3SF)	2	CRV	24953
8	25	0.5	Unknown/m	lgK:-8.99 (3SF)	2	CRV	24953
9	25	1	NaClO ₄	lgK:-9.1 (0.2SD)	5	MGL	6586
10	25	1	Unknown/m	lgK:-9.11 (3SF)	2	CRV	24953
11	25	2	Unknown/m	lgK:-9.35 (3SF)	2	CRV	24953
12	25	3	Unknown/m	lgK:-9.59 (3SF)	1	CRV	24953
13	50	0	Inf. Dilution/m	lgK:-8.29 (3SF)	2	CRV	24953
14	75	0	Inf. Dilution/m	lgK:-7.86 (3SF)	2	CRV	24953
15	100	0	Inf. Dilution/m	lgK:-7.53 (3SF)	1	CRV	24953
16	150	0	Inf. Dilution/m	lgK:-7.11 (3SF)	1	CRV	24953
17	200	0	Inf. Dilution/m	lgK:-6.89 (3SF)	0	CRV	24953
18	250	0	Inf. Dilution/m	lgK:-6.80 (3SF)	0	CRV	24953
19	300	0	Inf. Dilution/m	lgK:-6.80 (3SF)	0	CRV	24953
20	25	0	Inf. Dilution	dH:64.78 (18.61SD) kJ	4	CRV	32222

Reaction No. 63809



Note: See comments in Altmaier et al. [13AGF_25812], p. 910

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:-20.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-22.57 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-23.54 (4SF)	0	EOD	8120
4	25	0	Inf. Dilution	lgK:-19.22 (4SF)	0	EOD	25812
5	25	0	Inf. Dilution	lgK:-12.21 (0.150SD)	0	CRV	32222
Note (5): estimated by correlation							
6	25	0	Inf. Dilution	dH:118.61 (5.22SD) kJ	2	CRV	32222

Reaction No. 63810							
$\text{NpO}_2+1 + 3\text{<H}_2\text{O>} = \text{NpO}_2+1_OH-1(3) + 3\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-34.88 (4SF)	1	EBU	6372

Reaction No. 63811							
$\text{NpO}_2+1 + 4\text{<H}_2\text{O>} = \text{NpO}_2+1_OH-1(4) + 4\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-47.78 (4SF)	1	EBU	6372

Reaction No. 63812							
$\text{NpO}_2+1 + 5\text{<H}_2\text{O>} = \text{NpO}_2+1_OH-1(5) + 5\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-61.24 (4SF)	1	EBU	6372

Reaction No. 63813							
$2\text{<F-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_F-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.14 (3SF)	1	EBU	6372

Reaction No. 63814							
$3\text{<F-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_F-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.74 (3SF)	1	EBU	6372

Reaction No. 63815							
$4\text{<F-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_F-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.47 (2SF)	1	EBU	6372

Reaction No. 63816							
5<F-1> + NpO2+1 = NpO2+1_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.67(3SF)	1	EBU	6372

Reaction No. 63817							
3<Cl-1> + NpO2+1 = NpO2+1_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.43(2SF)	1	EBU	6372

Reaction No. 63818							
4<Cl-1> + NpO2+1 = NpO2+1_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.99(2SF)	1	EBU	6372

Reaction No. 63819							
5<Cl-1> + NpO2+1 = NpO2+1_Cl-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.68(3SF)	1	EBU	6372

Reaction No. 63820							
Br-1 + NpO2+1 = NpO2+1_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.13(2SF)	1	EBU	6372

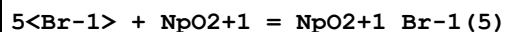
Reaction No. 63821							
2<Br-1> + NpO2+1 = NpO2+1_Br-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.05(1SF)	1	EBU	6372

Reaction No. 63822							
3<Br-1> + NpO2+1 = NpO2+1_Br-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.41(2SF)	1	EBU	6372

Reaction No. 63823							
4<Br-1> + NpO2+1 = NpO2+1_Br-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

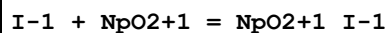
1	25	0	Inf. Dilution	lgK:-0.91(2SF)	1	EBU	6372
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Reaction No. 63824



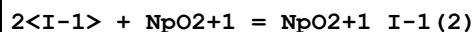
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.51(3SF)	1	EBU	6372

Reaction No. 63825



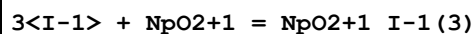
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.12(2SF)	1	EBU	6372

Reaction No. 63826



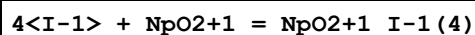
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.07(1SF)	1	EBU	6372

Reaction No. 63827



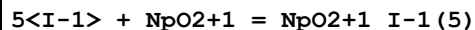
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.43(2SF)	1	EBU	6372

Reaction No. 63828



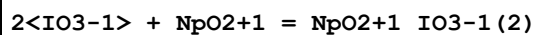
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.91(2SF)	1	EBU	6372

Reaction No. 63829



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.50(3SF)	1	EBU	6372

Reaction No. 63830



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.05(1SF)	1	EBU	6372

Reaction No. 63831

3<IO3-1> + NpO2+1 = NpO2+1_IO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.93(2SF)	1	EBU	6372

Reaction No. 63832							
4<IO3-1> + NpO2+1 = NpO2+1_IO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.21(3SF)	1	EBU	6372

Reaction No. 63833							
5<IO3-1> + NpO2+1 = NpO2+1_IO3-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.82(3SF)	1	EBU	6372

Reaction No. 63834							
2<SO4-2> + NpO2+1 = NpO2+1_SO4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.28(3SF)	1	EBU	6372

Reaction No. 63835							
3<SO4-2> + NpO2+1 = NpO2+1_SO4-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.19(2SF)	1	EBU	6372

Reaction No. 63836							
4<SO4-2> + NpO2+1 = NpO2+1_SO4-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.88(4SF)	1	EBU	6372

Reaction No. 63837							
NH3 + NpO2+1 = NpO2+1_NH3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.48(2SF)	1	EBU	6372

Reaction No. 63838							
2<NH3> + NpO2+1 = NpO2+1_NH3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.33(2SF)	1	EBU	6372

Reaction No. 63839							
$3\text{<NH3>} + \text{NpO2+1} = \text{NpO2+1_NH3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.34 (2SF)	1	EBU	6372

Reaction No. 63840							
$4\text{<NH3>} + \text{NpO2+1} = \text{NpO2+1_NH3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.47 (3SF)	1	EBU	6372

Reaction No. 63841							
$5\text{<NH3>} + \text{NpO2+1} = \text{NpO2+1_NH3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.03 (3SF)	1	EBU	6372

Reaction No. 63842							
$3\text{<NO3-1>} + \text{NpO2+1} = \text{NpO2+1_NO3-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.84 (3SF)	1	EBU	6372

Reaction No. 63843							
$4\text{<NO3-1>} + \text{NpO2+1} = \text{NpO2+1_NO3-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.72 (3SF)	1	EBU	6372

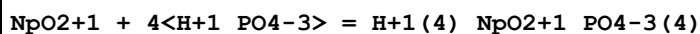
Reaction No. 63844							
$5\text{<NO3-1>} + \text{NpO2+1} = \text{NpO2+1_NO3-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.08 (3SF)	1	EBU	6372

Reaction No. 63845							
$\text{NpO2+1} + 2\text{<H+1_PO4-3>} = \text{H+1(2)_NpO2+1_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.07 (3SF)	1	EBU	6372

Reaction No. 63846							
$\text{NpO2+1} + 3\text{<H+1_PO4-3>} = \text{H+1(3)_NpO2+1_PO4-3(3)}$							

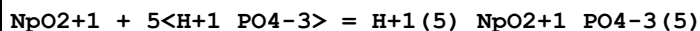
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.62 (3SF)	1	EBU	6372

Reaction No. 63847



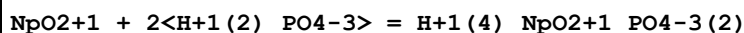
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.03 (3SF)	1	EBU	6372

Reaction No. 63848



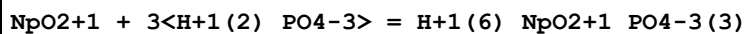
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.35 (3SF)	1	EBU	6372

Reaction No. 63849



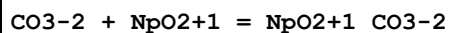
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.32 (2SF)	1	EBU	6372

Reaction No. 63850



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.55 (3SF)	1	EBU	6372

Reaction No. 63851



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.31 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:5.03 (3SF)	4	EOD	8120
3	25	0	Inf. Dilution	lgK:4.6 (2SF)	2	MEF	10566
4	25	0	Inf. Dilution	lgK:4.96 (0.06SD)	5	CRV	25812
5	25	0	Inf. Dilution/m	lgK:5.90 (3SF)	2	CRV	24953
6	25	0.01	Unknown/m	lgK:5.71 (3SF)	2	CRV	24953
7	25	0.1	Unknown	lgK:4.13 (3SF)	0	CRV	552
8	25	0.1	Unknown/m	lgK:5.37 (3SF)	2	CRV	24953
9	25	0.2	Unknown/m	lgK:5.19 (3SF)	2	CRV	24953
10	25	0.5	Unknown/m	lgK:4.86 (3SF)	2	CRV	24953
11	25	1	NaClO4	lgK:4.5 (0.06SD)	6	MGL	6586

12	25	1	Unknown/m	lgK:4.49 (3SF)	2	CRV	24953
13	25	2	Unknown/m	lgK:3.92 (3SF)	1	CRV	24953
14	25	3	Unknown/m	lgK:3.44 (3SF)	0	CRV	24953
15	50	0	Inf. Dilution/m	lgK:6.59 (3SF)	2	CRV	24953
16	75	0	Inf. Dilution/m	lgK:7.25 (3SF)	2	CRV	24953
17	100	0	Inf. Dilution/m	lgK:7.86 (3SF)	1	CRV	24953
18	150	0	Inf. Dilution/m	lgK:8.99 (3SF)	0	CRV	24953
19	200	0	Inf. Dilution/m	lgK:9.99 (3SF)	0	CRV	24953
20	25	0	Inf. Dilution	dH:59.90 (59.72SD) kJ	4	CRV	32222

Reaction No. 63852							
2<CO3-2> + NpO2+1 = NpO2+1_CO3-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.44 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:6.47 (3SF)	4	EOD	8120
3	25	0	Inf. Dilution	lgK:7.0 (2SF)	5	MEF	10566
4	25	0.1	Unknown	lgK:7.06 (3SF)	6	CRV	552
5	25	1	NaClO4	lgK:7.1 (0.07SD)	6	MGL	6586
6	25	0	Inf. Dilution	dH:39.03 (59.72SD) kJ	4	CRV	32222

Reaction No. 63853							
3<CO3-2> + NpO2+1 = NpO2+1_CO3-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.63 (3SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:5.37 (3SF)	0	EOD	8120
3	25	1	NaClO4	lgK:8.5 (0.09SD)	6	MGL	6586

Reaction No. 63854							
4<CO3-2> + NpO2+1 = NpO2+1_CO3-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.99 (4SF)	1	EBU	6372

Reaction No. 63855							
5<CO3-2> + NpO2+1 = NpO2+1_CO3-2 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.58 (4SF)	1	EBU	6372

Reaction No. 63856							
$\text{NpO}_2+1 + 2\text{<H+1_CO}_3-2\text{>} = \text{H+1(2)_NpO}_2+1_ \text{CO}_3-2(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.89(2SF)	1	EBU	6372

Reaction No. 63857							
$\text{NpO}_2+1 + 3\text{<H+1_CO}_3-2\text{>} = \text{H+1(3)_NpO}_2+1_ \text{CO}_3-2(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.48(2SF)	1	EBU	6372

Reaction No. 63858							
$\text{NpO}_2+1 + 4\text{<H+1_CO}_3-2\text{>} = \text{H+1(4)_NpO}_2+1_ \text{CO}_3-2(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.88(3SF)	1	EBU	6372

Reaction No. 63859							
$\text{CN-1} + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{CN-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.75(3SF)	1	EBU	6372

Reaction No. 63860							
$2\text{<CN-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{CN-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.80(3SF)	1	EBU	6372

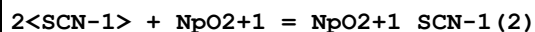
Reaction No. 63861							
$3\text{<CN-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{CN-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.30(3SF)	1	EBU	6372

Reaction No. 63862							
$4\text{<CN-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{CN-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.28(3SF)	1	EBU	6372

Reaction No. 63863							
$5\text{<CN-1>} + \text{NpO}_2+1 = \text{NpO}_2+1_ \text{CN-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

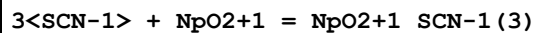
1	25	0	Inf. Dilution	lgK:2.78 (3SF)	1	EBU	6372
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Reaction No. 63864



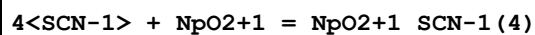
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.08 (1SF)	1	EBU	6372
2	25	5	Unknown	lgK:1.05 (3SF)	6	CRV	552

Reaction No. 63865



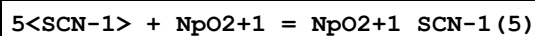
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.84 (2SF)	1	EBU	6372

Reaction No. 63866



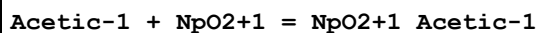
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.00 (3SF)	1	EBU	6372

Reaction No. 63867



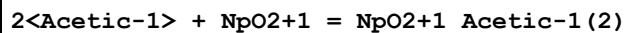
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.52 (3SF)	1	EBU	6372

Reaction No. 63868



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.85 (2SF)	1	EBU	6372
2	25	0	Inf. Dilution	lgK:1.29 (0.14MD)	5	MSP	30807
3	25	0	Inf. Dilution	lgK:1.34 (0.26MD)	5	MSP	30807
4	25	0	Inf. Dilution	lgK:1.56 (0.03SD)	2	EDT	31309
5	25	0.3	NaClO ₄	lgK:1.29 (3SF)	5	MEP	31309
6	25	0	Inf. Dilution	dH:13.6 (1.0MD) kJ	5	MTD	30807
7	25	0	Inf. Dilution	dH:15.9 (1.3MD) kJ	5	MTD	30807

Reaction No. 63869



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:0.40 (2SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:1.32 (0.29MD)	5	MSP	30807
3	25	0	Inf. Dilution	lgK:1.34 (0.14MD)	5	MSP	30807
4	25	0	Inf. Dilution	dH:19.4 (1.6MD) kJ	5	MTD	30807
5	25	0	Inf. Dilution	dH:18.6 (1.4MD) kJ	5	MTD	30807

Reaction No. 63870							
3<Acetic-1> + NpO2+1 = NpO2+1_Acetic-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.10 (3SF)	1	EBU	6372

Reaction No. 63871							
4<Acetic-1> + NpO2+1 = NpO2+1_Acetic-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.55 (3SF)	1	EBU	6372

Reaction No. 63872							
Benzoic-1 + NpO2+1 = NpO2+1_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.03 (3SF)	1	EBU	6372
2	25	2	NaClO4	lgK:0.82 (2SF)	5	MSP	11516
Note (2): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113							

Reaction No. 63873							
2<Benzoic-1> + NpO2+1 = NpO2+1_Benzoic-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.92 (2SF)	1	EBU	6372

Reaction No. 63874							
3<Benzoic-1> + NpO2+1 = NpO2+1_Benzoic-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.07 (1SF)	1	EBU	6372

Reaction No. 63875							
4<Benzoic-1> + NpO2+1 = NpO2+1_Benzoic-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.84 (3SF)	1	EBU	6372

Reaction No. 63876							
5<Benzoic-1> + NpO2+1 = NpO2+1_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.33(3SF)	1	EBU	6372

Reaction No. 63877							
Cat-2 + NpO2+1 = NpO2+1_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.55(3SF)	1	EBU	6372

Reaction No. 63878							
2<Cat-2> + NpO2+1 = NpO2+1_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.58(4SF)	1	EBU	6372

Reaction No. 63879							
EtDiAm + NpO2+1 = NpO2+1_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.86(3SF)	1	EBU	6372

Reaction No. 63880							
2<EtDiAm> + NpO2+1 = NpO2+1_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.42(3SF)	1	EBU	6372

Reaction No. 63881							
3<Glycolic-1> + NpO2+1 = NpO2+1_Glycolic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.73(3SF)	1	EBU	6372

Reaction No. 63882							
4<Glycolic-1> + NpO2+1 = NpO2+1_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.53(3SF)	1	EBU	6372

Reaction No. 63883							
Phenol-1 + NpO2+1 = NpO2+1_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:1.94 (3SF)	1	EBU	6372
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Reaction No. 63884							
2<Phenol-1> + NpO2+1 = NpO2+1_Phenol-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.94 (3SF)	1	EBU	6372

Reaction No. 63885							
3<Phenol-1> + NpO2+1 = NpO2+1_Phenol-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.13 (3SF)	1	EBU	6372

Reaction No. 63886							
4<Phenol-1> + NpO2+1 = NpO2+1_Phenol-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.56 (3SF)	1	EBU	6372

Reaction No. 63887							
5<Phenol-1> + NpO2+1 = NpO2+1_Phenol-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.26 (3SF)	1	EBU	6372

Reaction No. 63888							
NpO2+2 + H2O = NpO2+2_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.47 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution/m	lgK:-5.15 (3SF)	3	CRV	24953
3	25	0.01	Unknown/m	lgK:-5.23 (3SF)	2	CRV	24953
4	25	0.1	Unknown/m	lgK:-5.22 (3SF)	2	CRV	24953
5	25	0.2	Unknown/m	lgK:-5.12 (3SF)	2	CRV	24953
6	25	0.5	Unknown/m	lgK:-4.70 (3SF)	2	CRV	24953
7	25	1	Unknown/m	lgK:-3.92 (3SF)	2	CRV	24953
8	25	2	Unknown/m	lgK:-2.27 (3SF)	2	CRV	24953
9	25	3	Unknown/m	lgK:-0.58 (2SF)	0	CRV	24953
10	50	0	Inf. Dilution/m	lgK:-4.57 (3SF)	2	CRV	24953
11	75	0	Inf. Dilution/m	lgK:-4.07 (3SF)	1	CRV	24953
12	100	0	Inf. Dilution/m	lgK:-3.65 (3SF)	1	CRV	24953

13	150	0	Inf. Dilution/m	lgK:-2.97(3SF)	1	CRV	24953
14	200	0	Inf. Dilution/m	lgK:-2.44(3SF)	0	CRV	24953
15	250	0	Inf. Dilution/m	lgK:-2.03(3SF)	0	CRV	24953
16	300	0	Inf. Dilution/m	lgK:-1.69(3SF)	0	CRV	24953
17	25	0	Inf. Dilution	dH:42.96(15.40SD)kJ	4	CRV	32222

Reaction No. 63889							
$\text{NpO}_2+2 + 2\langle\text{H}_2\text{O}\rangle = \text{NpO}_2+2_OH-1(2) + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-11.42(4SF)	0	EBU	6372

Reaction No. 63890							
$\text{NpO}_2+2 + 3\langle\text{H}_2\text{O}\rangle = \text{NpO}_2+2_OH-1(3) + 3\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-17.75(4SF)	0	EBU	6372

Reaction No. 63891							
$\text{NpO}_2+2 + 4\langle\text{H}_2\text{O}\rangle = \text{NpO}_2+2_OH-1(4) + 4\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-24.39(4SF)	0	EBU	6372

Reaction No. 63892							
$\text{NpO}_2+2 + 5\langle\text{H}_2\text{O}\rangle = \text{NpO}_2+2_OH-1(5) + 5\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.32(4SF)	1	EBU	6372

Reaction No. 63893							
$2\langle\text{NpO}_2+2\rangle + 2\langle\text{H}_2\text{O}\rangle = \text{NpO}_2+2(2)_OH-1(2) + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.59(3SF)	0	EBU	6372
2	25	0	Inf. Dilution/m	lgK:-6.39(3SF)	3	CRV	24953
3	25	0.01	Unknown/m	lgK:-6.48(3SF)	2	CRV	24953
4	25	0.1	Unknown/m	lgK:-6.63(3SF)	2	CRV	24953
5	25	0.2	Unknown/m	lgK:-6.70(3SF)	2	CRV	24953

6	25	0.5	Unknown/m	lgK:-6.81(3SF)	2	CRV	24953
7	25	1	Unknown/m	lgK:-6.89(3SF)	2	CRV	24953
8	25	2	Unknown/m	lgK:-6.97(3SF)	1	CRV	24953
9	25	3	Unknown/m	lgK:-7.01(3SF)	0	CRV	24953
10	50	0	Inf. Dilution/m	lgK:-5.73(3SF)	2	CRV	24953
11	75	0	Inf. Dilution/m	lgK:-5.29(3SF)	1	CRV	24953
12	100	0	Inf. Dilution/m	lgK:-4.96(3SF)	1	CRV	24953
13	150	0	Inf. Dilution/m	lgK:-4.54(3SF)	0	CRV	24953
14	200	0	Inf. Dilution/m	lgK:-4.34(3SF)	0	CRV	24953
15	250	0	Inf. Dilution/m	lgK:-4.28(3SF)	0	CRV	24953
16	300	0	Inf. Dilution/m	lgK:-4.33(3SF)	0	CRV	24953
17	25	0	Inf. Dilution	dH:45.00(16.20SD)kJ	4	CRV	32222

Reaction No. 63894



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.48(4SF)	1	EBU	6372

Reaction No. 63895



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.5(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-19.43(4SF)	0	EBU	6372
3	25	0	Inf. Dilution/m	lgK:-17.49(4SF)	3	CRV	24953
4	25	0.01	Unknown/m	lgK:-17.77(4SF)	2	CRV	24953
5	25	0.1	Unknown/m	lgK:-18.21(4SF)	2	CRV	24953
6	25	0.2	Unknown/m	lgK:-18.40(4SF)	2	CRV	24953
7	25	0.5	Unknown/m	lgK:-18.68(4SF)	2	CRV	24953
8	25	1	Unknown/m	lgK:-18.86(4SF)	2	CRV	24953
9	25	2	Unknown/m	lgK:-18.95(4SF)	2	CRV	24953
10	25	3	Unknown/m	lgK:-18.93(4SF)	2	CRV	24953
11	50	0	Inf. Dilution/m	lgK:-16.03(4SF)	2	CRV	24953
12	75	0	Inf. Dilution/m	lgK:-14.89(4SF)	2	CRV	24953
13	100	0	Inf. Dilution/m	lgK:-13.99(4SF)	1	CRV	24953
14	150	0	Inf. Dilution/m	lgK:-12.75(4SF)	1	CRV	24953
15	200	0	Inf. Dilution/m	lgK:-12.03(4SF)	1	CRV	24953

16	250	0	Inf. Dilution/m	lgK:-11.66(4SF)	0	CRV	24953
17	300	0	Inf. Dilution/m	lgK:-11.52(4SF)	0	CRV	24953
18	25	0	Inf. Dilution	dH:110.67(17.65SD)kJ	4	CRV	32222

Reaction No. 63896



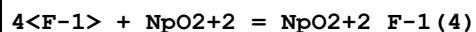
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.68(4SF)	1	EBU	6372

Reaction No. 63897



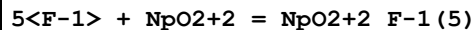
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.62(4SF)	1	EBU	6372

Reaction No. 63898



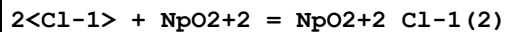
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.92(4SF)	1	EBU	6372

Reaction No. 63899



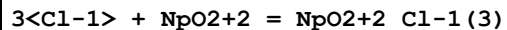
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.07(4SF)	1	EBU	6372

Reaction No. 63900



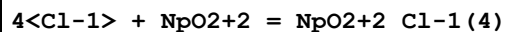
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.3(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.57(2SF)	0	EBU	6372

Reaction No. 63901



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.52(2SF)	1	EBU	6372

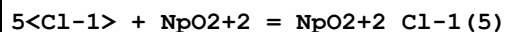
Reaction No. 63902



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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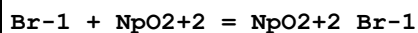
1	25	0	Inf. Dilution	lgK:0.34 (2SF)	1	EBU	6372
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Reaction No. 63903



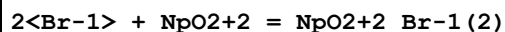
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.05 (1SF)	1	EBU	6372

Reaction No. 63904



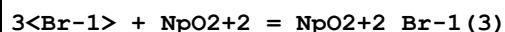
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.28 (2SF)	1	EBU	6372

Reaction No. 63905



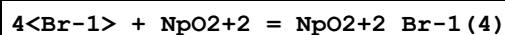
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.27 (2SF)	1	EBU	6372

Reaction No. 63906



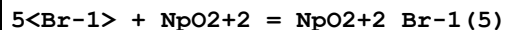
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.07 (1SF)	1	EBU	6372

Reaction No. 63907



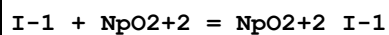
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.25 (2SF)	1	EBU	6372

Reaction No. 63908



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.67 (2SF)	1	EBU	6372

Reaction No. 63909



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.20 (2SF)	1	EBU	6372

Reaction No. 63910

2<I-1> + NpO2+2 = NpO2+2_I-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.10(2SF)	1	EBU	6372

Reaction No. 63911							
3<I-1> + NpO2+2 = NpO2+2_I-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.18(2SF)	1	EBU	6372

Reaction No. 63912							
4<I-1> + NpO2+2 = NpO2+2_I-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.58(2SF)	1	EBU	6372

Reaction No. 63913							
5<I-1> + NpO2+2 = NpO2+2_I-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.08(3SF)	1	EBU	6372

Reaction No. 63914							
2<IO3-1> + NpO2+2 = NpO2+2_IO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.82(3SF)	1	EBU	6372

Reaction No. 63915							
3<IO3-1> + NpO2+2 = NpO2+2_IO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.03(3SF)	1	EBU	6372

Reaction No. 63916							
4<IO3-1> + NpO2+2 = NpO2+2_IO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.94(3SF)	1	EBU	6372

Reaction No. 63917							
5<IO3-1> + NpO2+2 = NpO2+2_IO3-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.63(3SF)	1	EBU	6372

Reaction No. 63918							
$3\text{<SO}_4\text{-2>} + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_SO}_4\text{-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.12(3SF)	1	EBU	6372

Reaction No. 63919							
$4\text{<SO}_4\text{-2>} + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_SO}_4\text{-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.39(3SF)	1	EBU	6372

Reaction No. 63920							
$5\text{<SO}_4\text{-2>} + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_SO}_4\text{-2(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.75(3SF)	1	EBU	6372

Reaction No. 63921							
$\text{NH}_3 + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_NH}_3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.92(3SF)	1	EBU	6372

Reaction No. 63922							
$2\text{<NH}_3> + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_NH}_3(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.42(3SF)	1	EBU	6372

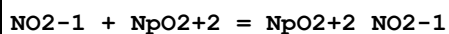
Reaction No. 63923							
$3\text{<NH}_3> + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_NH}_3(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.62(4SF)	1	EBU	6372

Reaction No. 63924							
$4\text{<NH}_3> + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_NH}_3(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.58(4SF)	1	EBU	6372

Reaction No. 63925							
$5\text{<NH}_3> + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_NH}_3(5)$							

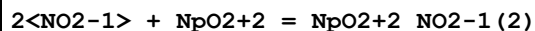
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.32 (4SF)	1	EBU	6372

Reaction No. 63926



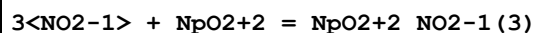
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.64 (2SF)	1	EBU	6372

Reaction No. 63927



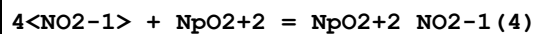
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.57 (2SF)	1	EBU	6372

Reaction No. 63928



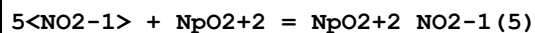
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.06 (1SF)	1	EBU	6372

Reaction No. 63929



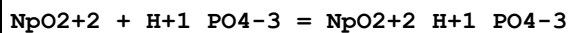
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.81 (2SF)	1	EBU	6372

Reaction No. 63930



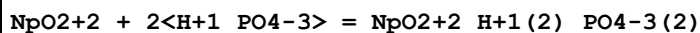
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.96 (3SF)	1	EBU	6372

Reaction No. 63931



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.55 (3SF)	1	EBU	6372

Reaction No. 63932



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.51 (4SF)	1	EBU	6372

Reaction No. 63933							
$\text{NpO}_2+2 + 3\text{<H+1_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(3)_PO}_4-3(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.11(4SF)	1	EBU	6372

Reaction No. 63934							
$\text{NpO}_2+2 + 4\text{<H+1_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(4)_PO}_4-3(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.48(4SF)	1	EBU	6372

Reaction No. 63935							
$\text{NpO}_2+2 + 5\text{<H+1_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(5)_PO}_4-3(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.65(4SF)	1	EBU	6372

Reaction No. 63936							
$\text{NpO}_2+2 + \text{H+1(2)_PO}_4-3 = \text{NpO}_2+2\text{_H+1(2)_PO}_4-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.00(3SF)	1	EBU	6372

Reaction No. 63937							
$\text{NpO}_2+2 + 2\text{<H+1(2)_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(4)_PO}_4-3(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.43(3SF)	1	EBU	6372

Reaction No. 63938							
$\text{NpO}_2+2 + 3\text{<H+1(2)_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(6)_PO}_4-3(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.54(3SF)	1	EBU	6372

Reaction No. 63939							
$\text{NpO}_2+2 + 4\text{<H+1(2)_PO}_4-3\text{>} = \text{NpO}_2+2\text{_H+1(8)_PO}_4-3(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.58(2SF)	1	EBU	6372

Reaction No. 63940							
$\text{CO}_3-2 + \text{NpO}_2+2 = \text{NpO}_2+2\text{_CO}_3-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:9.8 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.18 (3SF)	0	EBU	6372

Reaction No. 63941							
NpO ₂ +2 + 2<CO ₃ -2> = NpO ₂ +2_CO ₃ -2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.9 (3SF)	0	CRV	552
2	25	0	Inf. Dilution	lgK:15.49 (4SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:14.0 (3SF)	3	MEF	10566
4	25	0	Inf. Dilution/m	lgK:14.27 (4SF)	2	CRV	24953
5	25	1	Unknown	lgK:16.51 (4SF)	6	CRV	552
6	25	1	Unknown/m	lgK:16.51 (4SF)	2	CRV	24953
7	50	0	Inf. Dilution/m	lgK:14.73 (4SF)	1	CRV	24953
8	75	0	Inf. Dilution/m	lgK:15.18 (4SF)	1	CRV	24953
9	100	0	Inf. Dilution/m	lgK:15.62 (4SF)	0	CRV	24953
10	150	0	Inf. Dilution/m	lgK:16.48 (4SF)	0	CRV	24953
11	200	0	Inf. Dilution/m	lgK:17.29 (4SF)	0	CRV	24953
12	25	0	Inf. Dilution	dH:-2.4 (2SF)	4	CRV	552

Reaction No. 63942							
NpO ₂ +2 + 3<CO ₃ -2> = NpO ₂ +2_CO ₃ -2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.18 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution/m	lgK:20.21 (4SF)	1	CRV	24953
3	25	1	Unknown	lgK:21.1 (3SF)	6	CRV	552
4	50	0	Inf. Dilution/m	lgK:19.93 (4SF)	1	CRV	24953
5	75	0	Inf. Dilution/m	lgK:19.99 (4SF)	1	CRV	24953
6	100	0	Inf. Dilution/m	lgK:20.29 (4SF)	0	CRV	24953
7	150	0	Inf. Dilution/m	lgK:21.43 (4SF)	0	CRV	24953
8	200	0	Inf. Dilution/m	lgK:23.01 (4SF)	0	CRV	24953
9	25	0	Inf. Dilution	dH:-10.0 (3SF)	4	CRV	552

Reaction No. 63943							
4<CO ₃ -2> + NpO ₂ +2 = NpO ₂ +2_CO ₃ -2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.35 (4SF)	1	EBU	6372

Reaction No. 63944							
$5\text{<CO}_3\text{-2>} + \text{NpO}_2\text{+2} = \text{NpO}_2\text{+2_CO}_3\text{-2(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.05 (4SF)	1	EBU	6372

Reaction No. 63945							
$\text{NpO}_2\text{+2} + \text{CO}_3\text{-2} + \text{H}_2\text{O} = \text{NpO}_2\text{+2_OH-1_CO}_3\text{-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.18 (3SF)	1	EBU	6372

Reaction No. 63946							
$\text{NpO}_2\text{+2} + \text{CO}_3\text{-2} + 2\text{<H}_2\text{O>} = \text{NpO}_2\text{+2_OH-1(2)_CO}_3\text{-2} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.19 (3SF)	1	EBU	6372

Reaction No. 63947							
$2\text{<NpO}_2\text{+2>} + \text{CO}_3\text{-2} + 3\text{<H}_2\text{O>} = \text{NpO}_2\text{+2(2)_OH-1(3)_CO}_3\text{-2} + 3\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.65 (3SF)	1	EBU	6372
2	25	1	Unknown	lgK:-1.49 (3SF)	6	CRV	552

Reaction No. 63948							
$3\text{<NpO}_2\text{+2>} + \text{CO}_3\text{-2} + 3\text{<H}_2\text{O>} = \text{NpO}_2\text{+2(3)_OH-1(3)_CO}_3\text{-2} + 3\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.19 (2SF)	1	EBU	6372

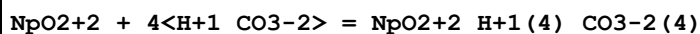
Reaction No. 63949							
$\text{NpO}_2\text{+2} + \text{H+1_CO}_3\text{-2} = \text{NpO}_2\text{+2_H+1_CO}_3\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.57 (3SF)	1	EBU	6372

Reaction No. 63950							
$\text{NpO}_2\text{+2} + 2\text{<H+1_CO}_3\text{-2>} = \text{NpO}_2\text{+2_H+1(2)_CO}_3\text{-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.11 (3SF)	1	EBU	6372

Reaction No. 63951							
$\text{NpO}_2\text{+2} + 3\text{<H+1_CO}_3\text{-2>} = \text{NpO}_2\text{+2_H+1(3)_CO}_3\text{-2(3)}$							

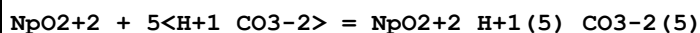
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.86(3SF)	1	EBU	6372

Reaction No. 63952



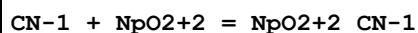
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.92(3SF)	1	EBU	6372

Reaction No. 63953



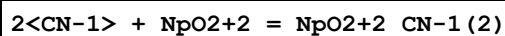
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.36(3SF)	1	EBU	6372

Reaction No. 63954



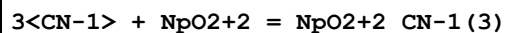
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.18(3SF)	1	EBU	6372

Reaction No. 63955



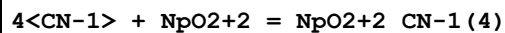
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.91(3SF)	1	EBU	6372

Reaction No. 63956



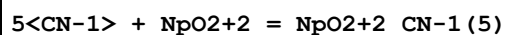
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.32(4SF)	1	EBU	6372

Reaction No. 63957



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.46(4SF)	1	EBU	6372

Reaction No. 63958



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.36(4SF)	1	EBU	6372

Reaction No. 63959							
SCN-1 + NpO2+2 = NpO2+2_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.85 (2SF)	1	EBU	6372

Reaction No. 63960							
2<SCN-1> + NpO2+2 = NpO2+2_SCN-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.32 (3SF)	1	EBU	6372

Reaction No. 63961							
3<SCN-1> + NpO2+2 = NpO2+2_SCN-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.52 (3SF)	1	EBU	6372

Reaction No. 63962							
4<SCN-1> + NpO2+2 = NpO2+2_SCN-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.51 (3SF)	1	EBU	6372

Reaction No. 63963							
5<SCN-1> + NpO2+2 = NpO2+2_SCN-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.31 (3SF)	1	EBU	6372

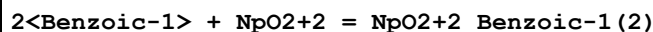
Reaction No. 63964							
4<Acetic-1> + NpO2+2 = NpO2+2_Acetic-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.22 (3SF)	1	EBU	6372

Reaction No. 63965							
5<Acetic-1> + NpO2+2 = NpO2+2_Acetic-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.16 (3SF)	1	EBU	6372

Reaction No. 63966							
Benzoic-1 + NpO2+2 = NpO2+2_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

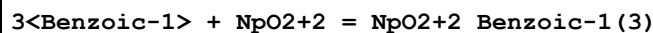
1	25	0	Inf. Dilution	lgK:3.28 (3SF)	1	EBU	6372
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Reaction No. 63967



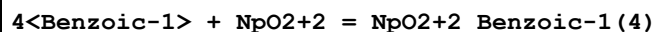
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.72 (3SF)	1	EBU	6372

Reaction No. 63968



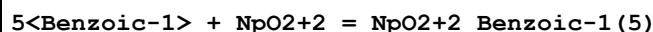
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.58 (3SF)	1	EBU	6372

Reaction No. 63969



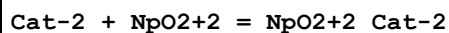
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.96 (3SF)	1	EBU	6372

Reaction No. 63970



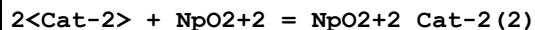
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.90 (3SF)	1	EBU	6372

Reaction No. 63971



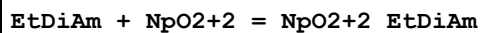
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.35 (4SF)	1	EBU	6372

Reaction No. 63972



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.50 (4SF)	1	EBU	6372

Reaction No. 63973



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.29 (4SF)	1	EBU	6372

Reaction No. 63974

2<EtDiAm> + NpO2+2 = NpO2+2_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.47 (4SF)	1	EBU	6372

Reaction No. 63975							
Gly-1 + NpO2+2 = NpO2+2_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.72 (3SF)	1	EBU	6372

Reaction No. 63976							
2<Gly-1> + NpO2+2 = NpO2+2_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.02 (3SF)	1	EBU	6372

Reaction No. 63977							
4<Glycolic-1> + NpO2+2 = NpO2+2_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.19 (3SF)	1	EBU	6372

Reaction No. 63978							
5<Glycolic-1> + NpO2+2 = NpO2+2_Glycolic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.46 (3SF)	1	EBU	6372

Reaction No. 63979							
Phenol-1 + NpO2+2 = NpO2+2_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.67 (3SF)	1	EBU	6372

Reaction No. 63980							
2<Phenol-1> + NpO2+2 = NpO2+2_Phenol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.72 (4SF)	1	EBU	6372

Reaction No. 63981							
3<Phenol-1> + NpO2+2 = NpO2+2_Phenol-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.30 (4SF)	1	EBU	6372

Reaction No. 63982							
4<Phenol-1> + NpO ₂ +2 = NpO ₂ +2_Phenol-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.44 (4SF)	1	EBU	6372

Reaction No. 63983							
5<Phenol-1> + NpO ₂ +2 = NpO ₂ +2_Phenol-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.17 (4SF)	1	EBU	6372

Reaction No. 65525							
2<Np+3> + 3<CO ₃ -2> = Np+3(2)_CO ₃ -2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.63 (4SF)	0	EBU	6372

Reaction No. 65526							
Np+4 + 2<CO ₃ -2> = Np+4_CO ₃ -2(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.50 (4SF)	4	EBU	6372

Reaction No. 65641							
Np+4 + H+1_CrO ₄ -2 = Np+4_CrO ₄ -2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.2	LiClO ₄	lgK:1.76 (3SF)	5	MGL	590
2	17	0.2	LiClO ₄	lgK:1.78 (3SF)	5	MGL	590
3	25	0.1	Unknown	lgK:1.81 (3SF)	6	CRV	552
4	25	0.2	LiClO ₄	lgK:1.80 (3SF)	5	MGL	590
5	0:30	0.1	Unknown	dH:1. (1SF)	6	CRV	552
6	25	0.2	LiClO ₄	dH:1.17 (0.14SD)	5	MCL	590

Reaction No. 65753							
NpO ₂ +1(2)_CO ₃ -2_(s) = 2<NpO ₂ +1> + CO ₃ -2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.14 (4SF)	4	CRV	552
2	25	1	NaClO ₄	lgK:-10.1 (0.04SD)	5	MGL	6586

Reaction No. 65755							
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$3\text{<NpO}_2+2\text{ _CO}_3-2(3)\text{>} = \text{NpO}_2+2(3)\text{ _CO}_3-2(6) + 3\text{<CO}_3-2\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.0(3SF)	1	ESC	
2	25	3	NaClO ₄	lgK:-10.2(3SF)	6	CRV	552

Reaction No. 65783							
$\text{N}_3-1 + \text{NpO}_2+1 = \text{NpO}_2+1\text{ _N}_3-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	Unknown	lgK:1.08(3SF)	6	CRV	552

Reaction No. 65784							
$2\text{<N}_3-1\text{>} + \text{NpO}_2+1 = \text{NpO}_2+1\text{ _N}_3-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	Unknown	lgK:1.85(3SF)	6	CRV	552

Reaction No. 65785							
$3\text{<N}_3-1\text{>} + \text{NpO}_2+1 = \text{NpO}_2+1\text{ _N}_3-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	Unknown	lgK:2.23(3SF)	6	CRV	552

Reaction No. 66866							
$\text{NpO}_2+2\text{ _OH}-1(2)\text{ _}(s) = \text{NpO}_2+2 + 2\text{<OH}-1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.33(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-21.4(3SF)	0	ENS	773
Note (2): Seems not to accord with other reactions; low wgt to reduce its role							
3	25	1	Unknown	lgK:-20.7(3SF)	0	ENS	773

Reaction No. 67122							
$1.5\text{<Cl}_2(g)\text{>} + \text{Np}(s) = \text{Np}+3\text{ _Cl}-1(3)\text{ _}(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:146.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:145.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:106.3(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:82.78(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:67.07(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-200.0(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-216.0 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-215.5 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-215.2 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-216.2 (4SF)	0	MTD	6422

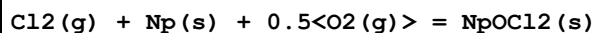
Reaction No. 67123							
$2\text{Cl}_2(\text{g}) + \text{Np}(\text{s}) = \text{NpCl}_4(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:157.5 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:156.4 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:113.5 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:87.77 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:70.62 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-214.8 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-236.0 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-235.4 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-235.0 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-235.8 (4SF)	0	MTD	6422

Reaction No. 67124							
$1.5\text{F}_2(\text{g}) + \text{Np}(\text{s}) = \text{NpF}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:251.5 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:249.9 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:184.4 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:145.2 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:119.0 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-343.2 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-360.0 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-359.3 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-358.7 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-359.5 (4SF)	4	MTD	6422

Reaction No. 67125							
$3\text{F}_2(\text{g}) + \text{Np}(\text{s}) = \text{NpF}_6(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:325.4 (4SF)	4	ENS	6422

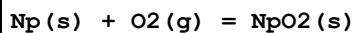
2	27	0	Inf. Dilution	lgK:323.2 (4SF)	4	ENS	6422
3	54	0	Inf. Dilution	lgK:293.8 (4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-443.9 (4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-474.5 (4SF)	4	MTD	6422
6	54	0	Inf. Dilution	dH:-474.2 (4SF)	4	MTD	6422

Reaction No. 67126



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:167.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:166.7 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:122.0 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:95.20 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:77.31 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-229.0 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-245.8 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-245.4 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-245.1 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-246.1 (4SF)	0	MTD	6422

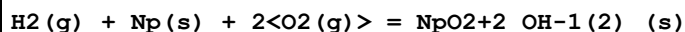
Reaction No. 67127



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:176.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:171.2 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:170.1 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:125.3 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:98.42 (4SF)	0	ENS	6422
6	300	0	Inf. Dilution/m	lgK:97.95 (4SF)	0	CRV	24953
7	327	0	Inf. Dilution	lgK:80.48 (4SF)	0	ENS	6422
8	25	0	Inf. Dilution	dG:-233.5 (4SF)	0	EFE	6422
9	25	0	Inf. Dilution/m	dG:-1021.80 (6SF) kJ	0	CRV	24953
10	25	0	Inf. Dilution	dH:-246.0 (4SF)	0	MTD	6422
11	25	0	Inf. Dilution/m	dH:-1074.00 (6SF) kJ	2	CRV	24953
12	127	0	Inf. Dilution	dH:-245.8 (4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-245.6 (4SF)	0	MTD	6422

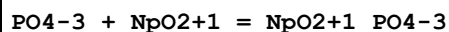
14	327	0	Inf. Dilution	dH:-246.8 (4SF)	0	MTD	6422
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Reaction No. 67128



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:214.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:219.9 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:218.4 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:157.9 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:121.6 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:97.41 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-300.0 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution/m	dG:-1236.00 (6SF) kJ	0	CRV	24953
9	25	0	Inf. Dilution	dH:-332.4 (4SF)	0	MTD	6422
10	25	0	Inf. Dilution/m	dH:-1377.00 (6SF) kJ	2	CRV	24953
11	127	0	Inf. Dilution	dH:-332.2 (4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-331.9 (4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-332.8 (4SF)	0	MTD	6422

Reaction No. 70679

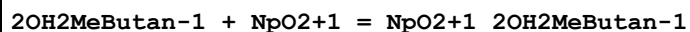


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.1	HClO ₄	lgK:3.11 (0.19SD)	4	MIX	2395
2	20	1	Unknown	lgK:5.78 (3SF)	5	MSL	11516

Note (2): Moskvina A et al., Zhur. Neorg. Khim., 1979, 24, 2449

3	25	0	Inf. Dilution	lgK:2.663 (4SF)	1	EOR	
4	25	0.1	HClO ₄	lgK:2.36 (0.42SD)	4	MIX	2395
5	35	0.1	HClO ₄	lgK:2.06 (0.39SD)	4	MIX	2395
6	10	0.1	HClO ₄	dG:-16.7 (3SF) kJ	4	EFE	2395
7	25	0.1	HClO ₄	dG:-13.4 (3SF) kJ	4	EFE	2395
8	35	0.1	HClO ₄	dG:-12.1 (3SF) kJ	4	EFE	2395
9	25	0.1	HClO ₄	dH:-69.9 (3SF) kJ	4	MTD	2395

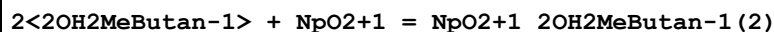
Reaction No. 70826



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:1.60 (3SF)	5	MDS	7703

2	25	1	NaClO4	lgK:1.77 (3SF)	5	MDS	7703
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Reaction No. 70827



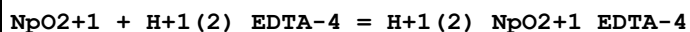
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.21 (3SF)	5	MDS	7703
2	25	1	NaClO4	lgK:2.25 (3SF)	5	MDS	7703

Reaction No. 71066



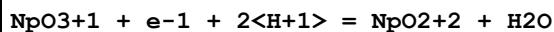
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.962 (3SF)	0	CRV	6330

Reaction No. 71447



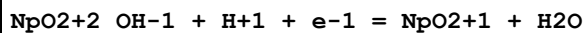
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.42 (3SF)	0	EOD	8120
2	25	0	Inf. Dilution	lgK:4.470 (0.07SD)	6	CRV	20830

Reaction No. 71980



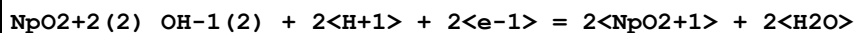
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.04 (3SF)	5	CRV	7761
2	25	0	Inf. Dilution	E:2.04 (3SF)	4	CRV	22551

Reaction No. 71981



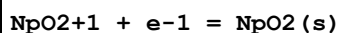
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.54 (3SF)	5	CRV	7761

Reaction No. 71982



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.43 (3SF)	5	CRV	7761

Reaction No. 71983



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.7(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:1.092(4SF)	3	CRV	7761
3	25	0	Inf. Dilution	dH:-95.9(3SF) kJ	3	CRV	7761

Reaction No. 71984							
$\text{NpO}_2 + 1 + 3\langle\text{H}+1\rangle + e-1 = \text{Np}+4_OH-1 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.5(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.479(3SF)	0	CRV	7761

Reaction No. 71985							
$\text{Np}+4_OH-1 + \text{H}+1 + e-1 = \text{Np}+3 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.245(3SF)	5	CRV	7761

Reaction No. 71986							
$\text{NpO}_2(\text{s}) + 4\langle\text{H}+1\rangle + e-1 = \text{Np}+3 + 2\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.369(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:-24.8(3SF) kJ	1	CRV	7761

Reaction No. 71987							
$\text{NpO}_2(\text{s}) + 4\langle\text{H}+1\rangle + 4\langle e-1\rangle = \text{Np}(\text{s}) + 2\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-93.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.418(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:502.2(4SF) kJ	1	CRV	7761

Reaction No. 71988							
$\text{NpO}_4-1_OH-1(2) + 2\langle\text{H}_2\text{O}\rangle + e-1 = \text{NpO}_2+2_OH-1(2)_(\text{s}) + 4\langle OH-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.61(2SF)	0	CRV	7761
Note (2): Low wgt for inter-reaction consistency							

Reaction No. 71989							
$\text{NpO}_4\text{-1_OH-1(2)} + 2\text{<H}_2\text{O>} + \text{e-1} = \text{NpO}_2\text{+2_OH-1(4)} + 2\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.58(2SF)	4	CRV	7761

Reaction No. 71990							
$\text{NpO}_2\text{+2_OH-1(4)} + \text{e-1} = \text{NpO}_2\text{+1_OH-1(s)} + 3\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.1(1SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.5(1SF)	0	CRV	7761

Reaction No. 71991							
$\text{NpO}_2\text{+2_OH-1(2)}\text{(s)} + \text{e-1} = \text{NpO}_2\text{+1_OH-1(s)} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.5(1SF)	0	CRV	7761
2	25	0	Inf. Dilution	dH:-79.9(3SF) kJ	3	CRV	7761

Reaction No. 71992							
$\text{NpO}_2\text{+2_OH-1(4)} + 2\text{<e-1>} = \text{NpO}_2\text{(s)} + 4\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.49(2SF)	0	CRV	7761
Note (2): Low wgt for inter-reaction consistency							

Reaction No. 71993							
$\text{NpO}_2\text{+2_OH-1(2)}\text{(s)} + 2\text{<e-1>} = \text{NpO}_2\text{(s)} + 2\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.48(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:-155.9(4SF) kJ	1	CRV	7761

Reaction No. 71994							
$\text{NpO}_2\text{+1_OH-1(s)} + \text{e-1} = \text{NpO}_2\text{(s)} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.4(1SF)	0	CRV	7761
Note (2): Low wgt for inter-reaction consistency							

3	25	0	Inf. Dilution	dH:-70.2(3SF)kJ	1	CRV	7761
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Reaction No. 71995							
$\text{NpO}_2 + 2\text{OH}^- (4) + e^- = \text{NpO}_2 + 1\text{OH}^- (2) + 2\text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.4(1SF)	0	CRV	7761

Reaction No. 71996							
$\text{NpO}_2 + 2\text{OH}^- (4) + 2e^- = \text{NpO}_2 (\text{am.}, s) + 4\text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.22(2SF)	0	CRV	7761

Reaction No. 71997							
$\text{NpO}_2 + 1\text{OH}^- (2) + e^- = \text{NpO}_2 (\text{am.}, s) + 2\text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.1(1SF)	0	CRV	7761

Reaction No. 71998							
$\text{NpO}_2 (\text{am.}, s) + 2\text{H}_2\text{O} + e^- = \text{Np} + 3\text{OH}^- (3) + \text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.9(2SF)	0	CRV	7761
Note (1): Low wgt for inter-reaction consistency							

Reaction No. 71999							
$\text{NpO}_2 (\text{am.}, s) + 2\text{H}_2\text{O} + 4e^- = \text{Np} (s) + 4\text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.11(3SF)	0	CRV	7761
Note (1): Low wgt for inter-reaction consistency							

Reaction No. 72000							
$\text{Np} + 3\text{OH}^- (3) (\text{am.}, s) + 3e^- = \text{Np} (s) + 3\text{OH}^- (1)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.18(3SF)	3	CRV	7761

Reaction No. 72001							
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Np+3_OH-1(3)_(s) + 3<e-1> = Np(s) + 3<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.22(3SF)	4	CRV	7761
2	25	0	Inf. Dilution	dH:548.5(4SF) kJ	3	CRV	7761

Reaction No. 72002							
NpO2(s) + 2<H2O> + 4<e-1> = Np(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-149.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.246(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:727.5(4SF) kJ	1	CRV	7761

Reaction No. 72003							
NpO2(s) + 2<H2O> + e-1 = Np+3_OH-1(3)_(s) + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.33(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:178.8(4SF) kJ	1	CRV	7761

Reaction No. 72575							
NpO2+1_Acac-1 + Acac-1 = NpO2+1_Acac-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	dH:-3.2(1.0SD)	5	MTD	5767

Reaction No. 72576							
F*3Acac-1 + NpO2+1 = NpO2+1_F*3Acac-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.38(3SF)	5	ENS	2556
Note (1): NIST: 'Value adjusted to listed protonation constant'							
2	25	0.1	NaClO4	lgK:2.57(0.03SD)	4	MPS	5767

Reaction No. 74050							
Np+3_OH-1(3)_(s) = Np+3 + 3<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-22.4(3SF)	0	CRV	22551

Reaction No. 75037							
CrO4-2 + Np+4 = Np+4_CrO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0.01	0	Inf. Dilution	lgK:9.7537 (5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:9.8267 (5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:10.2100 (6SF)	3	EOD	22958
4	100	0	Inf. Dilution	lgK:10.8380 (6SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:11.8000 (6SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:12.9300 (6SF)	3	EOD	22958

Reaction No. 75805							
NpO2+1_Glu-2 + Glu-2 = NpO2+1_Glu-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.4 (2SF)	5	MSE	1586

Reaction No. 75836							
NpO2+1_Asp-2 + Asp-2 = NpO2+1_Asp-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:2.69 (3SF)	5	MSE	1586

Reaction No. 75922							
NpO2+1_Lactic-1 + Lactic-1 = NpO2+1_Lactic-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.67 (2SF)	5	MDS	1586

Reaction No. 75942							
Np+4_Oxalic-2 + Oxalic-2 = Np+4_Oxalic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.5 (2SF)	1	ELC	
2	25	1	NaClO4	lgK:7.41 (3SF)	3	MDS	4706
3	26	1	NaClO4	lgK:7.25 (3SF)	3	MSL	3261

Reaction No. 75978							
NpO2+1_CO3-2 + CO3-2 = NpO2+1_CO3-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9 (2SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:2.40 (3SF)	1	MEF	10566

Reaction No. 76120							
NpO ₂ +2_SO ₄ -2 + SO ₄ -2 = NpO ₂ +2_SO ₄ -2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.7(2SF)	1	ELC	
2	25	2	NaClO ₄	lgK:-0.47(2SF)	0	MDS	5595

Reaction No. 76170							
NpO ₂ +1_Succinic-2 + Succinic-2 = NpO ₂ +1_Succinic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaClO ₄	lgK:0.91(2SF)	2	MSP	3899
2	25	1	NaClO ₄	lgK:0.37(2SF)	3	MDS	1586

Reaction No. 76195							
Np+4_NTA-3 + NTA-3 = Np+4_NTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO ₄	lgK:14.78(4SF)	5	MSP	7555

Reaction No. 76208							
Np+4_HOEDTA-3 + HOEDTA-3 = Np+4_HOEDTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.7(3SF)	1	ELC	
2	25	1	NaClO ₄	lgK:10.75(4SF)	3	MSP	7555

Reaction No. 76242							
Formic-1 + Np+4 = Np+4_Formic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.078(4SF)	1	EOR	
2	25	1	NaClO ₄	lgK:2.88(3SF)	5	MSP	11516
Note (2): Ananyev A et al., Radiokhim., 1984, 26, 635							

Reaction No. 76294							
NpO ₂ +2_Propanoic-1 + Propanoic-1 = NpO ₂ +2_Propanoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO ₄	lgK:2.01(3SF)	5	MGL	2500
2	25	0	Inf. Dilution	lgK:2.0(2SF)	1	ESC	

Reaction No. 76321							
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NpO2+1_Glutaric-2 + Glutaric-2 = NpO2+1_Glutaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.17(2SF)	5	MDS	1586

Reaction No. 76347							
NpO2+1 + Tartaric-2 = NpO2+1_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	NH4ClO4	lgK:2.32(3SF)	5	MIX	11516
Note (1): Moskvina A et al., Zhur. Neorg. Khim., 1961, 6, 926							

Reaction No. 76348							
NpO2+1_Tartaric-2 + Tartaric-2 = NpO2+1_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	NH4ClO4	lgK:1.98(3SF)	5	MIX	11516
Note (1): Moskvina A et al., Zhur. Neorg. Khim., 1961, 6, 926							

Reaction No. 76362							
NpO2+1_Glycolic-1 + Glycolic-1 = NpO2+1_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.49(2SF)	5	MDS	1586
2	25	2	NaClO4	lgK:0.47(2SF)	5	MSP	11516
Note (2): Rizkalla E et al., Radiochim. Acta, 1990, 51, 113							

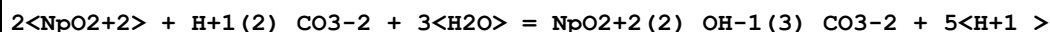
Reaction No. 76387							
Np+3_IButanoic-1 + IButanoic-1 = Np+3_IButanoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:2.50(3SF)	5	MSC	13362

Reaction No. 76517							
Np+4_Oxalic-2(2)_H2O(6)_(s) = Np+4 + 2<Oxalic-2> + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.49(4SF)	1	EOR	
2	26	1	HClO4	lgK:-18.89(0.015SD)	3	EOD	20830
Note (2): p. 626 By solubility in pure and mixed Na+ medium							

Reaction No. 76610							
2<Np(s)> + 2.5<O2(g)> = Np2O5(s)							

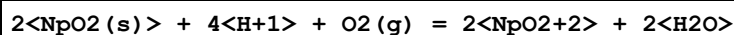
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:348.9(4SF)	1	ELC	
2	25	0	Inf. Dilution/m	dG:-2013.00(6SF)kJ	0	CRV	24953
3	25	0	Inf. Dilution/m	dH:-2148.30(6SF)kJ	1	CRV	24953

Reaction No. 76643



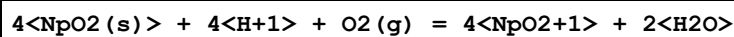
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.3(3SF)	1	ELC	
2	25	1	Unknown/m	lgK:-18.60(4SF)	2	CRV	24953

Reaction No. 76644



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:3.88(3SF)	1	CRV	24953
2	50	0	Inf. Dilution/m	lgK:1.92(3SF)	0	CRV	24953
3	75	0	Inf. Dilution/m	lgK:0.24(2SF)	0	CRV	24953
4	100	0	Inf. Dilution/m	lgK:-1.21(3SF)	0	CRV	24953
5	150	0	Inf. Dilution/m	lgK:-3.59(3SF)	0	CRV	24953
6	200	0	Inf. Dilution/m	lgK:-5.47(3SF)	0	CRV	24953
7	250	0	Inf. Dilution/m	lgK:-6.98(3SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:-8.22(3SF)	0	CRV	24953

Reaction No. 76645



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.2(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:8.47(3SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:5.95(3SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:3.86(3SF)	1	CRV	24953
5	100	0	Inf. Dilution/m	lgK:2.10(3SF)	1	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-0.65(2SF)	1	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-2.67(3SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-4.18(3SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-5.34(3SF)	0	CRV	24953

Reaction No. 76646

2<NpO2+1> + 4<H+1> = 2<Np+3> + O2(g) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.6(3SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:-56.37(4SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-51.92(4SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:-48.13(4SF)	0	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-44.89(4SF)	0	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-39.63(4SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-35.56(4SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-32.34(4SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-29.73(4SF)	0	CRV	24953

Reaction No. 76647							
4<NpO2+1> + 12<H+1> = 4<Np+4> + O2(g) + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40.7(3SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:-39.65(4SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-40.03(4SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:-40.38(4SF)	0	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-40.72(4SF)	1	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-41.33(4SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-41.89(4SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-42.39(4SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-42.85(4SF)	0	CRV	24953

Reaction No. 76648							
4<NpO2+1> + 4<H+1> + O2(g) = 4<NpO2+2> + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:-0.71(2SF)	2	CRV	24953
2	50	0	Inf. Dilution/m	lgK:-2.12(3SF)	2	CRV	24953
3	75	0	Inf. Dilution/m	lgK:-3.38(3SF)	1	CRV	24953
4	100	0	Inf. Dilution/m	lgK:-4.53(3SF)	1	CRV	24953
5	150	0	Inf. Dilution/m	lgK:-6.54(3SF)	0	CRV	24953
6	200	0	Inf. Dilution/m	lgK:-8.26(3SF)	0	CRV	24953
7	250	0	Inf. Dilution/m	lgK:-9.77(3SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:-11.10(4SF)	0	CRV	24953

Reaction No. 76649							
$2\text{<NpO}_2(\text{s})\text{>} + \text{O}_2(\text{g}) + 2\text{<H}_2\text{O}> = 2\text{<NpO}_2+2\text{_{OH-1}(2)_{(s)}}\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:-7.0(2SF)	2	CRV	24953
2	50	0	Inf. Dilution/m	lgK:-8.53(3SF)	0	CRV	24953
3	75	0	Inf. Dilution/m	lgK:-8.97(3SF)	1	CRV	24953
4	100	0	Inf. Dilution/m	lgK:-9.37(3SF)	0	CRV	24953
5	150	0	Inf. Dilution/m	lgK:-10.09(4SF)	0	CRV	24953
6	200	0	Inf. Dilution/m	lgK:-10.71(4SF)	0	CRV	24953
7	250	0	Inf. Dilution/m	lgK:-11.25(4SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:-11.74(4SF)	0	CRV	24953

Reaction No. 76650							
$2\text{<NpO}_2(\text{s})\text{>} + \text{O}_2(\text{g}) + 2\text{<H}_2\text{O}> = 2\text{<NpO}_2+2\text{_{OH-1}(2)_{(am.,s)}}\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:-8.32(3SF)	2	CRV	24953
2	50	0	Inf. Dilution/m	lgK:-8.74(3SF)	1	CRV	24953
3	75	0	Inf. Dilution/m	lgK:-9.11(3SF)	1	CRV	24953
4	100	0	Inf. Dilution/m	lgK:-9.45(3SF)	1	CRV	24953
5	150	0	Inf. Dilution/m	lgK:-10.02(4SF)	0	CRV	24953
6	200	0	Inf. Dilution/m	lgK:-10.50(4SF)	0	CRV	24953
7	250	0	Inf. Dilution/m	lgK:-10.92(4SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:-11.28(4SF)	0	CRV	24953

Reaction No. 76651							
$4\text{<NpO}_2(\text{s})\text{>} + \text{O}_2(\text{g}) = 2\text{<Np}_2\text{O}_5(\text{s})\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.3(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:-10.37(4SF)	2	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-10.38(4SF)	2	CRV	24953
4	75	0	Inf. Dilution/m	lgK:-10.41(4SF)	1	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-10.43(4SF)	1	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-10.50(4SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-10.58(4SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-10.66(4SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-10.74(4SF)	0	CRV	24953

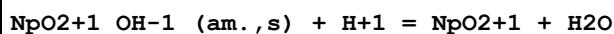
Reaction No. 76652							
$\text{NpO}_2(\text{s}) + 2\text{H}_2\text{O} = \text{Np}^{4+} + \text{OH}^{-}(4)_{-}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	$\lg K: -8.59(3\text{SF})$	2	CRV	24953
2	50	0	Inf. Dilution/m	$\lg K: -8.27(3\text{SF})$	1	CRV	24953
3	75	0	Inf. Dilution/m	$\lg K: -8.02(3\text{SF})$	1	CRV	24953
4	100	0	Inf. Dilution/m	$\lg K: -7.82(3\text{SF})$	1	CRV	24953
5	150	0	Inf. Dilution/m	$\lg K: -7.55(3\text{SF})$	0	CRV	24953
6	200	0	Inf. Dilution/m	$\lg K: -7.39(3\text{SF})$	0	CRV	24953
7	250	0	Inf. Dilution/m	$\lg K: -7.31(3\text{SF})$	0	CRV	24953
8	300	0	Inf. Dilution/m	$\lg K: -7.28(3\text{SF})$	0	CRV	24953

Reaction No. 76653							
$4\text{NpO}_2(\text{s}) + 12\text{H}^{+} = 4\text{Np}^{3+} + \text{O}_2(\text{g}) + 6\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -103.6(4\text{SF})$	1	ELC	
2	50	0	Inf. Dilution/m	$\lg K: -97.88(4\text{SF})$	0	CRV	24953
3	75	0	Inf. Dilution/m	$\lg K: -92.41(4\text{SF})$	0	CRV	24953
4	100	0	Inf. Dilution/m	$\lg K: -87.68(4\text{SF})$	0	CRV	24953
5	150	0	Inf. Dilution/m	$\lg K: -79.91(4\text{SF})$	0	CRV	24953
6	200	0	Inf. Dilution/m	$\lg K: -73.80(4\text{SF})$	0	CRV	24953
7	250	0	Inf. Dilution/m	$\lg K: -68.86(4\text{SF})$	0	CRV	24953
8	300	0	Inf. Dilution/m	$\lg K: -64.80(4\text{SF})$	0	CRV	24953

Reaction No. 76654							
$\text{Np}_2\text{O}_5(\text{s}) + 2\text{H}^{+} = 2\text{NpO}_2 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 9.6(2\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 3.70(2.78\text{SD})$	0	CRV	32222
3	25	0	Inf. Dilution/m	$\lg K: 9.42(3\text{SF})$	0	CRV	24953
4	50	0	Inf. Dilution/m	$\lg K: 8.17(3\text{SF})$	0	CRV	24953
5	75	0	Inf. Dilution/m	$\lg K: 7.13(3\text{SF})$	0	CRV	24953
6	100	0	Inf. Dilution/m	$\lg K: 6.27(3\text{SF})$	0	CRV	24953
7	150	0	Inf. Dilution/m	$\lg K: 4.93(3\text{SF})$	0	CRV	24953
8	200	0	Inf. Dilution/m	$\lg K: 3.95(3\text{SF})$	0	CRV	24953
9	250	0	Inf. Dilution/m	$\lg K: 3.24(3\text{SF})$	0	CRV	24953

10	300	0	Inf. Dilution/m	lgK:2.70 (3SF)	0	CRV	24953
11	25	0	Inf. Dilution	dH:-79.49 (13.26SD) kJ	1	CRV	32222

Reaction No. 76655



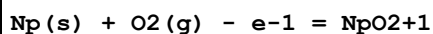
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:4.20 (3SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:3.67 (3SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:3.25 (3SF)	0	CRV	24953
5	100	0	Inf. Dilution/m	lgK:2.91 (3SF)	0	CRV	24953
6	150	0	Inf. Dilution/m	lgK:2.39 (3SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:2.03 (3SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:1.78 (3SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:1.62 (3SF)	0	CRV	24953

Reaction No. 76710



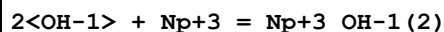
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:91.56 (4SF)	0	CRV	32224
2	25	0	Inf. Dilution	dG:-502.9 (4SF) kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-491.770 (2.8SD) kJ	4	CRV	27292
4	25	0	Inf. Dilution	dG:-489.08 (5SF) kJ	3	CRV	29375
5	25	0	Inf. Dilution	dH:-556.020 (2.1SD) kJ	4	CRV	27292

Reaction No. 76711



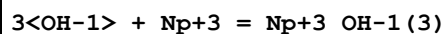
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:158.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-915. (3SF) kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-907.770 (2.8SD) kJ	0	CRV	27292
4	25	0	Inf. Dilution	dG:-921.28 (5SF) kJ	0	CRV	29375
5	25	0	Inf. Dilution	dH:-978.180 (2.3SD) kJ	2	CRV	27292
6	25	0	Inf. Dilution	Cp:-4.000 (12.5SD) J/K	2	CRV	27292

Reaction No. 77646



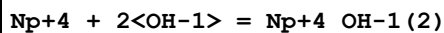
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.74 (4SF)	1	ELC	

Reaction No. 77653



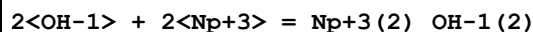
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.0 (3SF)	1	ELC	

Reaction No. 77654



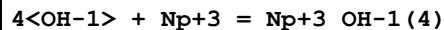
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:28.35 (4SF)	0	EOD	29848

Reaction No. 77656



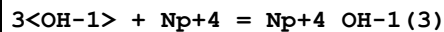
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.0 (4SF)	1	ELC	

Reaction No. 77664



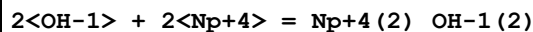
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.0 (4SF)	1	ELC	

Reaction No. 77665



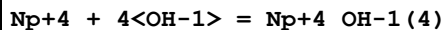
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.30 (4SF)	1	ELC	

Reaction No. 77666



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.0 (4SF)	1	ELC	

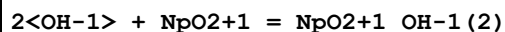
Reaction No. 77673



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.7 (3SF)	1	ELC	

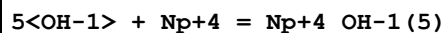
2	25	0	Inf. Dilution	lgK:47.7(3SF)	1	EOD	29848
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Reaction No. 77674



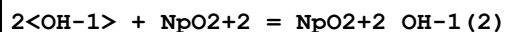
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.1(2SF)	1	ELC	

Reaction No. 77684



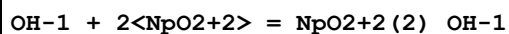
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.7(3SF)	1	ELC	

Reaction No. 77685



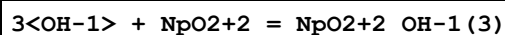
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.6(4SF)	1	ELC	

Reaction No. 77686



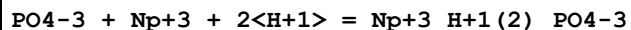
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.5(2SF)	1	ELC	

Reaction No. 77691



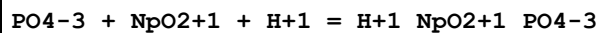
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.4(4SF)	1	ELC	

Reaction No. 77745



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.74(4SF)	1	ELC	

Reaction No. 77758



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.85(4SF)	1	ELC	

Reaction No. 77765

PO4-3 + NpO2+2 + H+1 = NpO2+2_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.85 (4SF)	1	ELC	

Reaction No. 77772							
PO4-3 + NpO2+2 + 2<H+1> = NpO2+2_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.34 (4SF)	1	ELC	

Reaction No. 77793							
3<PO4-3> + Np+3 + 6<H+1> = Np+3_H+1(6)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.96 (4SF)	1	ELC	

Reaction No. 77826							
3<OH-1> + Np+4 + CO3-2 = Np+4_OH-1(3)_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.8 (3SF)	1	ELC	

Reaction No. 77990							
PO4-3 + NpO2+1 + 2<H+1> = H+1(2)_NpO2+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.55 (4SF)	1	ELC	

Reaction No. 78170							
EDTA-4 + NpO2+1 + H+1 = H+1_NpO2+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.99 (4SF)	1	ELC	

Reaction No. 78381							
EDTA-4 + OH-1 + NpO2+1 = NpO2+1_OH-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.89 (4SF)	1	ELC	

Reaction No. 78769							
5<OH-1> + Np+3 = Np+3_OH-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.14 (4SF)	1	ELC	

Reaction No. 78770							
$6\text{<OH-1>} + \text{Np+3} = \text{Np+3_OH-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.12(4SF)	1	ELC	

Reaction No. 78771							
$5\text{<OH-1>} + 3\text{<Np+3>} = \text{Np+3(3)_OH-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.0(4SF)	1	ELC	

Reaction No. 78772							
$\text{PO4-3} + \text{Np+3} + \text{H+1} = \text{Np+3_H+1_PO4-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.7(3SF)	1	ELC	

Reaction No. 78773							
$2\text{<PO4-3>} + \text{Np+3} + 2\text{<H+1>} = \text{Np+3_H+1(2)_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.6(3SF)	1	ELC	

Reaction No. 78774							
$3\text{<PO4-3>} + \text{Np+3} + 3\text{<H+1>} = \text{Np+3_H+1(3)_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.55(4SF)	1	ELC	

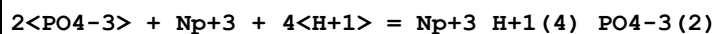
Reaction No. 78775							
$4\text{<PO4-3>} + \text{Np+3} + 4\text{<H+1>} = \text{Np+3_H+1(4)_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:72.01(4SF)	1	ELC	

Reaction No. 78776							
$5\text{<PO4-3>} + \text{Np+3} + 5\text{<H+1>} = \text{Np+3_H+1(5)_PO4-3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:87.57(4SF)	1	ELC	

Reaction No. 78777							
$6\text{<PO4-3>} + \text{Np+3} + 6\text{<H+1>} = \text{Np+3_H+1(6)_PO4-3(6)}$							

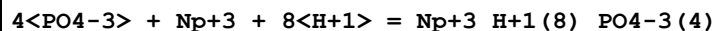
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:102.3 (4SF)	1	ELC	

Reaction No. 78778



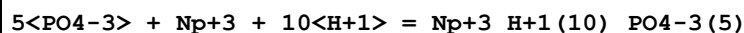
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.79 (4SF)	1	ELC	

Reaction No. 78779



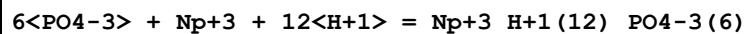
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:80.95 (4SF)	1	ELC	

Reaction No. 78780



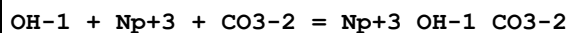
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:98.56 (4SF)	1	ELC	

Reaction No. 78781



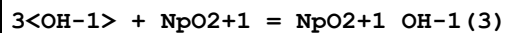
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:115.2 (4SF)	1	ELC	

Reaction No. 78782



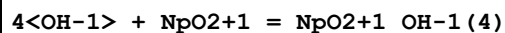
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.69 (4SF)	1	ELC	

Reaction No. 78783



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.12 (4SF)	1	ELC	

Reaction No. 78784



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.22 (4SF)	1	ELC	

Reaction No. 78785							
$5\text{<OH-1>} + \text{NpO2+1} = \text{NpO2+1_OH-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.76(4SF)	1	ELC	

Reaction No. 78786							
$2\text{<PO4-3>} + \text{NpO2+1} + 2\text{<H+1>} = \text{H+1(2)_NpO2+1_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.72(4SF)	1	ELC	

Reaction No. 78787							
$3\text{<PO4-3>} + \text{NpO2+1} + 3\text{<H+1>} = \text{H+1(3)_NpO2+1_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.6(4SF)	1	ELC	

Reaction No. 78788							
$4\text{<PO4-3>} + \text{NpO2+1} + 4\text{<H+1>} = \text{H+1(4)_NpO2+1_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:54.34(4SF)	1	ELC	

Reaction No. 78789							
$5\text{<PO4-3>} + \text{NpO2+1} + 5\text{<H+1>} = \text{H+1(5)_NpO2+1_PO4-3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:64.98(4SF)	1	ELC	

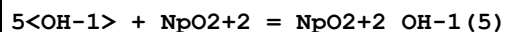
Reaction No. 78790							
$2\text{<PO4-3>} + \text{NpO2+1} + 4\text{<H+1>} = \text{H+1(4)_NpO2+1_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.72(4SF)	1	ELC	

Reaction No. 78791							
$3\text{<PO4-3>} + \text{NpO2+1} + 6\text{<H+1>} = \text{H+1(6)_NpO2+1_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.01(4SF)	1	ELC	

Reaction No. 78792							
$4\text{<OH-1>} + \text{NpO2+2} = \text{NpO2+2_OH-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

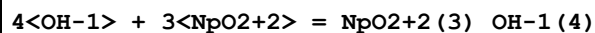
1	25	0	Inf. Dilution	lgK:30.4(3SF)	1	ELC	
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Reaction No. 78793



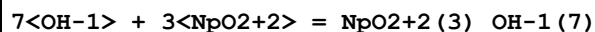
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.68(4SF)	1	ELC	

Reaction No. 78794



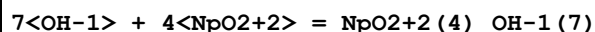
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.52(4SF)	1	ELC	

Reaction No. 78795



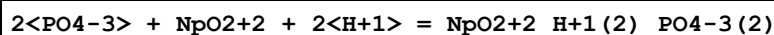
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:66.32(4SF)	1	ELC	

Reaction No. 78796



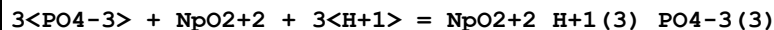
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:71.38(4SF)	1	ELC	

Reaction No. 78797



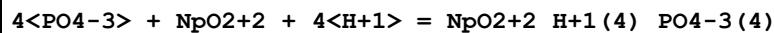
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.16(4SF)	1	ELC	

Reaction No. 78798



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.09(4SF)	1	ELC	

Reaction No. 78799



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.79(4SF)	1	ELC	

Reaction No. 78800

5<PO4-3> + NpO2+2 + 5<H+1> = NpO2+2_H+1(5)_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:85.28(4SF)	1	ELC	

Reaction No. 78801							
2<PO4-3> + NpO2+2 + 4<H+1> = NpO2+2_H+1(4)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.47(4SF)	1	ELC	

Reaction No. 78802							
3<PO4-3> + NpO2+2 + 6<H+1> = NpO2+2_H+1(6)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.1(4SF)	1	ELC	

Reaction No. 78803							
4<PO4-3> + NpO2+2 + 8<H+1> = NpO2+2_H+1(8)_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.5(4SF)	1	ELC	

Reaction No. 78804							
OH-1 + NpO2+2 + CO3-2 = NpO2+2_OH-1_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.18(4SF)	1	ELC	

Reaction No. 78805							
2<OH-1> + NpO2+2 + CO3-2 = NpO2+2_OH-1(2)_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.81(4SF)	1	ELC	

Reaction No. 78806							
3<OH-1> + 2<NpO2+2> + CO3-2 = NpO2+2(2)_OH-1(3)_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.4(3SF)	1	ELC	

Reaction No. 78807							
3<OH-1> + 3<NpO2+2> + CO3-2 = NpO2+2(3)_OH-1(3)_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.19(4SF)	1	ELC	

Reaction No. 79149							
EDTA-4 + NpO ₂ +1 + 2<H+1> = H+1(2)_NpO ₂ +1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.08(4SF)	1	ELC	

Reaction No. 79167							
3<OH-1> + Np+3 = Np+3_OH-1(3)_(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.8(3SF)	1	ELC	

Reaction No. 79455							
2<OH-1> + NpO ₂ +2 = NpO ₂ +2_OH-1(2)_(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.69(4SF)	1	ELC	

Reaction No. 79456							
OH-1 + NpO ₂ +1 = NpO ₂ +1_OH-1_(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.5(3SF)	1	ELC	

Reaction No. 79929							
0.5<H ₂ (g)> + Np(s) + 0.5<O ₂ (g)> - 3<e-1> = Np+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:126.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-738.21(5SF)kJ	0	CRV	29375

Reaction No. 79930							
H ₂ (g) + Np(s) + O ₂ (g) - 2<e-1> = Np+4_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:169.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-978.43(5SF)kJ	0	CRV	29375

Reaction No. 79931							
1.5<H ₂ (g)> + Np(s) + 1.5<O ₂ (g)> - e-1 = Np+4_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:214.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1222.9(5SF)kJ	4	CRV	29375

Reaction No. 79932							
$2\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 2\text{O}_2(\text{g}) = \text{NpO}_2 + \text{H}_2\text{O}(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:243.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1423.2(5SF) kJ	0	CRV	29375

Reaction No. 79933							
$2.5\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 2.5\text{O}_2(\text{g}) + \text{e}^- = \text{NpO}_2 + \text{H}_2\text{O}(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1578.1(5SF) kJ	4	CRV	29375

Reaction No. 79934							
$3\text{C}(\text{s}) + \text{H}_2(\text{g}) + \text{Np}(\text{s}) + 5.5\text{O}_2(\text{g}) + 4\text{e}^- = \text{NpO}_2 + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2686.8(5SF) kJ	4	CRV	29375

Reaction No. 79935							
$5\text{C}(\text{s}) + \text{Np}(\text{s}) + 7.5\text{O}_2(\text{g}) + 6\text{e}^- = \text{NpO}_2 + 5\text{CO}_2(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:585.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3370.9(5SF) kJ	0	CRV	29375

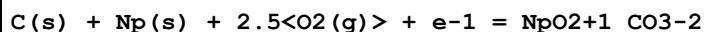
Reaction No. 79936							
$0.5\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 1.5\text{O}_2(\text{g}) = \text{NpO}_2 + \text{H}_2\text{O}(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:190.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1095.6(5SF) kJ	0	CRV	29375

Reaction No. 79937							
$\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 2\text{O}_2(\text{g}) + \text{e}^- = \text{NpO}_2 + \text{H}_2\text{O}(\text{l})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1262.5(5SF) kJ	4	CRV	29375

Reaction No. 79938							
$0.5\text{Cl}_2(\text{g}) + \text{Np}(\text{s}) + \text{O}_2(\text{g}) = \text{NpO}_2 + \text{Cl}_2(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:181.2(4SF)	1	ELC	

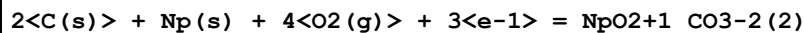
2	25	0	Inf. Dilution	dG:-1028.7 (5SF) kJ	0	CRV	29375
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Reaction No. 79939



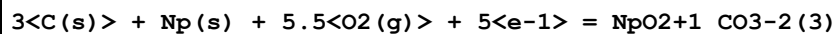
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:256.3 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1474.0 (5SF) kJ	0	CRV	29375

Reaction No. 79940



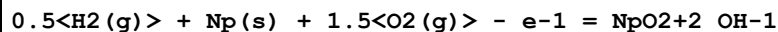
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:350.7 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2014.3 (5SF) kJ	0	CRV	29375

Reaction No. 79941



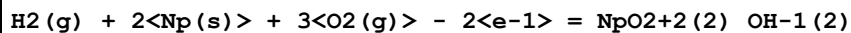
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:446.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2536.4 (5SF) kJ	0	CRV	29375

Reaction No. 79942



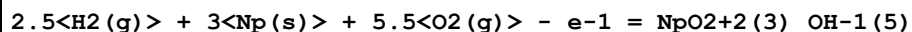
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:174.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1027.9 (5SF) kJ	0	CRV	29375

Reaction No. 79943



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:351.8 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2053.2 (5SF) kJ	0	CRV	29375

Reaction No. 79944



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:603.2 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3497.9 (5SF) kJ	0	CRV	29375

Reaction No. 79945							
$1.5\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 2.5\text{O}_2(\text{g}) + \text{e}^- = \text{NpO}_2 + 2\text{OH}^- (3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:243.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1409.4(5SF) kJ	0	CRV	29375

Reaction No. 79946							
$2\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 3\text{O}_2(\text{g}) + 2\text{e}^- = \text{NpO}_2 + 2\text{OH}^- (4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1584.9(5SF) kJ	4	CRV	29375

Reaction No. 79947							
$\text{C}(\text{s}) + \text{Np}(\text{s}) + 2.5\text{O}_2(\text{g}) = \text{NpO}_2 + 2\text{CO}_3^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:240.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1404.6(5SF) kJ	0	CRV	29375

Reaction No. 79948							
$2\text{C}(\text{s}) + \text{Np}(\text{s}) + 4\text{O}_2(\text{g}) + 2\text{e}^- = \text{NpO}_2 + 2\text{CO}_3^{2-} (2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:337.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1967.3(5SF) kJ	0	CRV	29375

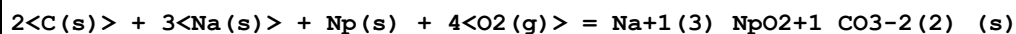
Reaction No. 79949							
$3\text{C}(\text{s}) + \text{Np}(\text{s}) + 5.5\text{O}_2(\text{g}) + 4\text{e}^- = \text{NpO}_2 + 2\text{CO}_3^{2-} (3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:436.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-6.498(4SF) kJ	0	CRV	29375
Note (2): Looks odd!							

Reaction No. 79975							
$2\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 2\text{O}_2(\text{g}) = \text{Np} + 4\text{OH}^- (4)_{(\text{am.}, \text{s})}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1470.5(5SF) kJ	4	CRV	29375

Reaction No. 79976							
$0.5\text{H}_2(\text{g}) + \text{Np}(\text{s}) + 1.5\text{O}_2(\text{g}) = \text{NpO}_2 + \text{OH}^- (1)_{(\text{am.}, \text{s})}$							

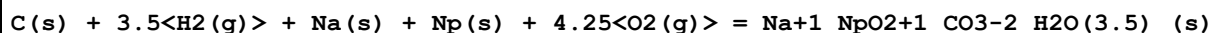
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1130.6(5SF)kJ	4	CRV	29375

Reaction No. 79977



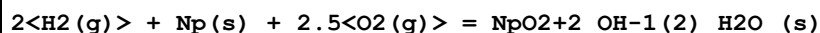
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2839.2(5SF)kJ	4	CRV	29375

Reaction No. 79978



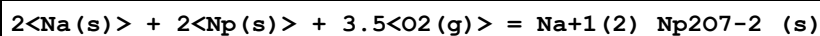
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2603.8(5SF)kJ	4	CRV	29375

Reaction No. 79979



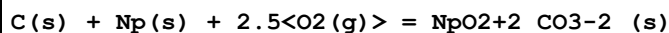
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1499.5(5SF)kJ	4	CRV	29375

Reaction No. 79980



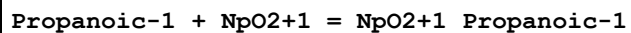
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2729.2(5SF)kJ	4	CRV	29375

Reaction No. 79981



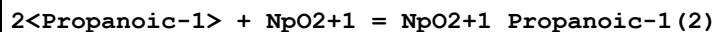
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1433.7(5SF)kJ	4	CRV	29375

Reaction No. 80389



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.31(0.27MD)	5	MSP	30807
2	25	0	Inf. Dilution	dH:14.8(1.6MD)kJ	5	MTD	30807

Reaction No. 80390



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.33(0.29MD)	5	MSP	30807

2	25	0	Inf. Dilution	dH:19.0(2.1MD)kJ	5	MTD	30807
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Reaction No. 80597							
Np+4 + 4<OH-1> = NpO2(am.,s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.40(0.5SD)	3	CRV	31301
2	25	0	Inf. Dilution	lgK:56.70(0.25SD)	2	EOD	31301

Reaction No. 80598							
Np+4 + CO3-2 + 4<OH-1> = Np+4_OH-1(4)_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.20(0.5SD)	4	CRV	31301

Reaction No. 80641							
Citric-3 + Np+4 = Np+4_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.98(4SF)	4	CRV	31301

Reaction No. 80642							
Np+4 + H+1_Citric-3 = Np+4_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.35(3SF)	4	CRV	31301

Reaction No. 80643							
NpO2+1 + H+1_Citric-3 = H+1_NpO2+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.56(3SF)	4	CRV	31301

Reaction No. 81110							
NpO2(am.,s) + 4<H+1> = Np+4 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.2(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.70(0.50SD)	0	CRV	32222
3	25	0	Inf. Dilution	dH:-81.16(8.79SD)kJ	3	CRV	32222

Reaction No. 81111							
Np+3 + CO3-2 + H2O = Np+3_OH-1_CO3-2(s) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

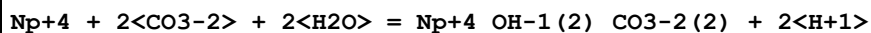
1	25	0	Inf. Dilution	lgK:6.06(3SF)	4	CRV	32222
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Reaction No. 81112



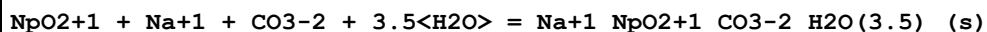
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.83(1.13SD)	4	CRV	32222

Reaction No. 81113



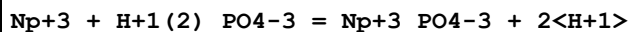
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.17(1.50SD)	4	CRV	32222

Reaction No. 81114



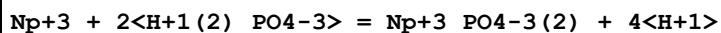
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-30.99(15.31SD) kJ	4	CRV	32222

Reaction No. 81115



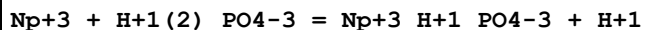
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.83(0.61SD)	4	CRV	32222

Reaction No. 81116



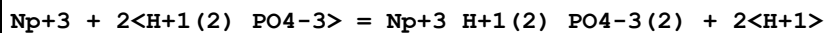
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.57(1.01SD)	4	CRV	32222

Reaction No. 81117



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.78(0.67SD)	4	CRV	32222

Reaction No. 81118



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.38(1.01SD)	4	CRV	32222

Reaction No. 81119

Np+3 + 2<H+1(2)_PO4-3> = Np+3_H+1(2)_PO4-3(2)_(s) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.06(0.51SD)	4	CRV	32222

Reaction No. 81120							
NpO2+2 + H+1(2)_PO4-3 = NpO2+2_H+1_PO4-3_(s) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:92.21(4SF) kJ	4	CRV	32222

Reaction No. 81518							
NpO2+3 + 4<H+1> + 3<e-1> = Np+4 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.64(4SF)	4	CRV	32224

Reaction No. 81519							
Np+3_OH-1(3)_(s) + 3<H+1> = Np+4 + 3<H2O> + e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:19.42(4SF)	0	CRV	32224

Reaction No. 81520							
NpO2+1_OH-1_(s) + 5<H+1> + e-1 = Np+4 + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.81(4SF)	4	CRV	32224

Reaction No. 81521							
NpO2+2_OH-1(2)_(s) + 6<H+1> + 2<e-1> = Np+4 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:35.61(4SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability

7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EBU	Brown's Unified Theory
EDH	Debye-Hueckel corrections
EDT	Debye-Hueckel & Temperature corrections
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
EVK	Variation of lgK with T
MAX	Anion exchange
MCE	Cation exchange
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MEP	Electrophoresis
MGL	Glass electrode
MIX	Ion exchange

MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPH	pH method, not specified
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MQH	Quinhydrone electrode
MRX	Emf with redox electrode
MSC	Miscellaneous
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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